Braid Monodromy Rigidity on One-Dimensional Symmetric Cyclic Prym Slices: Adjoint $\mathfrak{sl}_4$-Dichotomy, Ramanujan-Sum Trace Averages, and Length-Three Cyclotomic Separation

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Abstract

Let $m \geq 3$ be odd and let $S = \mathbf{P}^1_\lambda \setminus \{0, \infty, 1, -1\} \subset \mathcal{M}_{0,6}$ be the one-dimensional symmetric slice of degree-m cyclic covers $C_\lambda \to \mathbf{P}^1$ branched at $(-1, -\lambda, 0, \lambda, 1, \infty)$ with a $(2, 2, 1, 1)$-symmetric weight tuple $\mathbf{a}$ under an involution $h \in (\mathbf{Z}/m\mathbf{Z})^\times$ ($H = \{1, h\}$). Unlike on $\mathcal{M}_{0,6}$, under the local non-resonance conditions of Lemma 2.1 (satisfied by all units $t \in (\mathbf{Z}/m\mathbf{Z})^\times$ on our CRT families), every puncture of S is a simultaneous multi-point collision whose local monodromy generator $g_p(t)$ ($p \in \{1, -1, 0, \infty\}$) on $V_t = H_1(C_\lambda, \mathbf{C})_t$ and non-scalar powers $g_p(t)^k \notin \mathbf{C}^\times I_4$ have $\operatorname{rank}(g_p(t)^k - cI_4) \geq 2$ for all $c \in \mathbf{C}^\times$, excluding rank-1 pseudoreflections and transvections; moreover, for composite $m = a \cdot m'$ ($\gcd(a, m') = 1$), the multiplier $\upsilon \equiv (-1, +1) \pmod{(a, m')} \notin \pm H$ is locally isospectral with $t = 1$ and collides on the length-2 braid words $g_1 g_0$ and $g_1 g_{-1}$.

We establish an unconditional global braid-monodromy rigidity and cyclotomic separation theory on S: (1) cyclotomic linear disjointness in the $g_1(t)$-eigenbasis yields the adjoint dichotomy $\mathfrak{g} \in \{\{0\}, \mathfrak{sl}_4(\mathbf{C})\}$ (Theorem A); (2) for $m = 15N$ ($\gcd(N, 30) = 1$), Ramanujan-sum trace averages of $(g_1 g_0)^2$ prove factorwise Zariski density in $\mathrm{SU}(J(t))$ up to μ_{15N} scalars, and dual-basis expansion of $g_1^2 g_0$ over $\mathbf{Q}(\zeta_N)$ separates all non-trivial cosets $[t] \neq [\pm 1]$ (Theorem B); (3) $g_1^2 g_0$ uniformly separates υ from $\pm H$ for coprime odd $a, m' \geq 3$ (Theorem C); and (4) Galois involution projections extend density and length-3 separation to three-factor conductors $m = a \cdot b \cdot N$ ($\gcd(N, 30) = 1$), alongside four method-boundary examples (Theorem D). By Goursat's lemma, this determines the generic derived Mumford–Tate group $\mathrm{Hg}(B_{\eta,\mathrm{prim}})_{\mathrm{der}} = \prod_{[t]} \mathrm{SU}(J(t))$ and endomorphism algebra $\mathbf{Q}(\zeta_m)^H$, and on characterwise $(2, 2)$-balanced subfamilies ($d_t = 3$) gives $\mathrm{Hg}(B_{\eta,\mathrm{prim}}) = \prod_{[t]} \mathrm{SU}(2, 2)$, reducing all-powers Hodge algebraicity to the separate determinant-cycle input on individual factors (Corollary E).

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1. Introduction: One-Dimensional Symmetric Slices Beyond Rank-1 Reflection Generation

1.1 Monodromy on Full Configuration Spaces versus One-Dimensional Symmetric Slices

Let $m \geq 3$ be an odd integer. Over the full three-dimensional configuration space $\mathcal{M}_{0,6}$ of six distinct ordered points $(z_1, \ldots, z_6)$ on $\mathbf{P}^1$, smooth projective cyclic covers

$$C_z : y^m = \prod_{j=1}^{5} (x - z_j)^{a_j}, \qquad \sum_{j=1}^{6} a_j \equiv 0 \pmod{m}, \quad \gcd(m, a_1, \ldots, a_6) = 1,$$

give rise to variations of weight-1 Hodge structure whose character eigenspaces $V_t := H_1(C_z, \mathbf{C})_t$ ($t \in (\mathbf{Z}/m\mathbf{Z})^\times$) have dimension $n = 6 - 2 = 4$ whenever $m \nmid t a_j$ for all $1 \leq j \leq 6$. On the full space $\mathcal{M}_{0,6}$, the pure braid group P_5 acts

on V_t via the reduced Gassner representation. Because each codimension-1 boundary stratum of $\mathcal{M}_{0,6}$ corresponds to the simple coalescence of a single pair of branch points $z_i \to z_j$, the local monodromy around each boundary divisor is a **rank-1 unitary pseudoreflection** $R_{i,j}(t)$ satisfying $\operatorname{rank}(R_{i,j}(t) - I_4) = 1$. In the classical framework of Deligne–Mostow [1, Proposition 9.2], Rohde [2, Proposition 3.3.5], and the general balanced cyclic Prym theory on $\mathcal{M}_{0,n+2}$ [3, Lemma 6.2], this abundance of rank-1 pseudoreflections serves two indispensable purposes: 1. **Factorwise Zariski density**: By the classification of irreducible unitary reflection groups and inductive restriction to positive-dimensional boundary strata (e.g., [2, Theorem 5.1.1]), a connected group generated by rank-1 pseudoreflections preserving a non-degenerate Hermitian form $J(t)$ of signature (p_t, q_t) is automatically Zariski-dense in $\mathrm{SU}(J(t))$ as soon as it is infinite and irreducible. 2. **Pairwise projective non-equivalence via local puncture spectra**: On $\mathcal{M}_{0,6}$, the single-collision loop around $z_i = z_j$ has a unique non-trivial eigenvalue $\zeta_m^{t(a_i+a_j)}$ of multiplicity 1 alongside the eigenvalue 1 of multiplicity 3. Because the set of eigenvalue multiplicities $\{3, 1\}$ is asymmetric, no non-trivial scalar multiple of $R_{i,j}(t)$ can restore an eigenvalue of multiplicity 3. Consequently, any projective equivalence $\mathbf{P}\rho_t \simeq \mathbf{P}\rho_s$ on $\pi_1(\mathcal{M}_{0,6})$ forces literal equality of all pairwise exponents $t(a_i + a_j) \equiv s(a_i + a_j) \pmod m$ (cf. [3, Lemma 6.2]).

The situation changes fundamentally when we restrict the family to a **one-dimensional symmetric parameter slice**

$$S := \mathbf{P}^1_\lambda \setminus \{0, \infty, 1, -1\} \hookrightarrow \mathcal{M}_{0,6}, \qquad \lambda \longmapsto (-1, -\lambda, 0, \lambda, 1, \infty), \tag{1.1}$$

adapted to a cyclic Prym involution. Specifically, suppose there is an element $h \in (\mathbf{Z}/m\mathbf{Z})^\times$ with $h^2 \equiv 1 \pmod m$ such that the ordered branch weight tuple $\mathbf{a} = (a_0, a_\infty, a_1, a_{-1}, a_\lambda, a_{-\lambda}) \in \{1, \dots, m-1\}^6$ at $(0, \infty, 1, -1, \lambda, -\lambda)$ satisfies the $(2, 2, 1, 1)$-symmetric congruence

$$ha_0 \equiv a_0, \quad ha_\infty \equiv a_\infty, \quad ha_1 \equiv a_{-1}, \quad ha_{-1} \equiv a_1, \quad ha_\lambda \equiv a_{-\lambda}, \quad ha_{-\lambda} \equiv a_\lambda \pmod m. \tag{1.2}$$

Then for every $\lambda \in S$, the curve

$$C_\lambda : y^m = x^{a_0}(x-1)^{a_1}(x+1)^{a_{-1}}(x-\lambda)^{a_\lambda}(x+\lambda)^{a_{-\lambda}} \tag{1.3}$$

admits on the smooth projective relative normalization $\mathcal{C}_S \to S$ of (1.3) (cf. Rohde [2, Section 1.2]) the canonical lifted S-involution $\tau_h(x, y) = \big(-x,\, \varepsilon_h r_\lambda(x) y^h\big)$, where $A_{\mathrm{fin}} := a_0 + a_1 + a_{-1} + a_\lambda + a_{-\lambda}$, $\varepsilon_h := (-1)^{A_{\mathrm{fin}}} \in \{\pm 1\}$, and $r_\lambda(x) := x^{q_0}(x-1)^{q_1}(x+1)^{q_{-1}}(x-\lambda)^{q_\lambda}(x+\lambda)^{q_{-\lambda}} \in \mathbf{Q}(\lambda)(x)^\times$ with integer exponents $q_0 = (1-h)a_0/m$, $q_{\pm 1} = (a_{\mp 1} - ha_{\pm 1})/m$, and $q_{\pm\lambda} = (a_{\mp\lambda} - ha_{\pm\lambda})/m$ from (1.2). Because m is odd, $m\big((h+1)A_{\mathrm{fin}} + \sum_{z \neq \infty} q_z\big) = \big(h(m-1) + (m+1)\big)A_{\mathrm{fin}} \equiv 0 \pmod{2m}$, so $(h+1)A_{\mathrm{fin}} + \sum_{z \neq \infty} q_z$ is even and $\tau_h^2 = \mathrm{id}$ for every integer lift $h \in \mathbf{Z}$; in the infinity chart $(u, \tilde{y}) = (1/x, x^{-D}y)$ $(D = (A_{\mathrm{fin}} + a_\infty)/m)$, $D(1-h) - \sum_{z \neq \infty} q_z = (1-h)a_\infty/m =: q_\infty \in \mathbf{Z}$ and $(-1)^{D + A_{\mathrm{fin}}} = (-1)^{a_\infty}$, so τ_h extends biregularly across $x = \infty$ and commutes with the $\pi_1(S)$-monodromy action and the cyclic deck group μ_m ($\tau_h \circ \zeta = \zeta^h \circ \tau_h$) up to $H := \{1, h\} \subset (\mathbf{Z}/m\mathbf{Z})^\times$. The associated Prym abelian variety $B_\lambda := (1 + \tau_h)\operatorname{Jac}(C_\lambda)$ decomposes over $\mathbf{Q}$ up to isogeny into rational abelian subvarieties indexed by H-orbits of divisors of m, with its primitive isotypic summand $B_{\lambda,\mathrm{prim}}$ carrying endomorphism action by the H-fixed cyclotomic subfield $L := \mathbf{Q}(\zeta_m)^H$ of degree $\varphi(m)/2$ over $\mathbf{Q}$. After extending scalars from $\mathbf{Q}$ to $\mathbf{C}$, the homology local system $R_1\pi_*\mathbf{C}_{\mathrm{prim}}$ splits into 4-dimensional complex embedding eigenspaces $\mathbf{V}_{[t]}$ indexed by primitive character cosets $[t] \in \mathcal{T}_{\mathrm{prim}} := (\mathbf{Z}/m\mathbf{Z})^\times/H$ (which pair over $\mathbf{R}$ into conjugate real variations $(\mathbf{V}_{[t]} \oplus \mathbf{V}_{[-t]})_{\mathbf{R}}$ indexed by $[t] \in (\mathbf{Z}/m\mathbf{Z})^\times/(\pm H)$), where for each $t \in (\mathbf{Z}/m\mathbf{Z})^\times$ the 4-dimensional local system $\mathbf{V}_t$ over S is governed by the restriction of the reduced Gassner representation to the fundamental group

$$\pi_1(S, \lambda_0) = \langle g_1, g_0, g_{-1}, g_\infty \mid g_1 g_0 g_{-1} g_\infty = 1 \rangle \simeq F_3. \tag{1.4}$$

On the one-dimensional slice S, **both pillars of the full-configuration-space proof collapse simultaneously**: - **Absence of local puncture rank-1 pseudoreflections under non-resonance**: Because the involution $x \mapsto -x$ forces the moving branch points $\pm\lambda$ to collide simultaneously with ± 1 as $\lambda \to \pm 1$ ($g_{\pm 1} = \sigma_1\sigma_4$ up to braid conjugacy) and simultaneously with 0 as $\lambda \to 0$ ($g_0 = (\sigma_2\sigma_3\sigma_2)^2$, the full twist on the three inner strands $-\lambda, 0, \lambda$), every puncture of S is a 0-dimensional multi-point stratum. Under the local non-resonance conditions $b_1b_2 \neq 1$, $b_4b_5 \neq 1$, $b_1b_4 \neq 1$, $b_2b_5 \neq 1$, $w(t) \neq 1$, and $b_1b_3b_5 \neq 1$ of Lemma 2.1 (which hold for every unit $t \in (\mathbf{Z}/m\mathbf{Z})^\times$ on our CRT families (1.7)–(1.9), whereas resonant $(2, 2, 1, 1)$-symmetric tuples with $w(t) = 1$—such as $m = 15, h = 4, \mathbf{a} = (5, 10, 1, 4, 2, 8)$ at $t = 1$—can produce a rank-1 unipotent transvection $g_0(1) - I_4 \neq 0$ with $(g_0(1) - I_4)^2 = 0$), every local puncture generator $g_p(t) \in \mathrm{GL}_4(\mathbf{C})$ ($p \in \{1, -1, 0, \infty\}$) has two eigenvalues of multiplicity 2 or two 1-dimensional non-trivial eigenvalues alongside a 2-dimensional 1-eigenspace, so $\operatorname{rank}(g_p(t) - cI_4) \geq 2$ and $\operatorname{rank}(g_p(t)^k - cI_4) \geq 2$ for all non-scalar powers

$g_p(t)^k \notin \mathbf{C}^\times I_4$ and all $c \in \mathbf{C}^\times$. None of the local puncture generators or their non-scalar powers is a rank-1 pseudoreflection or transvection, so boundary-reflection generation is unavailable on S. - **Breakdown of local puncture-spectrum and length-2 rigidity**: Because $\mathrm{Spec}(g_0(t)) = \{1, 1, w(t), w(t)\}$ ($w(t) = \zeta_m^{t(a_0+a_\lambda+a_{-\lambda})}$) has balanced multiplicity $(2,2)$, scaling $g_0(t)$ by $w(t)^{-1} \in \mathbf{C}^\times$ yields $\{w(t)^{-1}, w(t)^{-1}, 1, 1\} = \mathrm{Spec}(g_0(vt))$ whenever $w(vt) = w(t)^{-1}$. For any odd composite conductor $m = a \cdot m'$ with $\gcd(a, m') = 1$, $a \mid a_{\pm 1}, a_{\pm\lambda}$, $m' \mid (a_0 + a_\lambda + a_{-\lambda})$, and $a_1 + a_{-1} \equiv a_\lambda + a_{-\lambda} \not\equiv 0 \pmod m$, the CRT multiplier $v \equiv (-1, +1) \pmod{(a, m')}$ lies outside $\pm H$, yet $\rho_v(g_p)$ and $\rho_1(g_p)$ have **identical projective spectra and Jordan canonical forms at all four punctures** $p \in \{1, -1, 0, \infty\}$, and their projective trace invariants collide identically on the length-2 braid words $g_1 g_0$ and $g_1 g_{-1}$.

1.2 Organization of Main Results

To overcome the simultaneous failure of local reflection generation and local spectrum rigidity on S, we develop a global, characteristic-zero method organized into four theorems (**Theorems A–D**) and a geometric application (**Corollary E**), stated precisely in Section 1.3 and proved in Sections 2–8: - **Theorem A (Sections 2–3)** establishes the rank-≥ 2 property of local puncture generators and their non-scalar powers ($g_p(t)^k \notin \mathbf{C}^\times I_4$) under non-resonance, the local-spectrum and length-2 trace collisions of $v \equiv (-1, +1) \pmod{(a, m')}$, the closed-form tridiagonal invariant Hermitian form $J(t) = -iH(\mathbf{b})$ of signature $(d_t - 1, 5 - d_t)$, and the exact adjoint dichotomy $\mathfrak{g} \in \{\{0\}, \mathfrak{sl}_4(\mathbf{C})\}$ in the $g_1(t)$-eigenbasis via $\mathbf{Q}(\zeta_a) \cap \mathbf{Q}(\zeta_{bN}) = \mathbf{Q}$. - **Theorem B (Sections 4–5)** proves unconditional factorwise Zariski density for $m = 15N$ ($\gcd(N, 30) = 1$) by evaluating the Galois mean of $|\mathrm{tr}((g_1(t)g_0(t))^2)|^2$ via Chebyshev–Laurent polynomials and Ramanujan sums $c_N(r)$, and proves complete length-3 separation by $W_1 = g_1^2 g_0$ via the dual basis $\mathcal{B}_{15}$ of $\mathbf{Q}(\zeta_{15N})$ over the field $\mathbf{Q}(\zeta_N)$ (bypassing $\Phi_N(z)$ for composite N). - **Theorem C (Section 6)** proves uniform-in-m length-3 separation of the local-collision multiplier $v \equiv (-1, +1) \pmod{(a, m')}$ from $\pm H$ for arbitrary coprime odd $a, m' \geq 3$. - **Theorem D (Section 7)** extends unconditional Zariski density and length-3 separation to the three-factor generalized primary CRT family $m = a \cdot b \cdot N$ ($\gcd(N, 30) = 1$) via four commuting Galois involution projectors $\Pi_{\epsilon_a, \epsilon_b}$ and a field-norm argument on $\mathbf{Q}(\zeta_a) \otimes \mathbf{Q}(\zeta_b) \otimes \mathbf{Q}(\zeta_N)$, and analyzes four method-boundary examples showing where complete-residue-block Kubert balance and companion (t_a, t_b)-independence break down on the companion-template examples (b.1)–(b.2), where the global-average shortcut fails inside Theorem D(1) on the generalized primary family example (b.3) at $(5, 3, 7)$, and where the 7-eigenvalue adjoint spectrum collapses at $N = 3$ on the primary-template example (b.4) outside $\gcd(N, 30) = 1$. - **Corollary E (Section 8)** combines Theorems A–D with Goursat's lemma to determine the generic derived Mumford–Tate group $\mathrm{Hg}(B_{\eta,\mathrm{prim}})_{\mathrm{der}} = \prod_{[t]} \mathrm{SU}(J(t))$ and endomorphism algebra $\mathbf{Q}(\zeta_m)^H$, contrasts unbalanced families with characterwise $(2,2)$-balanced subfamilies ($d_t = 3$), and obtains the balanced all-powers Hodge reduction. Section 9 tabulates the explicit cyclotomic invariants across representative conductors.

1.3 Precise Statements of Main Theorems

Throughout this paper, for any odd integer $m \geq 3$, a branch weight tuple $\mathbf{a} = (a_0, a_\infty, a_1, a_{-1}, a_\lambda, a_{-\lambda}) \in \{1, \ldots, m-1\}^6$ at the six ordered punctures $(0, \infty, 1, -1, \lambda, -\lambda)$ (reordered along the real axis as $(z_1, z_2, z_3, z_4, z_5, z_6) = (-1, -\lambda, 0, \lambda, 1, \infty)$) is called a $(2, 2, 1, 1)$**-symmetric tuple** with respect to an involution $h \in (\mathbf{Z}/m\mathbf{Z})^\times$ ($h^2 \equiv 1 \pmod m$) if $\sum_i a_i \equiv 0 \pmod m$, $\gcd(\mathbf{a}, m) = 1$, and (1.2) holds. For each unit $t \in (\mathbf{Z}/m\mathbf{Z})^\times$ and $\zeta := e^{2\pi i t/m}$, we set

$$(b_1, b_2, b_3, b_4, b_5) := \left(\zeta^{a_{-1}}, \zeta^{a_{-\lambda}}, \zeta^{a_0}, \zeta^{a_\lambda}, \zeta^{a_1}\right), \qquad w(t) := b_2 b_3 b_4 = \zeta^{a_0+a_\lambda+a_{-\lambda}}, \qquad P_{k+1}(t) := \prod_{j=1}^{k+1} b_j, \quad (1.5)$$

and for any braid word $W \in \pi_1(S, \lambda_0)$ we define the PGL_4-invariant normalized trace fourth power

$$I_t(W) := \frac{\mathrm{tr}\big(\rho_t(W)\big)^4}{\det\big(\rho_t(W)\big)} \in \mathbf{Q}(\zeta_m). \quad (1.6)$$

We consider three infinite Chinese Remainder Theorem (CRT) families of $(2, 2, 1, 1)$-symmetric tuples: - **Primary CRT Family** ($m = 15N$, $\gcd(N, 30) = 1$): With $h \equiv (+1, +1, -1) \pmod{(3, 5, N)}$ and $H = \{1, h\}$,

$$\mathbf{a}_{\mathrm{prim}} \equiv (1, 2, 0, 0, 0, 0) \pmod 3, \qquad \mathbf{a}_{\mathrm{prim}} \equiv (1, 1, 2, 2, 2, 2) \pmod 5, \qquad \mathbf{a}_{\mathrm{prim}} \equiv (0, 0, 1, -1, 2, -2) \pmod N. \quad (1.7)$$

- **Companion CRT Family** ($m = 15N, \gcd(N,30) = 1, c_1 \in (\mathbf{Z}/N\mathbf{Z})^\times$): With $h \equiv (+1,-1,+1) \pmod{(3,5,N)}$ and $H = \{1,h\}$,

$$\mathbf{a}_{\text{comp}} \equiv (2,1,0,0,0,0) \pmod 3, \qquad \mathbf{a}_{\text{comp}} \equiv (0,0,1,4,2,3) \pmod 5, \qquad \mathbf{a}_{\text{comp}} \equiv (-2c_1,-2c_1,c_1,c_1,c_1,c_1) \pmod N. \tag{1.8}$$

- **Generalized Three-Factor Primary CRT Family** ($m = a \cdot b \cdot N$**, pairwise coprime odd** $a \geq 3, b \geq 3, N \geq 7, \gcd(N,30) = 1, c_b \in (\mathbf{Z}/b\mathbf{Z})^\times$): With $h \equiv (+1,+1,-1) \pmod{(a,b,N)}$ and $H = \{1,h\}$,

$$\mathbf{a}_{\text{gen}} \equiv (1,a{-}1,0,0,0,0) \pmod a, \qquad \mathbf{a}_{\text{gen}} \equiv (-2c_b,-2c_b,c_b,c_b,c_b,c_b) \pmod b, \qquad \mathbf{a}_{\text{gen}} \equiv (0,0,1,-1,2,-2) \pmod N. \tag{1.9}$$

(Notice that (1.9) specializes to (1.7) when $(a,b,c_b) = (3,5,2)$.)

Convention on CRT Character Coordinates. For $m = a \cdot b \cdot N$ (with $(a,b) = (3,5)$ on $m = 15N$), the Chinese Remainder Theorem ring isomorphism $\mathbf{Z}/m\mathbf{Z} \xrightarrow{\sim} \mathbf{Z}/a\mathbf{Z} \times \mathbf{Z}/b\mathbf{Z} \times \mathbf{Z}/N\mathbf{Z}$ restricts on unit groups to the canonical multiplicative group isomorphism $t \mapsto (\bar t_a, \bar t_b, \bar t_N) := (t \bmod a, t \bmod b, t \bmod N)$, $(\mathbf{Z}/m\mathbf{Z})^\times \xrightarrow{\sim} (\mathbf{Z}/a\mathbf{Z})^\times \times (\mathbf{Z}/b\mathbf{Z})^\times \times (\mathbf{Z}/N\mathbf{Z})^\times$, sending $t = 1 \mapsto (1,1,1)$, under which $t \in \pm H \iff (\bar t_a, \bar t_b, \bar t_N) \equiv (s,s,\pm 1) \pmod{(a,b,N)}$ $(s \in \{\pm 1\})$ on (1.7) and (1.9), and $(\bar t_3, \bar t_5, \bar t_N) \equiv (s,\pm 1,s) \pmod{(3,5,N)}$ $(s \in \{\pm 1\})$ on (1.8). Relative to the fixed standard primitive roots $\zeta_m = e^{2\pi i/m}$ and $\zeta_q = e^{2\pi i/q}$ $(q \in \{a,b,N\})$, any weight a_j with residues $a_{j,q} := a_j \bmod q$ satisfies

$$\zeta_m^{t a_j} = \prod_{q \in \{a,b,N\}} \zeta_q^{t_q a_{j,q}}, \qquad \text{where } t_q := \equiv e_q\, \bar t_q \equiv e_q\, t \pmod q \in (\mathbf{Z}/q\mathbf{Z})^\times, \quad e_q := \equiv (m/q)^{-1} \pmod q \quad (q \in \{a,b,N\}). \tag{1.9a}$$

Multiplication by the fixed unit $e_q = (m/q)^{-1} \bmod q$ is a **bijection of the unit set** $(\mathbf{Z}/q\mathbf{Z})^\times$ (a $\mathbf{Z}/q\mathbf{Z}$-linear coset translation, which is not a multiplicative group homomorphism when $e_q \not\equiv 1 \pmod q$) that leaves all character averages over $(\mathbf{Z}/q\mathbf{Z})^\times$ invariant. Under (1.9a), the character $t = 1$ has factored coordinates (e_a, e_b, e_N) (for example, at $m = 105 = 3 \cdot 5 \cdot 7$, $(e_3,e_5,e_7) = (2,1,1)$, so $t = 1$ has $(t_3,t_5,t_7) = (2,1,1)$ and $w(1) = \zeta_3^2 = \omega_3^{t_3}$; at $(a,b,N) = (5,3,7)$, $m = 105$, $(e_5,e_3,e_7) = (1,2,1)$, so $t^\star = 92$ has ordinary residues $(\bar t_5^\star, \bar t_3^\star, \bar t_7^\star) = (2,2,1)$ and factored coordinates $(t_5^\star, t_3^\star, t_7^\star) = (2,1,1)$), whereas the unique unit $t_{\text{ref}} \in (\mathbf{Z}/m\mathbf{Z})^\times$ satisfying $t_{\text{ref}} \equiv m/q \equiv e_q^{-1} \pmod q$ has factored coordinates $(1,1,1)$. For any two units $t, s \in (\mathbf{Z}/m\mathbf{Z})^\times$, the translation factor e_q cancels in the ratio $t_q s_q^{-1} \equiv \bar t_q \bar s_q^{-1} \equiv t s^{-1} \pmod q$, so in the factored coordinates (1.9a) the subgroup $\pm H$ is given by $(t_a,t_b,t_N) \equiv (s e_a, s e_b, \pm e_N) \pmod{(a,b,N)}$ $(s \in \{\pm 1\})$ on (1.7) and (1.9), and $(t_3,t_5,t_N) \equiv (s e_3, \pm e_5, s e_N) \pmod{(3,5,N)}$ $(s \in \{\pm 1\})$ on (1.8). Equivalently, if one relabels the factor primitive roots to the CRT-compatible primitive roots $\tilde\zeta_q := \zeta_q^{e_q} = \zeta_m^{(m/q)e_q}$ (or applies the factorwise Galois automorphism $\sigma_{e_a}^{-1} \otimes \sigma_{e_b}^{-1} \otimes \sigma_{e_N}^{-1}$, which equals $\sigma_{t_{\text{ref}}} \in \mathrm{Gal}(\mathbf{Q}(\zeta_m)/\mathbf{Q})$ since $t_{\text{ref}} \equiv e_q^{-1} \pmod q$), then $\zeta_m^{t a_j} = \prod_q \tilde\zeta_q^{\bar t_q a_{j,q}}$ is expressed directly in the ordinary CRT residues $\bar t_q = e_q^{-1} t_q \equiv t \pmod q$, where $t = 1$ has coordinates $(1,1,1)$.

Theorem A (Obstruction to Local Reflection/Spectrum Methods on S and Adjoint $\mathfrak{sl}_4(\mathbf{C})$-Dichotomy).
Let $m \geq 3$ be an odd integer and $\mathbf{a}$ *a $(2,2,1,1)$-symmetric branch weight tuple on $S = \mathbf{P}^1_\lambda \setminus \{0,\infty,1,-1\}$.* 1. **(Absence of Local Puncture Rank-1 Pseudoreflections under Non-Resonance and Local-Spectrum Failure on S):** *- For every $t \in (\mathbf{Z}/m\mathbf{Z})^\times$ satisfying the local non-resonance conditions $b_1b_2 \neq 1$, $b_4b_5 \neq 1$, $b_1b_4 \neq 1$, $b_2b_5 \neq 1$, $w(t) \neq 1$, and $b_1b_3b_5 \neq 1$, every local puncture monodromy generator $g_p(t)$ ($p \in \{1,-1,0,\infty\}$) and every non-scalar power $g_p(t)^k \notin \mathbf{C}^\times I_4$ ($k \in \mathbf{Z}$) satisfies*

$$\min_{c \in \mathbf{C}^\times} \operatorname{rank}\big(g_p(t) - cI_4\big) = 2, \qquad \min_{c \in \mathbf{C}^\times} \operatorname{rank}\big(g_p(t)^k - cI_4\big) = 2.$$

In particular, none of the local puncture generators $g_p(t)$ ($p \in \{1,-1,0,\infty\}$) or their non-scalar powers $g_p(t)^k \notin \mathbf{C}^\times I_4$ is a rank-1 pseudoreflection or transvection. - If $m = a \cdot m'$ with coprime odd integers $a \geq 3, m' \geq 3$, $a_1 + a_{-1} \equiv a_\lambda + a_{-\lambda} \not\equiv 0 \pmod m$, $a \mid a_1, a_{-1}, a_\lambda, a_{-\lambda}$, and $m' \mid (a_0 + a_\lambda + a_{-\lambda})$, then the unique multiplier $v \in (\mathbf{Z}/m\mathbf{Z})^\times$ with $v \equiv -1 \pmod a$ and $v \equiv +1 \pmod{m'}$ satisfies $v \notin \pm H$, yet for every puncture $p \in \{1,-1,0,\infty\}$ the operators $\rho_v(g_p)$ and $\rho_1(g_p)$ have identical projective spectra and Jordan canonical forms, and the length-2 braid words g_1g_0 and g_1g_{-1} satisfy identically

$$I_v(g_1g_0) = I_{-1}(g_1g_0), \qquad I_v(g_1g_{-1}) = I_1(g_1g_{-1}) = I_{-1}(g_1g_{-1}).$$

2. **(Closed-Form Invariant Hermitian Form and Exact Adjoint $\mathfrak{sl}_4(\mathbf{C})$-Dichotomy):** *For every $t \in (\mathbf{Z}/m\mathbf{Z})^\times$ on the generalized primary CRT family (1.9) ($m = a \cdot b \cdot N$) and on the companion CRT family (1.8) ($m = 15N$),*

all four partial products satisfy $P_{k+1}(t) \neq 1$ ($1 \le k \le 4$), and $\rho_t(\pi_1(S))$ preserves a non-degenerate tridiagonal Hermitian form $J(t) = -iH(\mathbf{b}(t))$ of signature $(d_t - 1, 5 - d_t) \in \{(0,4),(1,3),(2,2),(3,1),(4,0)\}$, where $d_t = \sum_{j=1}^{6}\{ta_{z_j}/m\} \in \{1,2,3,4,5\}$ (and $d_t \in \{2,3,4\}$, $\mathrm{sig}(J(t)) \in \{(1,3),(2,2),(3,1)\}$ universally on the $m = 15N$ families (1.7) and (1.8)). In the $g_1(t)$-eigenbasis $C \in \mathrm{GL}_4(\mathbf{Q}(\zeta_m))$ where $C^{-1}g_1(t)C = \mathrm{diag}(\lambda_1, \lambda_2, 1, 1)$, all 12 off-diagonal entries, the top-left diagonal entries $M_{1,1}, M_{2,2}, (M^{-1})_{1,1}, (M^{-1})_{2,2}$, and all four off-diagonal 2×2 block minors $\det M_{(1,2),(3,4)}, \det M_{(3,4),(1,2)}, \det M^{-1}_{(1,2),(3,4)}, \det M^{-1}_{(3,4),(1,2)}$ of $M = C^{-1}g_0(t)C$ are non-zero by the linear disjointness $\mathbf{Q}(\zeta_a) \cap \mathbf{Q}(\zeta_{bN}) = \mathbf{Q}$, and $\mathrm{Ad}(g_1(t))$ has 7 pairwise distinct eigenvalues on $\mathfrak{sl}_4(\mathbf{C})$. Consequently, every $\mathrm{Ad}(\langle g_1(t), g_0(t)\rangle)$-invariant Lie subalgebra $\mathfrak{g} \subseteq \mathfrak{sl}_4(\mathbf{C})$ satisfies

$$\mathfrak{g} = \{0\} \qquad \text{or} \qquad \mathfrak{g} = \mathfrak{sl}_4(\mathbf{C}),$$

so the identity component $G_0(t)^\circ$ of $G_0(t) := \overline{\langle g_1(t), g_0(t)\rangle}^{\mathrm{Zar}} \cap \mathrm{SL}_4(\mathbf{C})$ satisfies $G_0(t)^\circ = \{I_4\}$ or $G_0(t)^\circ = \mathrm{SL}_4(\mathbf{C})$ (equivalently, the real Zariski closure is $\mathrm{SU}(J(t)) \simeq \mathrm{SU}(d_t - 1, 5 - d_t)$).

Theorem B (Unconditional General-N Zariski Density and Dual $\mathcal{B}_{15}$-Basis Length-3 Separation for $m = 15N$). *Let $N > 1$ be any integer with $\gcd(N, 30) = 1$ (including all primes $p \ge 7$, all prime powers p^k, and all odd composites coprime to 30), and set $m = 15N$. Then on both the primary CRT family (1.7) and the companion CRT family (1.8):*
1. **(Unconditional Factorwise Zariski Density via Ramanujan-Sum Trace Averages):** *Let $W(t) := (g_1(t)g_0(t))^2$. Writing $u := z^{t_N}$ on (1.7) (resp. $u := z^{c_1 t_N}$ on (1.8)) and $X_r := u^r + u^{-r} = 2\cos(2\pi r t_N/N)$ (with $z = e^{2\pi i/N}$), the partial average of $|\mathrm{tr}(W(t))|^2$ over the 8 classes $(t_3, t_5) \in (\mathbf{Z}/3\mathbf{Z})^\times \times (\mathbf{Z}/5\mathbf{Z})^\times$ is given by the exact Chebyshev–Laurent polynomials*

$$M_{\mathrm{prim}}(u) = 64 + 41X_1 + 37X_2 + 22X_3 + 15X_4 - X_5 + 2X_6 - 2X_7 + X_8, \tag{1.10}$$

$$M_{\mathrm{comp}}(u) = 26 + 18X_1 + 14X_2 + 8X_3 + 4X_4. \tag{1.11}$$

Averaging further over $t_N \in (\mathbf{Z}/N\mathbf{Z})^\times$ via Ramanujan's sum $c_N(r) := \sum_{t_N \in (\mathbf{Z}/N\mathbf{Z})^\times} z^{rt_N} = \varphi(N)\frac{\mu(N/\gcd(N,r))}{\varphi(N/\gcd(N,r))}$ yields: - On the primary family (1.7), for every $N > 1$ with $\gcd(N, 30) = 1$,

$$\frac{1}{\varphi(15N)} \sum_{t \in (\mathbf{Z}/15N\mathbf{Z})^\times} \left|\mathrm{tr}(W(t))\right|^2 = 64 + \frac{2}{\varphi(N)} \sum_{r=1}^{8} \alpha_r c_N(r) \ge 21 > 16;$$

*in particular, for every prime power $N = p^k$ with $k \ge 2$ and $p \ge 7$, the Galois mean equals $\frac{194}{3} = 64.\overline{6} > 16$ at $(p, k) = (7, 2)$ and **identically** $64 > 16$ for all $p \ge 11, k \ge 2$ and all $p = 7, k \ge 3$. - On the companion family (1.8), for every $N > 7$ with $\gcd(N, 30) = 1$,*

$$\frac{1}{\varphi(15N)} \sum_{t \in (\mathbf{Z}/15N\mathbf{Z})^\times} \left|\mathrm{tr}(W(t))\right|^2 = 26 + \frac{2}{\varphi(N)} \sum_{r=1}^{4} \beta_r c_N(r) \ge \frac{86}{5} = 17.2 > 16,$$

*and equals **identically** $26 > 16$ for every prime power $N = p^k$ with $k \ge 2$ and $p \ge 7$. At $N = 7$ (where the global mean is $34/3 < 16$), for every t with $c_1 t_7 \equiv \pm 1 \pmod 7$, the algebraic integer $S := 1 + X_1 + X_2 = 2\cos(\pi/7) \in (3/2, 2)$ (satisfying $S^3 - S^2 - 2S + 1 = 0$) gives the exact cyclotomic identity*

$$\left|\mathrm{tr}(W(t))\right|^2 - 16 = 2(2S - 3)(S + 2) > 0.$$

*Because $\mathrm{Gal}(\mathbf{Q}(\zeta_{15N})/\mathbf{Q}) \simeq (\mathbf{Z}/15N\mathbf{Z})^\times$ acts transitively on all primitive characters t, the element $W(t)^{15N} \in G_0(t)$ has infinite order for **every** $t \in (\mathbf{Z}/15N\mathbf{Z})^\times$, and $\langle g_1(t), g_0(t)\rangle$ is Zariski-dense in $\mathrm{SU}(J(t))$ up to μ_{15N} scalars.* 2. **(Unconditional Length-3 Separation via the Dual $\mathcal{B}_{15}$-Basis over $\mathbf{Q}(\zeta_N)$):** *Let $W_1 := g_1^2 g_0$. Viewing $\mathcal{B}_{15} = \{\omega^i \xi^j : i \in \{1,2\}, 1 \le j \le 4\}$ as a basis of $\mathbf{Q}(\zeta_{15N}) \simeq \mathbf{Q}(\zeta_{15}) \otimes_{\mathbf{Q}} \mathbf{Q}(\zeta_N)$ over the base field $\mathbf{Q}(\zeta_N)$ (and $\{1, \sqrt{5}, \sqrt{-3}, \sqrt{-15}\}$ on the companion family), for every $N > 1$ with $\gcd(N, 30) = 1$ and every $t \in (\mathbf{Z}/15N\mathbf{Z})^\times$:*

$$I_t(g_1^2 g_0) \in \{I_1(g_1^2 g_0),\ I_{-1}(g_1^2 g_0)\} \iff \tau_t(g_1^2 g_0) = \pm\tau_{\pm 1}(g_1^2 g_0) \iff t \in \pm H. \tag{1.12}$$

Hence $W_1 = g_1^2 g_0$ separates all $2\varphi(N) - 1$ non-trivial cosets of $(\mathbf{Z}/15N\mathbf{Z})^\times/(\pm H)$ from $\pm H$, and by Galois transitivity in $\mathrm{Gal}(\mathbf{Q}(\zeta_{15N})/\mathbf{Q})$, the projective representations $\mathbf{P}\rho_t$ and $\mathbf{P}\rho_{\pm s}$ on $\pi_1(S)$ are non-isomorphic for all $[t] \neq [\pm s] \in \mathcal{T}_{\mathrm{prim}}$.

Theorem C (Uniform-in-m Length-3 Separation of Symmetric Local-Collision Multipliers for Arbitrary Odd $m = a \cdot m'$).
Let $m = a \cdot m'$ be any odd composite integer factored into coprime odd integers $a \geq 3$ and $m' \geq 3$ with $\gcd(a, m') = 1$. Let $\mathbf{a} = (a_0, a_\infty, a_1, a_{-1}, a_\lambda, a_{-\lambda}) \in \{1, \dots, m-1\}^6$ be a $(2,2,1,1)$-symmetric tuple under $H = \{1, h\}$ with $\sum a_i \equiv 0 \pmod m$, $\gcd(\mathbf{a}, m) = 1$, $a_\lambda \not\equiv a_1, a_{-1} \pmod m$, and

$$a_1 + a_{-1} \equiv a_\lambda + a_{-\lambda} \not\equiv 0 \pmod m \qquad (\text{so } b_1 b_5 = b_2 b_4 =: \eta_+ \in \mu_{m'} \setminus \{1\} \iff v \notin \pm H). \tag{1.13}$$

Suppose $v \in (\mathbf{Z}/m\mathbf{Z})^\times \setminus (\pm H)$ is the local-monodromy collision multiplier with $v \equiv (-1, +1) \pmod{(a, m')}$ fixing $(a_1, a_{-1}, a_\lambda, a_{-\lambda}) \pmod m$ and inverting $w = \zeta_m^{a_0 + a_\lambda + a_{-\lambda}} = \omega \in \mu_a \setminus \{1\}$ (with $m' \mid (a_0 + a_\lambda + a_{-\lambda})$). Then, despite the identical puncture spectra and length-2 trace collisions of Theorem A(1), the length-3 braid word $W_1 = g_1^2 g_0$ satisfies unconditionally

$$I_v(g_1^2 g_0) \notin \{I_1(g_1^2 g_0),\ I_{-1}(g_1^2 g_0)\}. \tag{1.14}$$

—

Theorem D (Three-Factor Generalized Primary CRT Family $m = a \cdot b \cdot N$ and Four-Fold Method-Boundary Examples).
Let $a \geq 3$, $b \geq 3$, and $N \geq 7$ be pairwise coprime odd integers with $\gcd(N, 30) = 1$, and let $m = a \cdot b \cdot N$ carry the generalized primary CRT branch weight tuple $\mathbf{a}_{\text{gen}}$ of (1.9). 1. **(Unconditional Zariski Density and Length-3 Separation via Galois Involution Projections):** - **(Zariski Density)**: *For every (a, b, N) as above, there exists $t^\star \in (\mathbf{Z}/m\mathbf{Z})^\times$ such that $|\operatorname{tr}((g_1(t^\star) g_0(t^\star))^2)|^2 > 16$. Consequently, for every $t \in (\mathbf{Z}/m\mathbf{Z})^\times$, $\langle g_1(t), g_0(t) \rangle$ is Zariski-dense in $\mathrm{SU}(J(t))$ up to μ_m scalars.* - **(Length-3 Separation)**: *Let $\tau_{-1,a} \in \mathrm{Gal}(\mathbf{Q}(\zeta_m)/\mathbf{Q}(\zeta_{bN}))$ and $\tau_{-1,b} \in \mathrm{Gal}(\mathbf{Q}(\zeta_m)/\mathbf{Q}(\zeta_{aN}))$ denote the partial complex conjugations on $\mathbf{Q}(\zeta_a)$ and $\mathbf{Q}(\zeta_b)$, and set $\Pi_{\epsilon_a, \epsilon_b} := \frac{1+\epsilon_a \tau_{-1,a}}{2} \frac{1+\epsilon_b \tau_{-1,b}}{2}$ $(\epsilon_a, \epsilon_b \in \{+, -\})$. For every $t \in (\mathbf{Z}/m\mathbf{Z})^\times$, the projections $\Delta_{--}(t) := \Pi_{-,-}(\tau_t(g_1^2 g_0))$ and $\Delta_{+-}(t) := \Pi_{+,-}(\tau_t(g_1^2 g_0))$ are non-zero pure tensors in $\mathbf{Q}(\zeta_a) \otimes_{\mathbf{Q}} \mathbf{Q}(\zeta_b) \otimes_{\mathbf{Q}} \mathbf{Q}(\zeta_N)$, and*

$$I_t(g_1^2 g_0) \in \{I_1(g_1^2 g_0),\ I_{-1}(g_1^2 g_0)\} \iff \tau_t(g_1^2 g_0) = \pm \tau_{\pm 1}(g_1^2 g_0) \iff t \in \pm H. \tag{1.15}$$

2. **(Four-Fold Method-Boundary Examples for Template Shortcuts and the Coprimality Hypothesis $\gcd(N, 30) = 1$):** - **(b.1) Arithmetic Rigidity of $(a, b) = (3, 5)$ for the Complete-Residue-Block Kubert $(2,2)$-Balance Construction**: *In the complete-residue-block CRT construction where the 2 fixed weights form a complete non-zero residue block modulo a and the 4 paired weights form a complete non-zero residue block modulo b, Kubert distribution cancellation modulo a and b requires $a - 1 = 2 \iff a = 3$ and $b - 1 = 4 \iff b = 5$ (and $N = 7$ on the companion template for universal balance across all t; this characterizes the complete-residue-block template rather than classifying all balanced symmetric tuples). At $(a, b, N) = (9, 5, 7)$ $(m = 315)$, d_t is non-constant, taking the values $d_t = 3$ on 80 units, $d_t = 2$ on 32 units, and $d_t = 4$ on 32 units.* - **(b.2) Breakdown of Companion (t_a, t_b)-Independence for $b \neq 5$**: *On the companion template, the factorization $\prod_{j \in \{1,2,4,5\}} (b_j - 1) = \Phi_5(u)$ independent of t_b holds exclusively at $b = 5$. At $(a, b, N) = (3, 7, 11)$ $(m = 231)$, $|\operatorname{tr}(W(t))|^2$ takes 3 distinct values as $t_7 \in \{1, 2, 3\} \pmod 7$ varies.* - **(b.3) Breakdown of the Global Galois-Mean Shortcut Inside Theorem D(1) at $(a, b, N) = (5, 3, 7)$**: *The admissible triple $(a, b, N) = (5, 3, 7)$ $(m = 105)$ satisfies all hypotheses of Theorem D(1), but because $b = 3$ satisfies $\eta_3^2 + 1 = -\eta_3$ and collapses $C_1(\eta_b)$ to $2\eta_3^2(1 + X_1)(1 + X_2)$, the full Galois average of $|\operatorname{tr}(W(t))|^2$ over $(\mathbf{Z}/105\mathbf{Z})^\times$ drops to $\frac{35}{3} = 11.\overline{6} < 16$—showing that the global-average shortcut of Theorem B(1) fails and that the specialized character evaluation at $t^\star = 92$ (where $y = |\operatorname{tr}(W(92))|^2$ has minimal polynomial $y^6 - 70y^5 + 1924y^4 - 26257y^3 + 185064y^2 - 630364y + 809201 = 0$ and satisfies $y > \frac{48043}{2500} = 19.2172 > 16$) is essential.* - **(b.4) Collapse of the 7-Eigenvalue Adjoint Spectrum Outside Theorem D at $N = 3$**: *When $N = 3$ on the primary template—specifically at $(a, b, N) = (5, 7, 3)$ $(m = 105$, violating $\gcd(N, 30) = 1)$—the relation $z^{\pm 3 t_3} = 1$ in μ_3 forces $\lambda_1 = \lambda_2 = \eta_7^2$, collapsing $\mathrm{Spec}(\mathrm{Ad}(g_1(t)))$ on $\mathfrak{sl}_4(\mathbf{C})$ from 7 distinct eigenvalues to 3 eigenvalues $\{1$ (mult. 7), η_7^2 (mult. 4), η_7^{-2} (mult. 4)$\}$ and invalidating the root-space projection step of Theorem A(2).*

Corollary E (Generic Derived Mumford–Tate Rigidity, Endomorphism Rigidity, and Balanced All-Powers Hodge Reduction on S).

*For every conductor $m = 15N$ ($\gcd(N,30)=1$) on the primary or companion CRT family and every $m = a\cdot b\cdot N$ on the generalized primary CRT family (1.9), let B_η be the generic fiber of the Prym abelian scheme $B_\lambda = (1+\tau_h)\,\mathrm{Jac}(C_\lambda)$ over the one-dimensional slice S, and let $B_{\eta,\mathrm{prim}}$ be its primitive isotypic summand with endomorphism field $L = \mathbf{Q}(\zeta_m)^H$. 1. For every primitive character coset $[t] \in \mathcal{T}_{\mathrm{prim}} = (\mathbf{Z}/m\mathbf{Z})^\times/H$, the connected algebraic monodromy group of the 4-dimensional complex local system $\mathbf{V}_{[t]}$ over S is $\mathrm{SL}_4(\mathbf{C})$ (with real form $\mathrm{SU}(J(t)) \simeq \mathrm{SU}(d_t-1, 5-d_t)$ on $(\mathbf{V}_{[t]}\oplus\mathbf{V}_{[-t]})_{\mathbf{R}}$), and the projective representations $\mathbf{P}\rho_t$ and $\mathbf{P}\rho_{\pm s}$ are pairwise non-isomorphic for all $[t] \neq [\pm s] \in \mathcal{T}_{\mathrm{prim}}$ already upon restriction to $\pi_1(S)$. 2. By Goursat's lemma and the Deligne–André theorem, the connected algebraic monodromy group G°_{mon} and the generic **derived** Mumford–Tate (Hodge) group $\mathrm{Hg}(B_{\eta,\mathrm{prim}})_{\mathrm{der}}$ over S equal the full semisimple product*

$$\mathrm{Hg}(B_{\eta,\mathrm{prim}})_{\mathrm{der}} = G^\circ_{\mathrm{mon}} = \prod_{[t]\in(\mathbf{Z}/m\mathbf{Z})^\times/(\pm H)} \mathrm{SU}\big(J(t)\big).$$

Consequently, there are no diagonal identifications between the simple $\mathfrak{sl}_4$ factors on S, and Schur's lemma yields $\mathrm{End}_{\overline{\mathbf{Q}(S)}}(B_{\eta,\mathrm{prim}})\otimes\mathbf{Q} \simeq \mathbf{Q}(\zeta_m)^H$ (ruling out cross-factor endomorphisms). 3. On unbalanced families where $d_t \neq 3$ for some t, $\mathrm{Hg}(B_{\eta,\mathrm{prim}})_{\mathrm{der}}$ acts by the identity on each determinant line $D_{[t]} := \bigwedge^4 E_{[t]} \subset H^4(B_{\eta,\mathrm{prim}},\mathbf{C})$ (of Hodge type $(d_t-1, 5-d_t)$ up to convention) and therefore does not determine the central torus of $\mathrm{Hg}(B_{\eta,\mathrm{prim}})$ or exclude cross-orbit determinant Hodge classes $D_{[t]}\otimes D_{[s]}$ when $d_{tg}+d_{sg}=6$ across all $g\in(\mathbf{Z}/m\mathbf{Z})^\times$ (for instance, on the primary family (1.7) at $N=7, m=105$, $d_{8g}+d_{-13g}=6$ for all $g\in(\mathbf{Z}/105\mathbf{Z})^\times$, so the Galois-orbit sum of $D_{[8]}\otimes D_{[-13]}$ descends to a non-zero rational $(4,4)$ Hodge substructure in $H^8(B_{\eta,\mathrm{prim}},\mathbf{Q})$ coupling distinct $\pm H$ orbits). By contrast, on every characterwise $(2,2)$-balanced subfamily ($d_t=3$ for all $t\in(\mathbf{Z}/m\mathbf{Z})^\times$, such as the companion family at $m=105, N=7$), every determinant line $D_{[t]}$ already has Hodge type $(2,2)$, so the central torus acts by the identity on every $D_{[t]}$, the full generic Hodge group equals its derived group $\mathrm{Hg}(B_{\eta,\mathrm{prim}}) = \prod_{[t]}\mathrm{SU}(2,2)$, and divisor classes together with the L-determinant volume tensors $W_L(B_{\eta,\mathrm{prim}}) = \bigwedge^4_L H^1(B_{\eta,\mathrm{prim}},\mathbf{Q})$ generate all generic Hodge classes on every power $B^k_{\eta,\mathrm{prim}}$ ($k\geq 1$), reducing the all-powers rational Hodge conjecture on $B^k_{\eta,\mathrm{prim}}$ to the separate cycle-construction input $W_L(B_{\eta,\mathrm{prim}}) \subset \mathrm{cl}\ CH^2(B_{\eta,\mathrm{prim}})_{\mathbf{Q}}$ of [3, Theorem 1.1 and Theorem 7.1].

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2. The 4×4 Reduced Gassner Representation and Obstruction to Local Methods on S

2.1 Explicit Puncture Generators on the One-Dimensional Symmetric Slice

Let $m\geq 3$ be an odd integer and $\mathbf{a} = (a_0, a_\infty, a_1, a_{-1}, a_\lambda, a_{-\lambda}) \in \{1,\dots,m-1\}^6$ a $(2,2,1,1)$-symmetric branch weight tuple on $S = \mathbf{P}^1_\lambda\backslash\{0,\infty,1,-1\}$ ordered along the real axis at the six branch points $(z_1,z_2,z_3,z_4,z_5,z_6) = (-1,-\lambda,0,\lambda,1,\infty)$ $(0<\lambda<1)$. Fix $\zeta_m = e^{2\pi i/m}$. For each unit $t\in(\mathbf{Z}/m\mathbf{Z})^\times$, put $\zeta := \zeta_m^t$ and define the local monodromy parameters

$$(b_1,b_2,b_3,b_4,b_5) := \big(\zeta^{a_{-1}},\ \zeta^{a_{-\lambda}},\ \zeta^{a_0},\ \zeta^{a_\lambda},\ \zeta^{a_1}\big), \qquad w := b_2b_3b_4 = \zeta^{a_0+a_\lambda+a_{-\lambda}}. \tag{2.1}$$

In the standard homology chain basis (e_1,e_2,e_3,e_4) of the 4-dimensional character eigenspace $V_t = H_1(C_\lambda,\mathbf{C})_t \simeq \mathbf{C}^4$ constructed in [2, Section 3.3], the elementary braid generators $\sigma_1,\sigma_2,\sigma_3,\sigma_4\in B_5$ act by reduced Burau–Gassner reflections. As λ loops around the three finite punctures $\{1,0,-1\}$ of S: - Around $\lambda=1$, the branch points $(-\lambda,-1)$ and $(\lambda,1)$ coalesce simultaneously via the commuting braid $\sigma_1^2\sigma_4^2$; - Around $\lambda=0$, the three inner branch points $(-\lambda,0,\lambda)$ coalesce simultaneously via the full twist $(\sigma_2\sigma_3\sigma_2)^2$; - Around $\lambda=-1$, the pairs $(-\lambda,1)$ and $(\lambda,-1)$ coalesce simultaneously via the braid-conjugate double twist.

Evaluating the reduced Gassner representation on these three loops yields the explicit puncture generators $g_1(t), g_0(t), g_{-1}(t)\in \mathrm{GL}_4(\mathbf{Z}[\zeta_m])$:

$$g_1(t) = \begin{pmatrix} b_1b_2 & 1-b_1 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & b_4(1-b_5) & b_4b_5\end{pmatrix}, \qquad g_0(t) = \begin{pmatrix} 1 & 0 & 0 & 0\\ b_2-w & w & 0 & 1-b_2\\ b_2b_3-w & 0 & w & 1-b_2b_3\\ 0 & 0 & 0 & 1\end{pmatrix}, \tag{2.2}$$

and

$$g_{-1}(t) = D_1(t)^{-1}\begin{pmatrix} b_1b_4 & 1-b_1 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & b_2(1-b_5) & b_2b_5\end{pmatrix}D_1(t), \qquad D_1(t) := \begin{pmatrix} 1 & 0 & 0 & 0\\ b_4 & 0 & -b_4 & 1\\ b_3b_4 & -b_3b_4 & 0 & 1\\ 0 & 0 & 0 & 1\end{pmatrix}, \tag{2.3}$$

with infinity monodromy $g_\infty(t) = (g_1(t)g_0(t)g_{-1}(t))^{-1}$ and $b_\infty := (b_1b_2b_3b_4b_5)^{-1} = \zeta^{a_\infty}$. Here, following Rohde [2, Proposition 3.3.5, p. 42], matrices act from the left on column vectors ($v \mapsto Mv$): $D_1(t) = \Delta_{2,3,4}(b_2, b_3, b_4)$ is the colored Gassner half-twist along the upper semicircle δ_+, $D_{1,\text{swapped}}(t) := D_1(t)|_{b_2 \leftrightarrow b_4}$ is the half-twist along the lower semicircle δ_-, $g_0(t) = D_{1,\text{swapped}}(t)D_1(t)$ identically in $\mathrm{GL}_4(\mathbf{Z}[b_1^{\pm1}, \dots, b_5^{\pm1}])$, and in the boundary product $P_1(t) := g_1(t)g_0(t)g_{-1}(t) = g_1(t)D_{1,\text{swapped}}(t)D_1(t)D_1(t)^{-1}g_{1,\text{swapped}}(t)D_1(t)$ the backtracking half-twists cancel identically via $D_1(t)D_1(t)^{-1} = I_4$. Let $\Gamma(t) := \langle g_1(t), g_0(t), g_{-1}(t)\rangle = \rho_t(\pi_1(S))$ be the full one-dimensional slice monodromy group and $\Gamma_0(t) := \langle g_1(t), g_0(t)\rangle \subseteq \Gamma(t)$ its two-generator subgroup.

2.2 Absence of Local Puncture Rank-1 Pseudoreflections under Non-Resonance on S

We now prove the first assertion of Theorem A(1), demonstrating that under the local non-resonance conditions the boundary rank-1 reflection generation mechanism ceases to operate on S.

Lemma 2.1 (Absence of Local Puncture Rank-1 Pseudoreflections vs. Torsion Bisections).
Let $m \geq 3$ be an odd integer and $t \in (\mathbf{Z}/m\mathbf{Z})^\times$ any unit such that $\lambda_1 := b_1b_2 \neq 1$, $\lambda_2 := b_4b_5 \neq 1$, $b_1b_4 \neq 1$, $b_2b_5 \neq 1$, $w \neq 1$, and $b_1b_3b_5 \neq 1$ (in particular, this holds for every $t \in (\mathbf{Z}/m\mathbf{Z})^\times$ on the CRT families (1.7), (1.8), and (1.9)). Then: 1. *For every puncture generator $g_p(t)$ ($p \in \{1, -1, 0, \infty\}$), every non-scalar power $g_p(t)^k \notin \mathbf{C}^\times I_4$ ($k \in \mathbf{Z}$), and every scalar $c \in \mathbf{C}^\times$, we have*

$$\operatorname{rank}\big(g_p(t) - cI_4\big) \geq 2 \qquad \text{and} \qquad \operatorname{rank}\big(g_p(t)^k - cI_4\big) \geq 2. \tag{2.4}$$

In particular, no local puncture generator or non-scalar power of a puncture generator is a rank-1 pseudoreflection or a rank-1 transvection. 2. *Instead, $g_0(t)$ and $g_\infty(t)$ are semisimple rank-(2, 2) bisections satisfying $(g_0(t) - I_4)(g_0(t) - wI_4) = 0$ and $(g_\infty(t) - (b_2b_4)^{-1}I_4)(g_\infty(t) - b_\infty I_4) = 0$; $g_{-1}(t)$ has spectrum $\{b_1b_4, b_2b_5, 1, 1\}$ (satisfying $(g_{-1}(t) - I_4)(g_{-1}(t) - b_1b_4I_4) = 0$ whenever $b_1b_4 = b_2b_5$); and on the generalized primary CRT family (1.9) with $m = a \cdot b \cdot N$, the powers $g_1(t)^N$ and $g_1(t)^b$ satisfy $(g_1(t)^N - I_4)(g_1(t)^N - \eta_b^{2N}I_4) = 0$ (of order b) and $\mathrm{Spec}(g_1(t)^b) = (z^{-3bt_N}, z^{+3bt_N}, 1, 1)$ (in $\mathrm{SL}_4(\mathbf{C})$ of order N).*

Proof. From (2.2), $g_1(t)$ is block-diagonal with 2×2 blocks $\left(\begin{smallmatrix}\lambda_1 & 1-b_1\\ 0 & 1\end{smallmatrix}\right)$ and $\left(\begin{smallmatrix}1 & 0\\ b_4(1-b_5) & \lambda_2\end{smallmatrix}\right)$, so $\mathrm{Spec}(g_1(t)) = (\lambda_1, \lambda_2, 1, 1)$ with $\lambda_1, \lambda_2 \neq 1$. Similarly, from (2.2) and (2.3), $\mathrm{Spec}(g_0(t)) = (w, w, 1, 1)$ with $w \neq 1$ and $(g_0(t) - I_4)(g_0(t) - wI_4) = 0$, $\mathrm{Spec}(g_{-1}(t)) = (b_1b_4, b_2b_5, 1, 1)$ with $b_1b_4, b_2b_5 \neq 1$, and around $\lambda = \infty$ the product $P_1(t) := g_1(t)g_0(t)g_{-1}(t)$ satisfies $(P_1(t) - b_2b_4I_4)(P_1(t) - b_\infty^{-1}I_4) = 0$, so $g_\infty(t) = P_1(t)^{-1}$ is a semisimple rank-(2, 2) bisection satisfying $(g_\infty(t) - (b_2b_4)^{-1}I_4)(g_\infty(t) - b_\infty I_4) = 0$ with $\mathrm{Spec}(g_\infty(t)) = ((b_2b_4)^{-1}, (b_2b_4)^{-1}, b_\infty, b_\infty)$ and $(b_2b_4)^{-1}/b_\infty = b_1b_3b_5 \neq 1$. For each of these four operators (and hence for every non-scalar power $g_0(t)^k \notin \mathbf{C}^\times I_4$ and $g_\infty(t)^k \notin \mathbf{C}^\times I_4$), every eigenspace has dimension at most 2; hence $\operatorname{rank}(g_p(t) - cI_4) = 4 - \dim\ker(g_p(t) - cI_4) \geq 4 - 2 = 2$ for all $c \in \mathbf{C}^\times$.

On the CRT families (1.7), (1.8), and (1.9), writing $\omega_a = \zeta_a^{t_a}$, $\eta_b = \zeta_b^{c_bt_b}$ (with $c_5 = 2$ on (1.7)), and $z = \zeta_N$, we have $\lambda_1 = \eta_b^2 z^{-3t_N}, \lambda_2 = \eta_b^2 z^{+3t_N}$ on (1.7) and (1.9), and $\lambda_1 = \xi^{-3t_5}z^{2c_1t_N}, \lambda_2 = \xi^{+3t_5}z^{2c_1t_N}$ on (1.8). More generally, for any (2, 2, 1, 1)-symmetric tuple (1.2), $a_{-1} + a_{-\lambda} \equiv h(a_1 + a_\lambda) \pmod m$ and $a_{-1} + a_\lambda \equiv h(a_1 + a_{-\lambda}) \pmod m$ give $\lambda_1 = b_1b_2 = \lambda_2^h$ and $b_1b_4 = (b_2b_5)^h$ with $\gcd(h, m) = 1$, so $\mathrm{ord}(\lambda_1) = \mathrm{ord}(\lambda_2)$ and $\mathrm{ord}(b_1b_4) = \mathrm{ord}(b_2b_5)$ universally; thus whenever $g_1(t)^k \notin \mathbf{C}^\times I_4$ (resp. $g_{-1}(t)^k \notin \mathbf{C}^\times I_4$), both non-trivial eigenvalues λ_1^k, λ_2^k (resp. $(b_1b_4)^k, (b_2b_5)^k$) remain $\neq 1$, every eigenspace has dimension at most 2, and $\operatorname{rank}(g_{\pm1}(t)^k - cI_4) \geq 2$ for all $c \in \mathbf{C}^\times$. Part (2) follows immediately from (2.2)–(2.3) and these eigenvalue formulas. □

(Remark on the Non-Resonance Hypothesis: The non-resonance conditions in Lemma 2.1 are essential. If a (2, 2, 1, 1)-symmetric tuple has $w(t) = 1$ while $b_2 \neq 1$ or $b_2b_3 \neq 1$—for instance, at $m = 15$, $h = 4$, and $\mathbf{a} = (5, 10, 1, 4, 2, 8)$ with $t = 1$, where $\sum a_i = 30 \equiv 0 \pmod{15}$, $\gcd(\mathbf{a}, 15) = 1$, and $w(1) = \zeta_{15}^{5+2+8} = 1$ while $b_2 = \zeta_{15}^8 \neq 1$ and $b_2b_3 = \zeta_{15}^{13} \neq 1$—then (2.2) gives a non-zero rank-1 nilpotent operator $g_0(1) - I_4$ with $(g_0(1) - I_4)^2 = 0$, so $g_0(1)$ is a rank-1 unipotent transvection. On the CRT families (1.7), (1.8), and (1.9), $w(t) = \omega_a^{\pm1} \in \mu_a \setminus \{1\}$ for all $t \in (\mathbf{Z}/m\mathbf{Z})^\times$, excluding all such resonant degenerations.)

2.3 Closed-Form Invariant Hermitian Form $J(t)$ and Signature Classification

We next construct the explicit tridiagonal Hermitian intersection form $J(t)$ preserved by $\Gamma(t)$ and prove its non-degeneracy and signature formula via Jacobi principal minors.

Proposition 2.2 (Closed-Form Invariant Hermitian Form $J(t)$, Non-Degeneracy $P_{k+1} \neq 1$, and Signature Classification).
For any unit-circle parameters $\mathbf{b} = (b_1, b_2, b_3, b_4, b_5)$ *with* $b_j = e^{2\pi i\theta_j} \neq 1$, $\theta_j := \{ta_{z_j}/m\} \in (0,1)$, *and* $\theta_6 := \{ta_\infty/m\} \in (0,1)$, *define the* 4×4 *tridiagonal matrix* $H(\mathbf{b})$ *by*

$$H(\mathbf{b})_{\ell,\ell} := \frac{b_\ell b_{\ell+1} - 1}{(1-b_\ell)(1-b_{\ell+1})} \quad (1 \le \ell \le 4), \qquad H(\mathbf{b})_{\ell,\ell+1} := \frac{1}{1-b_{\ell+1}}, \qquad H(\mathbf{b})_{\ell+1,\ell} := \frac{b_{\ell+1}}{1-b_{\ell+1}} \quad (1 \le \ell \le 3), \tag{2.5}$$

and $H(\mathbf{b})_{\ell,k} = 0$ *for* $|\ell - k| \ge 2$, *and set* $J(t) := -iH(\mathbf{b})$. *Let* $d_t := \sum_{j=1}^{6} \theta_j \in \{1,2,3,4,5\}$ *(with* $d_t \in \{2,3,4\}$ *on (1.7) and (1.8)) and* $P_{k+1} := \prod_{j=1}^{k+1} b_j$ $(1 \le k \le 4)$. *Then:* 1. **(Hermitian Symmetry and Monodromy Invariance):** $H(\mathbf{b})^* = -H(\mathbf{b})$, so $J(t)^* = J(t)$ is Hermitian, and every generator $g \in \{g_1(t), g_0(t), g_{-1}(t), g_\infty(t)\}$ satisfies $g^* J(t) g = J(t)$. 2. **(Exact Leading Principal Minors, Non-Degeneracy $P_{k+1} \neq 1$, and Jacobi Signature Formula):** *For each* $k \in \{1,2,3,4\}$, *the* k*-th leading principal minor of* $H(\mathbf{b})$ *and* $J(t)$ *factors identically as*

$$\det H(\mathbf{b})_{1..k} = \frac{P_{k+1} - 1}{\prod_{j=1}^{k+1}(b_j - 1)}, \qquad \det J(t)_{1..k} = (-1)^k \frac{\sin(\pi S_{k+1})}{2^k \prod_{j=1}^{k+1} \sin(\pi\theta_j)}, \tag{2.6}$$

where $S_{k+1} := \sum_{j=1}^{k+1} \theta_j$. *On the generalized primary CRT family (1.9) (*$m = a \cdot b \cdot N$, *including (1.7)) and on the companion CRT family (1.8) (*$m = 15N$*), for every* $t \in (\mathbf{Z}/m\mathbf{Z})^\times$ *and every* $k \in \{1,2,3,4\}$, *the partial product satisfies* $P_{k+1} \neq 1$ *(equivalently* $S_{k+1} \notin \mathbf{Z}$ *and* $\det J(t)_{1..k} \neq 0$*). Consequently, all leading principal minors of* $J(t)$ *are non-zero,* $J(t)$ *is non-degenerate, and* $J(t)$ *has exact signature*

$$\operatorname{sig}\big(J(t)\big) = (n_+, n_-) = \big(d_t - 1,\ 5 - d_t\big) \in \big\{(0,4),\ (1,3),\ (2,2),\ (3,1),\ (4,0)\big\} \tag{2.7}$$

(ranging over $\{(1,3),(2,2),(3,1)\}$ *on the* $m = 15N$ *families (1.7)–(1.8), while also attaining the definite signatures* $(0,4)$ *and* $(4,0)$ *for certain* $(a,b) \neq (3,5)$ *on (1.9), such as* $(a,b,N,c_b) = (5,7,17,1), m = 595$ *at* $t = 288, 307$*).* 3. **(Signature Distribution on the Primary and Companion Families for $m = 15N$):** - **On the Primary CRT Family (1.7):** *Across all* $\varphi(15N) = 8\varphi(N)$ *units* $t \in (\mathbf{Z}/15N\mathbf{Z})^\times$, *exactly* $4\varphi(N)$ *characters (*50%*) have* $d_t = 3$ *and* $\mathrm{SU}(J(t)) \simeq \mathrm{SU}(2,2)$, $2\varphi(N)$ *characters (*25%*) have* $d_t = 2$ *and* $\mathrm{SU}(J(t)) \simeq \mathrm{SU}(1,3)$, *and* $2\varphi(N)$ *characters (*25%*) have* $d_t = 4$ *and* $\mathrm{SU}(J(t)) \simeq \mathrm{SU}(3,1)$. *At* $t = 1$, $(d_1, \operatorname{sig}(J(1)))$ *depends only on* $N \bmod 15$*: it equals* $(3,(2,2))$ *if* $N \equiv 7, 8 \pmod{15}$, $(2,(1,3))$ *if* $N \equiv 11, 13, 14 \pmod{15}$, *and* $(4,(3,1))$ *if* $N \equiv 1, 2, 4 \pmod{15}$. - **On the Companion CRT Family (1.8)** $(c_1 = 3)$**:** *For every* $t \in (\mathbf{Z}/15N\mathbf{Z})^\times$, *setting* $y_0 := \{5^{-1} t_N/N\} \in (0,1)$, *we have*

$$d_t = 3 + \{5y_0\} - \{6y_0\} + \{2y_0\} - \{y_0\} = \begin{cases} 4\ (\mathrm{SU}(3,1)) & \text{if } y_0 \in (\frac{1}{6}, \frac{1}{5}) \cup (\frac{1}{3}, \frac{2}{5}), \\ 2\ (\mathrm{SU}(1,3)) & \text{if } y_0 \in (\frac{2}{5}, \frac{2}{3}) \cup (\frac{4}{5}, \frac{5}{6}), \\ 3\ (\mathrm{SU}(2,2)) & \text{otherwise.} \end{cases} \tag{2.8}$$

In particular, at $N = 7$ $(m = 105)$, *all* 48 *primitive characters* $t \in (\mathbf{Z}/105\mathbf{Z})^\times$ *satisfy* $d_t = 3$ *and* $\mathrm{SU}(J(t)) \simeq \mathrm{SU}(2,2)$ *(universal characterwise* $(2,2)$*-balance).*

Proof. 1. Using $\overline{b_j} = b_j^{-1}$ for $|b_j| = 1$, we have

$$\overline{H_{\ell,\ell}} = \frac{b_\ell^{-1} b_{\ell+1}^{-1} - 1}{(1-b_\ell^{-1})(1-b_{\ell+1}^{-1})} = -\frac{b_\ell b_{\ell+1} - 1}{(1-b_\ell)(1-b_{\ell+1})} = -H_{\ell,\ell}, \qquad \overline{H_{\ell,\ell+1}} = \frac{1}{1-b_{\ell+1}^{-1}} = -\frac{b_{\ell+1}}{1-b_{\ell+1}} = -H_{\ell+1,\ell},$$

so $H(\mathbf{b})^* = -H(\mathbf{b})$ and $J(t)^* = J(t)$. Direct matrix multiplication with (2.2) and (2.3) shows $g^* H(\mathbf{b}) g = H(\mathbf{b})$ for $g \in \{g_1(t), g_0(t), g_{-1}(t)\}$. Indeed, for the upper-left 2×2 block $B_{1,2} = \left(\begin{smallmatrix} b_1 b_2 & 1-b_1 \\ 0 & 1 \end{smallmatrix}\right)$ of $g_1(t)$, $(B_{1,2}^* H_{1..2} B_{1,2})_{1,1} = |b_1 b_2|^2 H_{1,1} = H_{1,1}$, $(B_{1,2}^* H_{1..2} B_{1,2})_{1,2} = b_1^{-1} b_2^{-1}(1-b_1) H_{1,1} + b_1^{-1} b_2^{-1} H_{1,2} = \frac{1 - b_1 b_2 + 1 - b_1}{b_1 b_2 (1-b_2)} = \frac{1}{1-b_2} = H_{1,2}$, and $(B_{1,2}^* H_{1..2} B_{1,2})_{2,3} = H_{2,3}$, with identical identities on the lower-right block and on the central 3×3 block of $g_0(t)$. 2. Put $D_k := \det H(\mathbf{b})_{1..k}$

and $P_r := \prod_{j=1}^{r} b_j$. For $k = 1$, $D_1 = H_{1,1} = \frac{P_2 - 1}{(b_1-1)(b_2-1)}$. Proceeding by induction on $k \in \{2, 3, 4\}$ via the tridiagonal recurrence $D_k = H_{k,k}D_{k-1} - H_{k-1,k}H_{k,k-1}D_{k-2}$ (with $D_0 := 1$ and $P_1 := b_1$):

$$D_k = \frac{(b_k b_{k+1} - 1)(P_k - 1) - b_k(b_k - 1)(b_{k+1} - 1)\frac{P_{k-1}-1}{b_k - 1}}{(b_k - 1)\prod_{j=1}^{k+1}(b_j - 1)} = \frac{(b_k - 1)(P_{k+1} - 1)}{(b_k - 1)\prod_{j=1}^{k+1}(b_j - 1)} = \frac{P_{k+1} - 1}{\prod_{j=1}^{k+1}(b_j - 1)}.$$

Substituting $b_j - 1 = 2i\sin(\pi\theta_j)e^{\pi i\theta_j}$ and $P_{k+1} - 1 = 2i\sin(\pi S_{k+1})e^{\pi i S_{k+1}}$ into $\det J(t)_{1..k} = (-i)^k D_k$ cancels the phase $e^{\pi i S_{k+1}}$ and establishes (2.6).

To prove $P_{k+1} \neq 1$ for $k = 1, 2, 3, 4$, let $t \in (\mathbf{Z}/m\mathbf{Z})^\times$ have CRT residues (t_a, t_b, t_N) on (1.9) (or (t_3, t_5, t_N) on (1.8)): - $k = 1$ ($P_2 = b_1 b_2 = \zeta^{a_{-1}+a_{-\lambda}}$)**:** On the generalized primary family (1.9), $b_1 = \eta_b z^{-t_N}$ and $b_2 = \eta_b z^{-2t_N}$, so $P_2 = \eta_b^2 z^{-3t_N}$. Since $\gcd(3t_N, N) = 1$ and $N \geq 7 > 1$, the N-primary component $z^{-3t_N} \in \mu_N \setminus \{1\}$ is non-trivial (and for $b = 5$, $\eta_5^2 = \xi^{4t_5} \neq 1$ as well). On the companion family (1.8), $a_{-1} + a_{-\lambda} \equiv 4 + 3 \equiv 2 \not\equiv 0 \pmod 5$, so the 5-primary component of P_2 is $\xi^{2t_5} \in \mu_5 \setminus \{1\}$. Thus $P_2 \neq 1$. - $k = 2$ ($P_3 = b_1 b_2 b_3 = \zeta^{a_{-1}+a_{-\lambda}+a_0}$)**:** On (1.9), $a \mid a_{-1}, a_{-\lambda}$ while $a_0 \equiv 1 \pmod a$, so the a-primary component of P_3 is $\omega_a = \zeta_a^{t_a} \in \mu_a \setminus \{1\}$. On (1.8), $3 \mid a_{-1}, a_{-\lambda}$ and $a_0 \equiv 2 \pmod 3$, so the 3-primary component of P_3 is $\omega^{2t_3} \in \mu_3 \setminus \{1\}$. Thus $P_3 \neq 1$. - $k = 3$ ($P_4 = b_1 b_2 b_3 b_4 = b_1 w$)**:** Since $a \mid a_{-1}, a_{-\lambda}, a_\lambda$ and $a_0 \equiv 1 \pmod a$ on (1.9) (resp. $a_0 \equiv 2 \pmod 3$ on (1.8)), the a-primary component of P_4 is again $\omega_a \in \mu_a \setminus \{1\}$ (resp. $\omega^{2t_3} \in \mu_3 \setminus \{1\}$). Thus $P_4 \neq 1$. - $k = 4$ ($P_5 = b_1 b_2 b_3 b_4 b_5 = \zeta^{-a_\infty}$)**:** Since $a_\infty \equiv a - 1 \not\equiv 0 \pmod a$ on (1.9) and $a_\infty \equiv 1 \not\equiv 0 \pmod 3$ on (1.8), the a-primary (resp. 3-primary) component of P_5 is $\omega_a \neq 1$ (resp. $\omega^{2t_3} \neq 1$). Thus $P_5 \neq 1$.

Consequently, for every $k \in \{1, 2, 3, 4\}$, $S_{k+1} \notin \mathbf{Z}$ and the k-th leading principal minor of $-J(t) = iH(\mathbf{b})$ is non-zero with sign $\operatorname{sgn}(\sin(\pi S_{k+1})) = (-1)^{\lfloor S_{k+1} \rfloor}$ since $\sin(\pi\theta_j) > 0$. By Jacobi's signature theorem, the number of negative eigenvalues of $-J(t)$—which equals the number of positive eigenvalues n_+ of $J(t)$—is the number of sign changes in the sequence $1 = (-1)^{\lfloor S_1 \rfloor}, (-1)^{\lfloor S_2 \rfloor}, (-1)^{\lfloor S_3 \rfloor}, (-1)^{\lfloor S_4 \rfloor}, (-1)^{\lfloor S_5 \rfloor}$. Since $0 < \theta_j < 1$ and $S_5 = d_t - \theta_6 \in (d_t - 1, d_t)$, the partial sums $0 < S_1 < S_2 < S_3 < S_4 < S_5$ increase by steps in $(0, 1)$ and cross each integer $1, 2, \ldots, d_t - 1$ exactly once without landing on or skipping any integer. This produces $n_+ = d_t - 1$ sign changes and $n_- = 4 - n_+ = 5 - d_t$, proving (2.7). 3. On the primary family (1.7), let $s := N^{-1} \bmod 15 \in (\mathbf{Z}/15\mathbf{Z})^\times$. By CRT, for each weight a_i with residues $(a_{i,15}, a_{i,N}) \pmod{(15, N)}$ where $a_{\bullet,15} = (1, 11, 12, 12, 12, 12)$ and $a_{\bullet,N} = (0, 0, 1, -1, 2, -2)$, we have

$$\frac{ta_i}{15N} \equiv \frac{t_{15} s a_{i,15}}{15} + \frac{t_N(1 - 15s)/N \cdot a_{i,N}}{15} \equiv \frac{t_{15} s a_{i,15} - t_N s a_{i,N}}{15} + \frac{t_N a_{i,N}}{15N} \pmod 1.$$

At $t = 1$, none of $s(a_{i,15} - a_{i,N}) = s \cdot (1, 11, 11, 13, 10, 14)$ is divisible by 15, and $|a_{i,N}|/(15N) \leq \frac{2}{15N} < \frac{1}{15}$ for $N \geq 7$. Thus the $\pm\frac{1}{15N}$ and $\pm\frac{2}{15N}$ perturbations cancel in pairs without crossing an integer, yielding

$$d_1 = \left\{\frac{s}{15}\right\} + 2\left\{\frac{11s}{15}\right\} + \left\{\frac{13s}{15}\right\} + \left\{\frac{10s}{15}\right\} + \left\{\frac{14s}{15}\right\} = 1 + 2\left\{\frac{11s}{15}\right\} + \left\{\frac{13s}{15}\right\} + \left\{\frac{10s}{15}\right\},$$

which equals 3 for $N \equiv 7, 8 \pmod{15}$ ($s \in \{13, 2\}$), 2 for $N \equiv 11, 13, 14 \pmod{15}$ ($s \in \{11, 7, 14\}$), and 4 for $N \equiv 1, 2, 4 \pmod{15}$ ($s \in \{1, 8, 4\}$). Similarly, on the companion family (1.8), applying Kubert's distribution identity $\sum_{j=0}^{n-1}\{x + j/n\} = \{nx\} + \frac{n-1}{2}$ modulo 3 (to a_0, a_∞) and modulo 5 (to $a_1, a_{-1}, a_\lambda, a_{-\lambda}$) reduces $\sum_{i=1}^{6}\{ta_i/(15N)\}$ directly to (2.8). At $N = 7$, $y_0 \in \{1/7, 2/7, 3/7, 4/7, 5/7, 6/7\}$ never lies in $(\frac{1}{6}, \frac{1}{5}) \cup (\frac{1}{3}, \frac{2}{5}) \cup (\frac{3}{5}, \frac{2}{3}) \cup (\frac{4}{5}, \frac{5}{6})$, so $d_t = 3$ for all 48 units $t \in (\mathbf{Z}/105\mathbf{Z})^\times$. □

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3. Exact Adjoint $\mathfrak{sl}_4(\mathbf{C})$-Irreducibility Dichotomy via Cyclotomic Linear Disjointness

3.1 Diagonalization of $g_1(t)$ and Factored Structure of $M = C^{-1}g_0(t)C$

In all five signature cases $d_t \in \{1, 2, 3, 4, 5\}$ (both indefinite $\mathrm{SU}(p_t, q_t)$ for $d_t \in \{2, 3, 4\}$ and compact $\mathrm{SU}(4)$ for $d_t \in \{1, 5\}$), $\mathrm{SU}(J(t))$ is a real form of $\mathrm{SL}_4(\mathbf{C})$ with complexified Lie algebra $\mathfrak{su}(J(t))\otimes_{\mathbf{R}}\mathbf{C} = \mathfrak{sl}_4(\mathbf{C})$. Since $(g_0(t) - I_4)(g_0(t) - wI_4) = 0$ by Lemma 2.1(2), the inverse of $g_0(t)$ is a linear combination of $g_0(t)$ and I_4:

$$g_0(t)^{-1} = -w^{-1}g_0(t) + (1 + w^{-1})I_4. \tag{3.1}$$

Lemma 3.1 (Exact Diagonalization of $g_1(t)$ and Cyclotomic Binomial Factorization of $M = C^{-1}g_0(t)C$). *Put $\lambda_1 := b_1b_2$ and $\lambda_2 := b_4b_5$. For every unit $t \in (\mathbf{Z}/m\mathbf{Z})^\times$ on the generalized primary CRT family (1.9) ($m = a\cdot b\cdot N$, including (1.7)) and on the companion CRT family (1.8) ($m = 15N$), define the change-of-basis matrix*

$$C := \begin{pmatrix} 1 & 0 & 1-b_1 & 0 \\ 0 & 0 & 1-b_1b_2 & 0 \\ 0 & 0 & 0 & 1-b_4b_5 \\ 0 & 1 & 0 & b_4(1-b_5) \end{pmatrix} \in \mathrm{GL}_4\big(\mathbf{Q}(\zeta_m)\big), \qquad \det(C) = (1-\lambda_1)(1-\lambda_2) \neq 0, \tag{3.2}$$

with explicit inverse

$$C^{-1} = \begin{pmatrix} 1 & \frac{1-b_1}{\lambda_1-1} & 0 & 0 \\ 0 & 0 & \frac{b_4(1-b_5)}{\lambda_2-1} & 1 \\ 0 & -\frac{1}{\lambda_1-1} & 0 & 0 \\ 0 & 0 & -\frac{1}{\lambda_2-1} & 0 \end{pmatrix}. \tag{3.3}$$

Then: 1. **(Exact Diagonalization of $g_1(t)$):**

$$C^{-1}g_1(t)C = \mathrm{diag}\big(\lambda_1,\ \lambda_2,\ 1,\ 1\big). \tag{3.4}$$

2. **(Closed-Form Factorization of $M := C^{-1}g_0(t)C$ and $M^{-1} = C^{-1}g_0(t)^{-1}C$):** *In the $g_1(t)$-eigenbasis, $M = C^{-1}g_0(t)C$ has entries*

$$\begin{aligned}
M_{1,1} &= \frac{(b_2-1)+(b_1-1)w}{\lambda_1-1}, & M_{1,2} &= \frac{(b_1-1)(b_2-1)}{\lambda_1-1}, & M_{1,3} &= \frac{(b_1-1)(b_2-1)(wb_1-1)}{\lambda_1-1}, & M_{1,4} &= -\frac{b_4(b_1-1)(b_2-1)(b_5-}{\lambda_1-1} \\
M_{2,1} &= \frac{w(b_4-1)(b_5-1)}{\lambda_2-1}, & M_{2,2} &= \frac{(b_4-1)+(b_5-1)w}{\lambda_2-1}, & M_{2,3} &= -\frac{w(b_1-1)(b_4-1)(b_5-1)}{\lambda_2-1}, & M_{2,4} &= \frac{b_4(b_4-1)(b_5-1)(wb_5}{\lambda_2-1} \\
M_{3,1} &= \frac{b_2(b_3b_4-1)}{\lambda_1-1}, & M_{3,2} &= \frac{b_2-1}{\lambda_1-1}, & M_{3,3} &= \frac{b_2\big((b_1-1)+b_1(b_2-1)b_3b_4\big)}{\lambda_1-1}, & M_{3,4} &= -\frac{b_4(b_2-1)(b_5-1)}{\lambda_1-1}, \\
M_{4,1} &= \frac{b_2b_3(b_4-1)}{\lambda_2-1}, & M_{4,2} &= \frac{b_2b_3-1}{\lambda_2-1}, & M_{4,3} &= -\frac{b_2b_3(b_1-1)(b_4-1)}{\lambda_2-1}, & M_{4,4} &= \frac{b_4\big((b_5-1)+b_2b_3(b_4-}{\lambda_2-1}
\end{aligned} \tag{3.5}$$

while $M^{-1} = -w^{-1}M + (1+w^{-1})I_4$ has off-diagonal entries $(M^{-1})_{i,j} = -w^{-1}M_{i,j}$ ($i \neq j$) and top-left diagonal entries

$$(M^{-1})_{1,1} = \frac{(b_1-1)b_2 + b_1(b_2-1)w}{w(\lambda_1-1)}, \qquad (M^{-1})_{2,2} = \frac{(b_5-1)b_4 + b_5(b_4-1)w}{w(\lambda_2-1)}. \tag{3.6}$$

Moreover, the off-diagonal 2×2 block minors of M and M^{-1} factor identically as

$$\begin{aligned}
\det M_{(1,2),(3,4)} &= \frac{b_4(b_1-1)(b_2-1)(b_4-1)(b_5-1)(w-1)(wb_1b_5-1)}{(\lambda_1-1)(\lambda_2-1)}, & \det M^{-1}_{(1,2),(3,4)} &= w^{-2}\det M_{(1,2),(3,4)}, \\
\det M_{(3,4),(1,2)} &= \frac{b_2(b_3-1)(w-1)}{(\lambda_1-1)(\lambda_2-1)}, & \det M^{-1}_{(3,4),(1,2)} &= w^{-2}\det M_{(3,4),(1,2)}.
\end{aligned} \tag{3.7}$$

3. **(Unconditional Non-Vanishing via Cyclotomic Linear Disjointness):** *For every unit $t \in (\mathbf{Z}/m\mathbf{Z})^\times$ on (1.9) and (1.8), all 12 off-diagonal entries $M_{i,j}, (M^{-1})_{i,j}$ ($i \neq j$), the four top-left diagonal entries $M_{1,1}, M_{2,2}, (M^{-1})_{1,1}, (M^{-1})_{2,2}$, and all four off-diagonal 2×2 minors in (3.7) are **non-zero**.*

Proof. 1. Direct inspection of (2.2) and (3.2) shows that the columns of C are eigenvectors of $g_1(t)$ with eigenvalues $(\lambda_1, \lambda_2, 1, 1)$, and multiplying (3.3), (2.2), and (3.2) yields (3.4)–(3.7). 2. On both (1.9) ($m = a\cdot b\cdot N$) and (1.8) ($m = 3\cdot 5\cdot N$, where $a = 3, b = 5$), we have $a \mid a_1, a_{-1}, a_\lambda, a_{-\lambda}$ and $a \nmid a_0, a_\infty$ ($\gcd(a_0, a) = 1$). Consequently, $b_1, b_2, b_4, b_5 \in \mu_{bN}$, whereas every monomial containing b_3 to the first power—namely $b_3, b_2b_3, b_3b_4, w = b_2b_3b_4, wb_1, wb_5, wb_1b_5$—has non-trivial a-primary component $\omega_a = \zeta_a^{t_a} \in \mu_a \setminus \{1\}$ (or $\omega^{2t_3} \in \mu_3 \setminus \{1\}$ on (1.8)) and therefore cannot equal 1. Meanwhile, b_1, b_2, b_4, b_5 have non-trivial b-primary component $\eta_b = \zeta_b^{c_bt_b} \in \mu_b \setminus \{1\}$ on (1.9) and $\xi^{\pm t_5}, \xi^{\pm 2t_5} \in \mu_5 \setminus \{1\}$ on (1.8), while $\lambda_1, \lambda_2 \neq 1$ by Lemma 2.1. Hence all 12 off-diagonal entries $M_{i,j}, (M^{-1})_{i,j}$ ($i \neq j$) and all four 2×2

minors in (3.7) are products of non-zero cyclotomic binomials $(u-1) \neq 0$. 3. Finally, because $\gcd(a, bN) = 1$, the cyclotomic fields $\mathbf{Q}(\zeta_a)$ and $\mathbf{Q}(\zeta_{bN})$ have coprime discriminants and are linearly disjoint over $\mathbf{Q}$:

$$\mathbf{Q}(\zeta_a) \cap \mathbf{Q}(\zeta_{bN}) = \mathbf{Q}. \tag{3.8}$$

Since $w = \omega_a = \zeta_a^{t_a}$ (or ω^{2t_3}) has exact multiplicative order $a \geq 3$, $w \notin \mathbf{Q}$ and the pair $\{1, w\}$ is linearly independent over $\mathbf{Q}(\zeta_{bN})$. Because $b_1, b_2, b_4, b_5 \in \mathbf{Q}(\zeta_{bN}) \setminus \{1\}$, the coefficients of 1 and w in the numerators $(b_2-1)+(b_1-1)w$, $(b_4-1)+(b_5-1)w$, $(b_1-1)b_2+b_1(b_2-1)w$, and $(b_5-1)b_4+b_5(b_4-1)w$ are non-zero elements of $\mathbf{Q}(\zeta_{bN})$. Therefore none of these four linear combinations can vanish, proving $M_{1,1}, M_{2,2}, (M^{-1})_{1,1}, (M^{-1})_{2,2} \neq 0$. □

3.2 Proof of the Adjoint $\mathfrak{sl}_4(\mathbf{C})$-Dichotomy (Theorem A(2))

Theorem 3.2 (Exact Adjoint Dichotomy for $\langle g_1(t), g_0(t)\rangle$).
Let $m = a \cdot b \cdot N$ with pairwise coprime odd integers $a \geq 3, b \geq 3, N \geq 7$ and $\gcd(N, 30) = 1$ on the generalized primary CRT family (1.9) (including $m = 15N$ on (1.7)), or let $m = 15N$ with $\gcd(N, 30) = 1, N > 1$ on the companion CRT family (1.8). For every unit $t \in (\mathbf{Z}/m\mathbf{Z})^\times$, let $\mathfrak{g} \subseteq \mathfrak{sl}_4(\mathbf{C})$ be any Lie subalgebra invariant under $\mathrm{Ad}(g_1(t))$ and $\mathrm{Ad}(g_0(t))$. Then either

$$\mathfrak{g} = \{0\} \qquad \text{or} \qquad \mathfrak{g} = \mathfrak{sl}_4(\mathbf{C}). \tag{3.9}$$

Proof. Conjugate $\mathfrak{sl}_4(\mathbf{C})$ by the matrix C of Lemma 3.1 so that $g_1(t)$ acts via $\mathrm{diag}(\lambda_1, \lambda_2, 1, 1)$ and $g_0(t)$ acts via $M = C^{-1} g_0(t) C$. On the standard Chevalley basis $\{E_{i,j} (1 \leq i \neq j \leq 4), E_{i,i} - E_{i+1,i+1} (1 \leq i \leq 3)\}$ of $\mathfrak{sl}_4(\mathbf{C})$, the semisimple operator $\mathrm{Ad}(g_1(t))(E_{i,j}) = (\lambda_i/\lambda_j)E_{i,j}$ is diagonal with 7 eigenvalues:

$$\mathrm{Spec}\big(\mathrm{Ad}(g_1(t))\big) = \Big\{\lambda_1\lambda_2^{-1},\ \lambda_2\lambda_1^{-1},\ \lambda_1,\ \lambda_1^{-1},\ \lambda_2,\ \lambda_2^{-1},\ 1\Big\}. \tag{3.10}$$

We first verify that these 7 eigenvalues are pairwise distinct for every $t \in (\mathbf{Z}/m\mathbf{Z})^\times$: - **On the generalized primary CRT family (1.9) (including (1.7)):** Writing $\eta_b = \zeta_b^{c_b t_b} \in \mu_b \setminus \{1\}$ and $u = z^{t_N} \in \mu_N$ (of exact primitive order N), we have $\lambda_1 = \eta_b^2 u^{-3}$ and $\lambda_2 = \eta_b^2 u^{+3}$, so the 7 eigenvalues in (3.10) are

$$\{u^{-6},\ u^{+6},\ \eta_b^2 u^{-3},\ \eta_b^{-2} u^{+3},\ \eta_b^2 u^{+3},\ \eta_b^{-2} u^{-3},\ 1\}.$$

Since $b \geq 3$ is odd and $\gcd(c_b t_b, b) = 1$, η_b has odd multiplicative order $b \geq 3$, so $1, \eta_b^2, \eta_b^{-2}$ are three pairwise distinct roots of unity in μ_b. Since $\gcd(N, 30) = 1$ and $N \geq 7$, we have $\gcd(6, N) = 1$ and $\gcd(12, N) = 1$, so $u^6 \neq 1$, $u^{-6} \neq 1$, $u^6 \neq u^{-6}$ (as $u^{12} \neq 1$), and $u^3 \neq u^{-3}$ in μ_N. Because $\gcd(b, N) = 1$, the three η_b^0-fibers $\{1, u^{-6}, u^{+6}\}$, the two η_b^2-fibers $\{\eta_b^2 u^{\mp 3}\}$, and the two η_b^{-2}-fibers $\{\eta_b^{-2} u^{\pm 3}\}$ are all pairwise distinct. - **On the companion CRT family (1.8):** Here $\lambda_1 = \xi^{-3t_5} z^{2c_1 t_N}$ and $\lambda_2 = \xi^{+3t_5} z^{2c_1 t_N}$, so the 7 eigenvalues in (3.10) are

$$\{\xi^{4t_5},\ \xi^{t_5},\ \xi^{2t_5} z^{2c_1 t_N},\ \xi^{3t_5} z^{-2c_1 t_N},\ \xi^{3t_5} z^{2c_1 t_N},\ \xi^{2t_5} z^{-2c_1 t_N},\ 1\}.$$

Since $\gcd(2c_1 t_N, N) = 1$ and $\gcd(N, 30) = 1$ ($N \geq 7 > 4$), $z^{2c_1 t_N} \neq z^{-2c_1 t_N}$ in μ_N, while $1, \xi^{t_5}, \xi^{2t_5}, \xi^{3t_5}, \xi^{4t_5}$ are distinct in μ_5. Thus all 7 eigenvalues are pairwise distinct.

By Lagrange interpolation in the polynomial ring $\mathbf{C}[\mathrm{Ad}(g_1(t))]$, any $\mathrm{Ad}(g_1(t))$-invariant subspace $\mathfrak{g} \subseteq \mathfrak{sl}_4(\mathbf{C})$ decomposes into the direct sum of its intersections with the 7 eigenspaces of $\mathrm{Ad}(g_1(t))$:

$$\mathfrak{g} = \mathfrak{g}_{\lambda_1\lambda_2^{-1}} \oplus \mathfrak{g}_{\lambda_2\lambda_1^{-1}} \oplus \mathfrak{g}_{\lambda_1} \oplus \mathfrak{g}_{\lambda_1^{-1}} \oplus \mathfrak{g}_{\lambda_2} \oplus \mathfrak{g}_{\lambda_2^{-1}} \oplus \mathfrak{g}_1, \tag{3.11}$$

where $\mathfrak{g}_{\lambda_1\lambda_2^{-1}} \subseteq \mathbf{C}E_{1,2}$, $\mathfrak{g}_{\lambda_2\lambda_1^{-1}} \subseteq \mathbf{C}E_{2,1}$, $\mathfrak{g}_{\lambda_r} \subseteq \mathrm{span}(E_{r,3}, E_{r,4})$ ($r \in \{1,2\}$), $\mathfrak{g}_{\lambda_r^{-1}} \subseteq \mathrm{span}(E_{3,r}, E_{4,r})$ ($r \in \{1,2\}$), and $\mathfrak{g}_1 \subseteq \{\mathrm{diag}(d_1, d_2, A) \in \mathfrak{sl}_4(\mathbf{C}) : d_1, d_2 \in \mathbf{C}, A \in M_2(\mathbf{C})\}$. We complete the proof in three steps:

1. **Step 1 (If $\mathfrak{g} \subseteq \mathfrak{g}_1$, then $\mathfrak{g} = \{0\}$):**
   Suppose $\mathfrak{g} \subseteq \mathfrak{g}_1$. For any $X = \mathrm{diag}(d_1, d_2, A) \in \mathfrak{g}$, invariance under $\mathrm{Ad}(g_0(t))$ implies $MXM^{-1} \in \mathfrak{g} \subseteq \mathfrak{g}_1$, so $MXM^{-1} = \mathrm{diag}(d_1', d_2', A')$ for some $d_1', d_2' \in \mathbf{C}$ and $A' \in M_2(\mathbf{C})$, or equivalently

   $$M\,\mathrm{diag}(d_1, d_2, A) = \mathrm{diag}(d_1', d_2', A')M. \tag{3.12}$$

   Comparing entries of (3.12) and invoking Lemma 3.1(3):

- Entry $(1,1)$ gives $M_{1,1}d_1 = d_1'M_{1,1}$; since $M_{1,1} \neq 0$, $d_1' = d_1$.
- Entry $(1,2)$ gives $M_{1,2}d_2 = d_1'M_{1,2}$; since $M_{1,2} \neq 0$, $d_2 = d_1' = d_1$.
- Entry $(2,1)$ gives $M_{2,1}d_1 = d_2'M_{2,1}$; since $M_{2,1} \neq 0$, $d_2' = d_1$.
- The 2×2 block in rows $(1,2)$ and columns $(3,4)$ gives $M_{(1,2),(3,4)}A = \operatorname{diag}(d_1', d_2')M_{(1,2),(3,4)} = d_1 M_{(1,2),(3,4)}$; since $\det M_{(1,2),(3,4)} \neq 0$, $M_{(1,2),(3,4)}$ is invertible and hence $A = d_1 I_2$.

Thus $X = d_1 I_4$. Since $X \in \mathfrak{sl}_4(\mathbf{C})$, $0 = \operatorname{tr}(X) = 4d_1$, so $d_1 = 0$ and $X = 0$. Therefore $\mathfrak{g} = \{0\}$.

2. **Step 2 (If $\mathfrak{g} \neq \{0\}$, then $\mathfrak{g}$ contains $E_{1,2}$ or $E_{2,1}$):**
Suppose $\mathfrak{g} \neq \{0\}$. By Step 1, $\mathfrak{g} \not\subseteq \mathfrak{g}_1$, so at least one of the 6 non-trivial eigenspaces in (3.11) is non-zero:

   - If $\mathfrak{g}_{\lambda_1\lambda_2^{-1}} \neq \{0\}$ or $\mathfrak{g}_{\lambda_2\lambda_1^{-1}} \neq \{0\}$, then $E_{1,2} \in \mathfrak{g}$ or $E_{2,1} \in \mathfrak{g}$ directly since both root spaces are 1-dimensional.
   - If $\mathfrak{g}_{\lambda_r} \neq \{0\}$ for some $r \in \{1,2\}$, choose $0 \neq X = c_3E_{r,3} + c_4E_{r,4} \in \mathfrak{g}_{\lambda_r}$ with $(c_3, c_4) \neq (0,0)$. The $(1,2)$ and $(2,1)$ entries of $MXM^{-1} \in \mathfrak{g}$ are

   $$(MXM^{-1})_{1,2} = M_{1,r}\big(c_3(M^{-1})_{3,2} + c_4(M^{-1})_{4,2}\big), \qquad (MXM^{-1})_{2,1} = M_{2,r}\big(c_3(M^{-1})_{3,1} + c_4(M^{-1})_{4,1}\big).$$

   Since $M_{1,r} \neq 0, M_{2,r} \neq 0$, and $\det M^{-1}_{(3,4),(1,2)} \neq 0$ by Lemma 3.1(3), these two entries cannot both vanish. Projecting $MXM^{-1} \in \mathfrak{g}$ onto $\mathfrak{g}_{\lambda_1\lambda_2^{-1}} = \mathbf{C}E_{1,2}$ or $\mathfrak{g}_{\lambda_2\lambda_1^{-1}} = \mathbf{C}E_{2,1}$ yields $E_{1,2} \in \mathfrak{g}$ or $E_{2,1} \in \mathfrak{g}$.
   - If $\mathfrak{g}_{\lambda_r^{-1}} \neq \{0\}$ for some $r \in \{1,2\}$, choose $0 \neq X = c_3E_{3,r} + c_4E_{4,r} \in \mathfrak{g}_{\lambda_r^{-1}}$ with $(c_3, c_4) \neq (0,0)$. The $(1,2)$ and $(2,1)$ entries of $MXM^{-1} \in \mathfrak{g}$ are

   $$(MXM^{-1})_{1,2} = \big(c_3M_{1,3} + c_4M_{1,4}\big)(M^{-1})_{r,2}, \qquad (MXM^{-1})_{2,1} = \big(c_3M_{2,3} + c_4M_{2,4}\big)(M^{-1})_{r,1}.$$

   Since $(M^{-1})_{r,2} \neq 0, (M^{-1})_{r,1} \neq 0$, and $\det M_{(1,2),(3,4)} \neq 0$ by Lemma 3.1(3), these two entries cannot both vanish, again yielding $E_{1,2} \in \mathfrak{g}$ or $E_{2,1} \in \mathfrak{g}$.

3. **Step 3 ($E_{1,2} \in \mathfrak{g}$ or $E_{2,1} \in \mathfrak{g}$ forces $\mathfrak{g} = \mathfrak{sl}_4(\mathbf{C})$):**

   - For any elementary matrix $E_{p,q}$, we have $(ME_{p,q}M^{-1})_{i,j} = M_{i,p}(M^{-1})_{q,j}$. Taking $(p,q,i,j) = (1,2,2,1)$ and $(p,q,i,j) = (2,1,1,2)$ and applying Lemma 3.1(3) gives

   $$(ME_{1,2}M^{-1})_{2,1} = M_{2,1}(M^{-1})_{2,1} = -w^{-1}M_{2,1}^2 \neq 0, \qquad (ME_{2,1}M^{-1})_{1,2} = M_{1,2}(M^{-1})_{1,2} = -w^{-1}M_{1,2}^2 \neq 0. \tag{3.13}$$

   Projecting $ME_{1,2}M^{-1}$ onto $\mathfrak{g}_{\lambda_2\lambda_1^{-1}}$ or $ME_{2,1}M^{-1}$ onto $\mathfrak{g}_{\lambda_1\lambda_2^{-1}}$ shows that **both** $E_{1,2} \in \mathfrak{g}$ **and** $E_{2,1} \in \mathfrak{g}$.
   - Projecting $ME_{1,2}M^{-1} \in \mathfrak{g}$ and $ME_{2,1}M^{-1} \in \mathfrak{g}$ onto $\mathfrak{g}_{\lambda_1} \subseteq \operatorname{span}(E_{1,3}, E_{1,4})$ produces the two vectors

   $$u_1 = M_{1,1}\big((M^{-1})_{2,3}E_{1,3} + (M^{-1})_{2,4}E_{1,4}\big) \in \mathfrak{g}, \qquad u_2 = M_{1,2}\big((M^{-1})_{1,3}E_{1,3} + (M^{-1})_{1,4}E_{1,4}\big) \in \mathfrak{g}.$$

   Since $M_{1,1}M_{1,2}\det M^{-1}_{(1,2),(3,4)} \neq 0$ by Lemma 3.1(3), u_1 and u_2 are linearly independent, so $E_{1,3}, E_{1,4} \in \mathfrak{g}$.
   - Projecting $ME_{1,2}M^{-1} \in \mathfrak{g}$ and $ME_{2,1}M^{-1} \in \mathfrak{g}$ onto $\mathfrak{g}_{\lambda_1^{-1}} \subseteq \operatorname{span}(E_{3,1}, E_{4,1})$ produces the two vectors

   $$v_1 = (M^{-1})_{2,1}\big(M_{3,1}E_{3,1} + M_{4,1}E_{4,1}\big) \in \mathfrak{g}, \qquad v_2 = (M^{-1})_{1,1}\big(M_{3,2}E_{3,1} + M_{4,2}E_{4,1}\big) \in \mathfrak{g}.$$

   Since $(M^{-1})_{2,1}(M^{-1})_{1,1}\det M_{(3,4),(1,2)} \neq 0$ by Lemma 3.1(3), v_1 and v_2 are linearly independent, so $E_{3,1}, E_{4,1} \in \mathfrak{g}$.
   - Taking Lie brackets $[X,Y] = XY - YX$ in $\mathfrak{g}$ generates the remaining six root vectors and all three Cartan generators:

   $$E_{2,3} = [E_{2,1}, E_{1,3}], \quad E_{2,4} = [E_{2,1}, E_{1,4}], \quad E_{3,2} = [E_{3,1}, E_{1,2}], \quad E_{4,2} = [E_{4,1}, E_{1,2}],$$

   $$E_{3,4} = [E_{3,1}, E_{1,4}], \quad E_{4,3} = [E_{4,1}, E_{1,3}], \quad E_{i,i} - E_{j,j} = [E_{i,j}, E_{j,i}] \quad (1 \leq i < j \leq 4).$$

   Thus $\mathfrak{g}$ contains all 12 root vectors $E_{i,j}$ $(i \neq j)$ and all 3 Cartan generators $E_{i,i} - E_{i+1,i+1}$, proving $\mathfrak{g} = \mathfrak{sl}_4(\mathbf{C})$. □

4. Unconditional Zariski Density for General $m = 15N$ via Ramanujan-Sum Trace Averages

4.1 Explicit Three-Term Trace Formula for the Product-Square Word $W = (g_1 g_0)^2$

By Theorem 3.2, to prove that $\Gamma_0(t) = \langle g_1(t), g_0(t)\rangle$ is Zariski-dense in $\mathrm{SU}(J(t))$ up to μ_m scalars, it suffices to exhibit a single element of $\Gamma_0(t)$ whose SL_4-normalized power has infinite order, thereby ruling out the finite-group case $\mathfrak{g}_0(t) = \{0\}$. We use the length-4 braid word $W(t) := (g_1(t)g_0(t))^2$.

Lemma 4.1 (Three-Term Trace Formula for $(g_1(t)g_0(t))^2$).
For any odd integer $m \geq 3$, any branch weight tuple $\mathbf{a}$*, and any unit $t \in (\mathbf{Z}/m\mathbf{Z})^\times$, the word $W(t) = (g_1(t)g_0(t))^2 \in \Gamma_0(t)$ has determinant $\det(W(t)) = (b_1b_2b_4b_5)^2w^4 = \delta_2(t)^4$ with $\delta_2(t) := (b_1b_2b_4b_5)^{(m+1)/2}w \in \mu_m$ and exact trace*

$$\mathrm{tr}\big((g_1(t)g_0(t))^2\big) = (b_2^2 + b_4^2) + 2(b_1 - 1)(b_2 - 1)(b_4 - 1)(b_5 - 1)\,w + (b_1^2 + b_5^2)\,w^2. \tag{4.1}$$

Proof. Multiplying $A := g_1(t)g_0(t)$ directly from (2.2) and using the relation $b_4(1 - b_2b_3) = b_4 - b_2b_3b_4 = b_4 - w$ yields

$$A = \begin{pmatrix} b_2 + (b_1 - 1)w & (1 - b_1)w & 0 & (1 - b_1)(1 - b_2) \\ b_2 - w & w & 0 & 1 - b_2 \\ b_2b_3 - w & 0 & w & 1 - b_2b_3 \\ (1 - b_4)(1 - b_5)w & 0 & b_4(1 - b_5)w & b_4 + (b_5 - 1)w \end{pmatrix}. \tag{4.2}$$

Since $A_{1,3} = A_{2,3} = A_{3,2} = A_{4,2} = 0$, expanding $\mathrm{tr}(A^2) = \sum_{i=1}^4 A_{i,i}^2 + 2\sum_{1\leq i<j\leq 4} A_{i,j}A_{j,i}$ gives

$$\begin{aligned} \mathrm{tr}(A^2) &= \big(b_2 + (b_1 - 1)w\big)^2 + w^2 + w^2 + \big(b_4 + (b_5 - 1)w\big)^2 \\ &\quad + 2(1 - b_1)w(b_2 - w) + 2(b_1 - 1)(b_2 - 1)(b_4 - 1)(b_5 - 1)w + 2(b_4 - w)(1 - b_5)w \\ &= (b_2^2 + b_4^2) + 2(b_1 - 1)(b_2 - 1)(b_4 - 1)(b_5 - 1)w + (b_1^2 + b_5^2)w^2, \end{aligned}$$

where the cross terms $2b_2(b_1 - 1)w + 2(1 - b_1)b_2w = 0$ and $2b_4(b_5 - 1)w + 2b_4(1 - b_5)w = 0$ cancel identically, and the w^2 terms collect as $((b_1 - 1)^2 + 2(b_1 - 1) + 1 + (b_5 - 1)^2 + 2(b_5 - 1) + 1)w^2 = (b_1^2 + b_5^2)w^2$. Since $\det g_1(t) = b_1b_2b_4b_5$ and $\det g_0(t) = w^2$, we have $\det(W(t)) = (b_1b_2b_4b_5)^2w^4 = \delta_2(t)^4$. □

4.2 Partial Average over $(\mathbf{Z}/3\mathbf{Z})^\times \times (\mathbf{Z}/5\mathbf{Z})^\times$ and Chebyshev–Laurent Polynomials

Let $N > 1$ be any odd integer with $\gcd(N, 30) = 1$ and set $m = 15N$. Fix primitive roots of unity $\omega \in \mu_3\backslash\{1\}$, $\xi \in \mu_5\backslash\{1\}$, and $z \in \mu_N$ (of exact primitive order N). Write $\mathrm{tr}(W(t)) = C_0 + C_1w + C_2w^2$ with

$$C_0 := b_2^2 + b_4^2, \qquad C_1 := 2(b_1 - 1)(b_2 - 1)(b_4 - 1)(b_5 - 1), \qquad C_2 := b_1^2 + b_5^2. \tag{4.3}$$

On both CRT families (1.7) and (1.8), $3 \mid a_1, a_{-1}, a_\lambda, a_{-\lambda}$, so C_0, C_1, C_2 are independent of $t_3 \in (\mathbf{Z}/3\mathbf{Z})^\times = \{1, 2\}$, while $w = \omega^{\pm t_3}$ runs over $\{\omega, \omega^2\}$. Using the orthogonality relation $\frac{1}{2}\sum_{t_3\in\{1,2\}} w^{j-k} = \frac{3\delta_{j,k}-1}{2}$, averaging $|\mathrm{tr}(W(t))|^2$ over $t_3 \in (\mathbf{Z}/3\mathbf{Z})^\times$ yields

$$\frac{1}{2}\sum_{t_3\in\{1,2\}} \big|\mathrm{tr}(W(t))\big|^2 = |C_0|^2 + |C_1|^2 + |C_2|^2 - \mathrm{Re}\big(C_0\overline{C_1} + C_1\overline{C_2} + C_2\overline{C_0}\big). \tag{4.4}$$

We now average (4.4) over $t_5 \in (\mathbf{Z}/5\mathbf{Z})^\times = \{1, 2, 3, 4\}$ on each of the two CRT families:

1. **Primary CRT Family (1.7):**
   Put $\eta := \xi^{2t_5} \in \mu_5 \setminus \{1\}$, $u := z^{t_N} \in \mu_N$, and $X_k := u^k + u^{-k} \in \mathbf{R}$ (which satisfy $X_0 = 2$ and the Chebyshev product identity $X_jX_k = X_{j+k} + X_{|j-k|}$). Then $b_1 = \eta u^{-1}$, $b_5 = \eta u$, $b_2 = \eta u^{-2}$, and $b_4 = \eta u^2$, so

$$C_0 = X_4\eta^2, \qquad C_2 = X_2\eta^2, \qquad C_1 = 2(\eta^2 - X_1\eta + 1)(\eta^2 - X_2\eta + 1) = \sum_{j=0}^4 p_j\eta^j, \tag{4.5}$$

where $(p_0, p_1, p_2, p_3, p_4) = 2\big(1,\ -(X_1+X_2),\ 2+X_1X_2,\ -(X_1+X_2),\ 1\big)$. Since $\frac{1}{4}\sum_{t_5=1}^{4}\eta^{j-k} = \frac{5\delta_{j,k}-1}{4}$, any two polynomials $P(\eta)=\sum_{j=0}^{4}p_j\eta^j$ and $Q(\eta)=\sum_{j=0}^{4}q_j\eta^j$ with real coefficients satisfy

$$\langle P,Q\rangle_{t_5} := \frac{1}{4}\sum_{t_5=1}^{4}P(\eta)\overline{Q(\eta)} = \frac{5}{4}\sum_{j=0}^{4}p_jq_j - \frac{1}{4}\left(\sum_{j=0}^{4}p_j\right)\left(\sum_{j=0}^{4}q_j\right). \tag{4.6}$$

Evaluating the three terms of (4.4) via (4.6) and $X_jX_k = X_{j+k}+X_{|j-k|}$ gives:

- $\langle |C_0|^2+|C_2|^2-\mathrm{Re}(C_2\overline{C_0})\rangle_{t_5} = X_4^2+X_2^2-X_2X_4 = 4-X_2+X_4-X_6+X_8$;
- $-\langle C_1, C_0+C_2\rangle_{t_5} = -(X_2+X_4)\big(\frac{5}{4}p_2-\frac{1}{4}\sum_{j=0}^{4}p_j\big) = -(X_2+X_4)(3+3X_1+X_2+2X_3) = -2-7X_1-4X_2-6X_3-4X_4-5X_5-X_6-2X_7$;
- $\langle C_1,C_1\rangle_{t_5} = 14+16(X_1+X_2)+12(X_1+X_3)+6(X_1+X_2)^2+4(X_1+X_3)^2+4(X_1+X_2)(X_1+X_3) = 62+48X_1+42X_2+28X_3+18X_4+4X_5+4X_6$.

Summing these three contributions yields the exact degree-8 Chebyshev–Laurent polynomial (1.10):

$$M_{\mathrm{prim}}(u) := \frac{1}{8}\sum_{t_3,t_5}\big|\mathrm{tr}(W(t))\big|^2 = 64+41X_1+37X_2+22X_3+15X_4-X_5+2X_6-2X_7+X_8. \tag{4.7}$$

2. **Companion CRT Family (1.8):**
Put $u := z^{c_1t_N}\in\mu_N$ and $X_k := u^k+u^{-k}$. Then $b_1=\xi^{-t_5}u$, $b_5=\xi^{t_5}u$, $b_2=\xi^{-2t_5}u$, and $b_4=\xi^{2t_5}u$, so

$$\prod_{j\in\{1,2,4,5\}}(b_j-1) = \prod_{r=1}^{4}(u-\xi^r) = 1+u+u^2+u^3+u^4 = u^2(1+X_1+X_2). \tag{4.8}$$

Dividing C_0, C_1, C_2 by the unit-modulus factor u^2 yields three **real** algebraic integers:

$$a_0 := u^{-2}C_0 = \xi^{t_5}+\xi^{-t_5}, \qquad a_2 := u^{-2}C_2 = \xi^{2t_5}+\xi^{-2t_5}, \qquad a_1 := u^{-2}C_1 = 2(1+X_1+X_2) =: 2S(u).$$

Because $a_0+a_2 = \sum_{r=1}^{4}\xi^r = -1$ and $a_0a_2=-1$ (so $a_0^2+a_2^2-a_0a_2 = (a_0+a_2)^2-3a_0a_2 = 4$) for **every** $(t_3,t_5)\in(\mathbf{Z}/3\mathbf{Z})^\times\times(\mathbf{Z}/5\mathbf{Z})^\times$, the norm square $|\mathrm{tr}(W(t))|^2$ is completely independent of (t_3,t_5) and equals (1.11):

$$\big|\mathrm{tr}(W(t))\big|^2 = M_{\mathrm{comp}}(u) = 4+2S(u)+4S(u)^2 = 26+18X_1+14X_2+8X_3+4X_4 \qquad \big(S(u) := 1+X_1+X_2\big). \tag{4.9}$$

—

4.3 Ramanujan-Sum Evaluation over $(\mathbf{Z}/N\mathbf{Z})^\times$ and Proof of Theorem B(1)

Theorem 4.2 (Unconditional Zariski Density for General $m = 15N$ with $\gcd(N,30)=1$).
Let $N>1$ be any integer with $\gcd(N,30)=1$ and $m=15N$. Then on both the primary CRT family (1.7) and the companion CRT family (1.8), all assertions of Theorem B(1) hold. In particular, for every $t\in(\mathbf{Z}/15N\mathbf{Z})^\times$, the group $\Gamma_0(t)=\langle g_1(t), g_0(t)\rangle$ is Zariski-dense in $\mathrm{SU}(J(t))$ up to μ_{15N} scalars.

Proof. For each integer $r\ge 1$, the sum of $X_r = u^r+u^{-r}$ over $t_N\in(\mathbf{Z}/N\mathbf{Z})^\times$ is twice Ramanujan's sum $c_N(r) := \sum_{t_N\in(\mathbf{Z}/N\mathbf{Z})^\times} z^{rt_N}$. By multiplicativity and Hölder's formula,

$$c_N(r) = \prod_{p^k\parallel N}c_{p^k}(r) = \varphi(N)\,\frac{\mu\big(N/\gcd(N,r)\big)}{\varphi\big(N/\gcd(N,r)\big)}, \qquad c_{p^k}(r) = \begin{cases} p^{k-1}(p-1) & \text{if } p^k\mid r,\\ -p^{k-1} & \text{if } p^{k-1}\parallel r,\\ 0 & \text{if } p^{k-1}\nmid r.\end{cases} \tag{4.10}$$

Writing $(\alpha_1,\dots,\alpha_8) = (41,37,22,15,-1,2,-2,1)$ and $(\beta_1,\dots,\beta_4)=(18,14,8,4)$, averaging (4.7) and (4.9) over $t_N\in(\mathbf{Z}/N\mathbf{Z})^\times$ gives

$$\begin{aligned}\mathcal{M}_{\mathrm{prim}}(N) &:= \frac{1}{\varphi(15N)}\sum_{t\in(\mathbf{Z}/15N\mathbf{Z})^\times}\big|\mathrm{tr}(W(t))\big|^2 = 64+\frac{2}{\varphi(N)}\sum_{r=1}^{8}\alpha_rc_N(r),\\ \mathcal{M}_{\mathrm{comp}}(N) &:= \frac{1}{\varphi(15N)}\sum_{t\in(\mathbf{Z}/15N\mathbf{Z})^\times}\big|\mathrm{tr}(W(t))\big|^2 = 26+\frac{2}{\varphi(N)}\sum_{r=1}^{4}\beta_rc_N(r).\end{aligned} \tag{4.11}$$

We analyze the prime, prime-power, and general composite regimes in turn:

1. **Prime $N = p \geq 7$:**
   - On the primary family, if $p \geq 11$, then $p \nmid r$ for all $1 \leq r \leq 8$, so $c_p(r) = -1$ and
   $$\mathcal{M}_{\text{prim}}(p) = 64 - \frac{2\sum_{r=1}^{8} \alpha_r}{p-1} = 64 - \frac{230}{p-1} = \frac{64p - 294}{p-1} \geq \frac{64(11) - 294}{10} = 41 > 16.$$
   If $p = 7$, then $c_7(7) = 6$ while $c_7(r) = -1$ for $r \neq 7$, so $\mathcal{M}_{\text{prim}}(7) = 64 + \frac{2}{6}(-115 + (-2)(7)) = 64 - 43 = 21 > 16$.
   - On the companion family, since $p \geq 7 > 4$, $p \nmid r$ for all $1 \leq r \leq 4$, so $c_p(r) = -1$ and
   $$\mathcal{M}_{\text{comp}}(p) = 26 - \frac{2\sum_{r=1}^{4} \beta_r}{p-1} = 26 - \frac{88}{p-1} = \frac{26p - 114}{p-1},$$
   which equals $\frac{34}{3}$ at $p = 7$ and satisfies $\mathcal{M}_{\text{comp}}(p) \geq \frac{86}{5} = 17.2 > 16$ for all $p \geq 11$.

2. **Prime Powers $N = p^k$ ($p \geq 7$ prime, $k \geq 2$):**
   Here $p^{k-1} \geq p \geq 7$.
   - On the companion family ($1 \leq r \leq 4$), $1 \leq r \leq 4 < 7 \leq p^{k-1}$, so $p^{k-1} \nmid r$ and $c_{p^k}(r) = 0$ for all $1 \leq r \leq 4$ by (4.10). Thus $\mathcal{M}_{\text{comp}}(p^k) = \mathbf{26} > \mathbf{16}$ identically for all $k \geq 2$ and all primes $p \geq 7$ (including $p = 7$).
   - On the primary family ($1 \leq r \leq 8$), if $p \geq 11$ and $k \geq 2$, or if $p = 7$ and $k \geq 3$ ($p^{k-1} \geq 49 > 8$), then $p^{k-1} \nmid r$ for all $1 \leq r \leq 8$, so $c_{p^k}(r) = 0$ and $\mathcal{M}_{\text{prim}}(p^k) = \mathbf{64} > \mathbf{16}$ identically. At the single remaining prime-power pair $(p, k) = (7, 2)$ ($N = 49$, $p^{k-1} = 7$), only $r = 7$ is divisible by 7, with $7^1 \parallel 7$, $c_{49}(7) = -7$, and $\frac{2c_{49}(7)}{\varphi(49)} = \frac{-14}{42} = -\frac{1}{3}$; since $\alpha_7 = -2$, we obtain
   $$\mathcal{M}_{\text{prim}}(49) = 64 + (-2)\left(-\frac{1}{3}\right) = \frac{194}{3} = 64.\overline{6} > 16.$$

3. **Arbitrary Composite $N > 1$ with $\gcd(N, 30) = 1$:**
   Since every prime factor of N is ≥ 7:
   - On the primary family ($1 \leq r \leq 8$), for any $N \neq 7$, we have $\gcd(N, r) = 1$ for $r \neq 7$, so $N/\gcd(N, r) = N \geq 11$ has $\varphi(N) \geq 10$ and $|c_N(r)|/\varphi(N) \leq \frac{1}{10}$ (indeed, if $N = p_1 \cdots p_s$ is a squarefree composite of $s \geq 2$ primes ≥ 7, $\varphi(N) \geq (7-1)(11-1) = 60$, and if N is non-squarefree with $r \neq 7$, $c_N(r) = 0$ unless $N = 49$, already treated above). For $r = 7$ with $N \neq 7$, $N/\gcd(N, 7) \geq 7$ has $\varphi(N/\gcd(N, 7)) \geq 6$, so $|c_N(7)|/\varphi(N) \leq \frac{1}{6}$. Therefore, for every $N \neq 7$ with $\gcd(N, 30) = 1$,
   $$\mathcal{M}_{\text{prim}}(N) \geq 64 - \frac{2}{10}\sum_{r \neq 7} |\alpha_r| - \frac{2}{6}|\alpha_7| = 64 - \frac{2(119)}{10} - \frac{4}{6} = \frac{593}{15} = 39.5\overline{3} > 16$$
   (and for composite N with $s \geq 2$ distinct prime factors ≥ 7, $\varphi(N/\gcd(N, r)) \geq \varphi(11) = 10$ even at $r = 7$, giving $\mathcal{M}_{\text{prim}}(N) \geq 64 - \frac{242}{10} = 39.8 > 16$; in fact, since $\mu(p_1 p_2) = +1$ for $s = 2$, the minimum over all $N > 1$ is $\mathcal{M}_{\text{prim}}(7) = 21 > 16$, and among $N > 7$ is $\mathcal{M}_{\text{prim}}(11) = 41 > 16$).
   - On the companion family ($1 \leq r \leq 4$), since every prime factor of N is $\geq 7 > 4$, we have $\gcd(N, r) = 1$ for all $1 \leq r \leq 4$. Thus $c_N(r)/\varphi(N) = \mu(N)/\varphi(N)$ for all $1 \leq r \leq 4$. Whenever $N > 7$ with $\gcd(N, 30) = 1$, either $N \geq 11$ is prime ($\mu(N)/\varphi(N) = -\frac{1}{N-1} \geq -\frac{1}{10}$), N is non-squarefree ($\mu(N) = 0$), or $N = p_1 \cdots p_s$ with $s \geq 2$ primes ≥ 7 ($\varphi(N) \geq 60$, $|\mu(N)|/\varphi(N) \leq \frac{1}{60}$). In each of these three regimes for $N > 7$,
   $$\mathcal{M}_{\text{comp}}(N) = 26 + \frac{88\,\mu(N)}{\varphi(N)} \geq 26 - \frac{88}{10} = \frac{86}{5} = 17.2 > 16.$$

4. **Exact Cyclotomic Positivity Identity for the Companion Family at $N = 7$ ($m = 105$):**
   At $N = 7$, although $\mathcal{M}_{\text{comp}}(7) = \frac{34}{3} < 16$, we work directly in the maximal real subfield $\mathbf{Q}(\zeta_7 + \zeta_7^{-1})$. The

cyclotomic relation $\Phi_7(u) = \sum_{j=0}^{6} u^j = 0$ divides by u^3 to give $1+X_1+X_2+X_3 = 0$, so $S(u) = 1+X_1+X_2 = -X_3$. Substituting into (4.9) yields the exact factorization in $\mathbf{Z}[\zeta_7 + \zeta_7^{-1}]$:

$$\big|\mathrm{tr}(W(t))\big|^2 - 16 = M_{\mathrm{comp}}(u) - 16 = 4S(u)^2 + 2S(u) - 12 = 2\big(2S(u) - 3\big)\big(S(u) + 2\big). \tag{4.12}$$

Choose any of the 16 units $t \in (\mathbf{Z}/105\mathbf{Z})^\times$ satisfying $c_1 t_7 \equiv \pm 1 \pmod 7$ (for $c_1 = 3$, this is $t_7 \in \{2, 5\} \pmod 7$). Then $u = z^{c_1 t_7} = z^{\pm 1} = e^{\pm 2\pi i/7}$, and

$$S(z^{\pm 1}) = -(z^3 + z^{-3}) = -2\cos(6\pi/7) = 2\cos(\pi/7) > 0.$$

Since $z^3 + z^{-3}$ has minimal polynomial $X^3 + X^2 - 2X - 1 = 0$, $S = -X_3$ satisfies $f(S) := S^3 - S^2 - 2S + 1 = 0$. Because $f(3/2) = \frac{27}{8} - \frac{9}{4} - 3 + 1 = -\frac{7}{8} < 0$, $f(2) = 8 - 4 - 4 + 1 = +1 > 0$, and $f(-1) = 1 > 0$, f has a single root in $(3/2, 2)$, and since $\cos(\pi/7) > \cos(\pi/3) = 1/2 \implies S > 1$, we have $S(z^{\pm 1}) \in (3/2, 2)$. Therefore $2S(z^{\pm 1}) - 3 > 0$ and $S(z^{\pm 1}) + 2 > 0$, proving by (4.12) that $|\mathrm{tr}(W(t))|^2 > 16$.

5. **Galois Conjugation to All Primitive Characters $t \in (\mathbf{Z}/15N\mathbf{Z})^\times$:**
By Steps 1–4, for every $N > 1$ with $\gcd(N, 30) = 1$ on both families, there exists at least one unit $t_0 \in (\mathbf{Z}/15N\mathbf{Z})^\times$ such that $|\mathrm{tr}(W(t_0))| > 4$. Let $K := \mathbf{Q}(\zeta_{15N})$ and $\widetilde{W}(t) := \delta_2(t)^{-1} W(t) \in \mathrm{SL}_4(K)$, where $\delta_2(t) \in \mu_{15N}$ satisfies $\delta_2(t)^4 = \det(W(t))$ by Lemma 4.1. Suppose for contradiction that $\widetilde{W}(t)$ had finite order $\widetilde{W}(t)^M = I_4$ ($M \geq 1$) for some $t \in (\mathbf{Z}/15N\mathbf{Z})^\times$. Applying the Galois automorphism $\sigma_{t_0 t^{-1}} \in \mathrm{Gal}(K/\mathbf{Q}) \simeq (\mathbf{Z}/15N\mathbf{Z})^\times$ entry-by-entry to the polynomial matrix identity $\widetilde{W}(t)^M = I_4$ gives $\widetilde{W}(t_0)^M = I_4$. Then all four eigenvalues of $\widetilde{W}(t_0)$ would be M-th roots of unity, forcing $|\mathrm{tr}(W(t_0))| = |\mathrm{tr}(\widetilde{W}(t_0))| \leq 1 + 1 + 1 + 1 = 4$, contradicting $|\mathrm{tr}(W(t_0))| > 4$. Thus $W(t)^{15N} = \widetilde{W}(t)^{15N} \in \Gamma_0(t) \cap \mathrm{SL}_4(\mathbf{C}) \subseteq G_0(t)$ has infinite order for every $t \in (\mathbf{Z}/15N\mathbf{Z})^\times$, so $\dim G_0(t) = \dim \mathfrak{g}_0(t) \geq 1$. By Theorem 3.2, $\mathfrak{g}_0(t) = \mathfrak{sl}_4(\mathbf{C})$ and $G_0(t) = \mathrm{SL}_4(\mathbf{C})$. □

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5. Dual $\mathcal{B}_{15}$-Basis Length-3 Separation for General $m = 15N$

5.1 Determinant Normalization and Sign-Only Ambiguity for $W_1 = g_1^2 g_0$

We now turn to the length-3 braid word $W_1 := g_1^2 g_0$ and first establish the general determinant normalization lemma valid for all odd conductors $m \geq 3$.

Lemma 5.1 (Exact Linearity in w, Fourth-Power Determinant, and Sign-Only Reduction for Odd m).
For any odd integer $m \geq 3$, any $(2, 2, 1, 1)$-symmetric branch weight tuple $\mathbf{a} \in \{1, \dots, m-1\}^6$ under $H = \{1, h\}$, and any $t \in (\mathbf{Z}/m\mathbf{Z})^\times$, let $W_0 := g_1 g_0$ and $W_1 := g_1^2 g_0$. Then: 1. $\mathrm{tr}(\rho_t(W_r)) = A_r(b_1, b_2, b_4, b_5) + B_r(b_1, b_2, b_4, b_5)\, w$ ($r \in \{0, 1\}$), *where $A_0 = b_2 + b_4$, $B_0 = b_1 + b_5$, and*

$$A_1 = (b_1 b_2^2 - b_1 b_2 + b_2) + (b_4^2 b_5 - b_4 b_5 + b_4), \qquad B_1 = (b_1^2 b_2 - b_1 b_2 + b_1) + (b_4 b_5^2 - b_4 b_5 + b_5). \tag{5.1}$$

2. *Put $D(t) := b_1 b_2 b_4 b_5$ and $e_0 := a_0 + a_\lambda + a_{-\lambda} \pmod m$. Then $\det(\rho_t(W_1)) = D(t)^2 w(t)^2 = \delta(t)^4$, where*

$$\delta(t) := \zeta_m^{t(a_1 + a_{-1} + a_\lambda + a_{-\lambda} + e_0)(m+1)/2} \in \mu_m, \qquad \tau_t(W_1) := \delta(t)^{-1} \mathrm{tr}\big(\rho_t(W_1)\big) \in \mathbf{Z}[\zeta_m]^H. \tag{5.2}$$

3. *Because m is odd, $\sqrt{-1} \notin \mathbf{Q}(\zeta_m)$ and $\mu_4 \cap \mathbf{Q}(\zeta_m) = \{\pm 1\}$. Consequently, for any $t_1, t_2 \in (\mathbf{Z}/m\mathbf{Z})^\times$ with $\tau_{t_2}(W_1) \neq 0$, any projective equivalence $A\rho_{t_1}(W_1)A^{-1} = \chi(W_1)\rho_{t_2}(W_1)$ ($\chi(W_1) \in \mathbf{C}^\times$) or invariant collision $I_{t_1}(W_1) = I_{t_2}(W_1)$ satisfies*

$$I_{t_1}(W_1) := \frac{\mathrm{tr}(\rho_{t_1}(W_1))^4}{\det(\rho_{t_1}(W_1))} = I_{t_2}(W_1) \iff \tau_{t_1}(W_1) = \pm \tau_{t_2}(W_1) \quad \text{in } \mathbf{Z}[\zeta_m]. \tag{5.3}$$

Proof. Squaring $g_1(t)$ in (2.2) gives

$$g_1(t)^2 = \begin{pmatrix} (b_1 b_2)^2 & (1 - b_1)(1 + b_1 b_2) & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & b_4(1 - b_5)(1 + b_4 b_5) & (b_4 b_5)^2 \end{pmatrix}.$$

Multiplying $W_1 = g_1(t)^2 g_0(t)$ and using $b_4(1 - b_2 b_3) = b_4 - w$ gives the four diagonal entries $(W_1)_{1,1} = (b_1 b_2^2 - b_1 b_2 + b_2) + (b_1^2 b_2 - b_1 b_2 + b_1 - 1)w$, $(W_1)_{2,2} = w$, $(W_1)_{3,3} = w$, and $(W_1)_{4,4} = (b_4^2 b_5 - b_4 b_5 + b_4) + (b_4 b_5^2 - b_4 b_5 + b_5 - 1)w$. Summing these four diagonal entries cancels $-w + w + w - w$ and yields (5.1). Since $\det g_1(t) = D(t)$, $\det g_0(t) = w(t)^2$, and $2 \cdot \frac{m+1}{2} \equiv 1 \pmod{m}$ for odd m, we have $\delta(t)^4 = D(t)^2 w(t)^2 = \det(\rho_t(W_1))$ and $I_t(W_1) = \tau_t(W_1)^4$. Thus $I_{t_1}(W_1) = I_{t_2}(W_1) \iff (\tau_{t_1}(W_1)/\tau_{t_2}(W_1))^4 = 1$, and since $\tau_{t_1}(W_1)/\tau_{t_2}(W_1) \in \mu_4 \cap \mathbf{Q}(\zeta_m) = \{\pm 1\}$ for odd m, (5.3) follows. □

5.2 Proof of Theorem B(2): Dual $\mathcal{B}_{15}$-Basis Expansion over the Field $\mathbf{Q}(\zeta_N)$

When $N = p \geq 7$ is prime, one can expand $\tau_t(W_1)$ in powers of $z = \zeta_p$ over $\mathbf{Q}(\zeta_{15})$ because the cyclotomic polynomial $\Phi_p(z) = \sum_{r=0}^{p-1} z^r$ has a uniform support size p. When N is a general odd composite integer coprime to 30, however, $\Phi_N(z)$ has a complicated structure depending on the prime factorization of N. We bypass $\Phi_N(z)$ completely by **inverting the base field**: since $\gcd(15, N) = 1$, $\mathbf{Q}(\zeta_{15})$ and $\mathbf{Q}(\zeta_N)$ are linearly disjoint over $\mathbf{Q}$, so

$$\mathbf{Q}(\zeta_{15N}) \simeq \mathbf{Q}(\zeta_{15}) \otimes_{\mathbf{Q}} \mathbf{Q}(\zeta_N) \tag{5.4}$$

is a free module of rank $\varphi(15) = 8$ over the **field** $\mathbf{Q}(\zeta_N)$, with explicit $\mathbf{Q}(\zeta_N)$-basis

$$\mathcal{B}_{15} := \{\omega^i \xi^j \,:\, i \in \{1, 2\},\ j \in \{1, 2, 3, 4\}\}. \tag{5.5}$$

Theorem 5.2 (Unconditional Length-3 Separation for General $m = 15N$ with $\gcd(N, 30) = 1$).
Let $N > 1$ be any integer with $\gcd(N, 30) = 1$ and $m = 15N$. Then for every unit $t \in (\mathbf{Z}/15N\mathbf{Z})^\times$ on both the primary CRT family (1.7) and the companion CRT family (1.8):

$$I_t(g_1^2 g_0) \in \{I_1(g_1^2 g_0),\ I_{-1}(g_1^2 g_0)\} \iff \tau_t(g_1^2 g_0) = \pm \tau_{\pm 1}(g_1^2 g_0) \iff t \in \pm H. \tag{5.6}$$

Proof. Let $s \in \{\pm 1\}$ and suppose that $\tau_t(W_1) = \varepsilon \tau_s(W_1)$ with $\varepsilon \in \{\pm 1\}$ by Lemma 5.1(3). By the CRT coordinate convention (1.9a), applying the factorwise Galois automorphism $\sigma_{e_3}^{-1} \otimes \sigma_{e_5}^{-1} \otimes \sigma_{e_N}^{-1} \in \mathrm{Gal}(\mathbf{Q}(\zeta_{15N})/\mathbf{Q})$ (which sends the CRT-compatible primitive roots $\omega^{e_3} \mapsto \omega$, $\xi^{e_5} \mapsto \xi$, $z^{e_N} \mapsto z$ and fixes $\varepsilon \in \{\pm 1\} \subset \mathbf{Q}^\times$) replaces the factored coordinates $(t_3, t_5, t_N) \equiv (e_3 t, e_5 t, e_N t)$ by the ordinary CRT residues $(\bar{t}_3, \bar{t}_5, \bar{t}_N) = (t \bmod 3, t \bmod 5, t \bmod N)$, under which $s \in \{\pm 1\}$ has coordinates $(s, s, s) \pmod{(3, 5, N)}$; for notational simplicity we write (t_3, t_5, t_N) for $(\bar{t}_3, \bar{t}_5, \bar{t}_N)$ throughout this proof. Put $Z_k(x) := z^{kx} + z^{-kx} \in \mathbf{Q}(\zeta_N)$ for $k \geq 1$. Since $\gcd(t_N, N) = 1$, z^{t_N} is a primitive N-th root of unity in the field $\mathbf{Q}(\zeta_N)$.

Case 1: Primary CRT Family (1.7) On (1.7), $t \in \pm H \iff (t_3, t_5) \in \{(+1, +1), (-1, -1)\} \pmod{(3, 5)}$ and $t_N \equiv \pm 1 \pmod{N}$. Substituting $b_1 = \xi^{2t_5} z^{-t_N}$, $b_2 = \xi^{2t_5} z^{-2t_N}$, $b_4 = \xi^{2t_5} z^{2t_N}$, $b_5 = \xi^{2t_5} z^{t_N}$, $w = \omega^{t_3}$, and $\delta(t) = \omega^{-t_3} \xi^{-t_5}$ into (5.1)–(5.2) gives

$$\tau_t(W_1) = \omega^{2t_3} \xi^{3t_5} Z_1(t_N) + \omega^{t_3} \xi^{3t_5} Z_2(t_N) + Z_3(t_N) + \omega^{2t_3} \xi^{2t_5} Z_4(t_N) + \omega^{t_3} \xi^{2t_5} Z_5(t_N). \tag{5.7}$$

Using the identity $1 = (-\omega - \omega^2)(-\xi - \xi^2 - \xi^3 - \xi^4) = \sum_{i=1}^{2} \sum_{j=1}^{4} \omega^i \xi^j$, the unique $\mathcal{B}_{15}$-expansion of $\tau_t(W_1)$ over $\mathbf{Q}(\zeta_N)$ has: - coefficient $Z_3(t_N)$ at the 4 basis elements $\omega^i \xi^j$ with $j \in \{t_5, 4t_5\} \pmod 5$; - coefficients $Z_3(t_N) + Z_1(t_N)$, $Z_3(t_N) + Z_2(t_N)$, $Z_3(t_N) + Z_4(t_N)$, and $Z_3(t_N) + Z_5(t_N)$ at $(i, j) \equiv (2t_3, 3t_5), (t_3, 3t_5), (2t_3, 2t_5), (t_3, 2t_5) \pmod{(3, 5)}$, respectively.

At $s \in \{\pm 1\}$, we have $Z_k(s) = Z_k(1)$, and the 4 basis elements with $j \in \{1, 4\}$ carry coefficient $Z_3(1)$, while the 4 basis elements with $j \in \{2, 3\}$ carry $Z_3(1) + Z_k(1)$ ($k \in \{1, 2, 4, 5\}$). We proceed in three steps:

1. **Step 1 (Elimination of $t_5 \equiv \pm 2 \pmod 5$):**
   If $t_5 \equiv \pm 2 \pmod 5$, then $\{2t_5, 3t_5\} \equiv \{1, 4\} \pmod 5$. Matching the $\mathcal{B}_{15}$-coefficients at $j \in \{1, 4\}$ in $\tau_t(W_1) = \varepsilon \tau_s(W_1)$ over $\mathbf{Q}(\zeta_N)$ forces

   $$Z_3(t_N) + Z_k(t_N) = \varepsilon Z_3(1) \qquad \text{for all } k \in \{1, 2, 4, 5\}.$$

   In particular, $Z_1(t_N) = Z_2(t_N)$, which factors in the field $\mathbf{Q}(\zeta_N)$ as

   $$0 = Z_2(t_N) - Z_1(t_N) = z^{-2t_N}\big(z^{t_N} - 1\big)\big(z^{3t_N} - 1\big).$$

Since $\mathbf{Q}(\zeta_N)$ is a field, either $z^{t_N} = 1$ or $z^{3t_N} = 1$, contradicting $\gcd(3t_N, N) = 1$ and $N > 1$. Thus $t_5 \equiv \pm 1 \pmod 5$, and matching the $j \in \{1, 4\}$ components gives

$$Z_3(t_N) = \varepsilon Z_3(1). \tag{5.8}$$

2. **Step 2 (Elimination of $(t_3, t_5) \not\equiv (s, s) \pmod{(3,5)}$ via Chebyshev and Laurent Factorizations):** Subtracting $Z_3(t_N) = \varepsilon Z_3(1)$ from the 4 coefficients at $j \in \{2, 3\} \pmod 5$ equates the quadruple $\big(Z_1(t_N), Z_2(t_N), Z_4(t_N), Z_5(t_N)\big)$ to a permutation of $\varepsilon\big(Z_1(1), Z_2(1), Z_4(1), Z_5(1)\big)$ determined by $(t_3, t_5) \in \{(\pm s, \pm s)\} \pmod{(3,5)}$. Using the Chebyshev identities $Z_{2m} = Z_m^2 - 2$ and $Z_{3m} = Z_m^3 - 3Z_m$ in $\mathbf{Q}(\zeta_N)$, we rule out the three non-diagonal sign pairs:
   - **If $(t_3, t_5) \equiv (-s, s) \pmod{(3,5)}$:** Then $Z_1(t_N) = \varepsilon Z_2(1)$ and $Z_2(t_N) = \varepsilon Z_1(1)$, so
   $$\varepsilon Z_1(1) = Z_2(t_N) = Z_1(t_N)^2 - 2 = \varepsilon^2 Z_2(1)^2 - 2 = Z_4(1),$$
   which factors in $\mathbf{Q}(\zeta_N)$ as $0 = Z_4(1) - \varepsilon Z_1(1) = z^{-4}(z^3 - \varepsilon)(z^5 - \varepsilon)$. Thus $z^6 = 1$ or $z^{10} = 1$, impossible since $\gcd(N, 30) = 1$ and $N > 1$.
   - **If $(t_3, t_5) \equiv (s, -s) \pmod{(3,5)}$:** Then $Z_1(t_N) = \varepsilon Z_4(1)$. Applying $Z_3(x) = Z_1(x)^3 - 3Z_1(x)$ and (5.8) gives
   $$\varepsilon Z_3(1) = Z_3(t_N) = Z_1(t_N)^3 - 3Z_1(t_N) = \varepsilon^3\big(Z_4(1)^3 - 3Z_4(1)\big) = \varepsilon Z_{12}(1).$$
   Cancelling $\varepsilon \in \{\pm 1\}$ gives $Z_{12}(1) = Z_3(1)$, which factors identically in $\mathbf{Z}[z, z^{-1}]$ as
   $$0 = Z_{12}(1) - Z_3(1) = \big(z^{12} + z^{-12}\big) - \big(z^3 + z^{-3}\big) = z^{-12}\big(z^9 - 1\big)\big(z^{15} - 1\big). \tag{5.9}$$
   Thus $z^9 = 1$ or $z^{15} = 1$, forcing $\gcd(N, 9) > 1$ or $\gcd(N, 15) > 1$, which contradicts $\gcd(N, 30) = 1$ and $N > 1$.
   - **If $(t_3, t_5) \equiv (-s, -s) \pmod{(3,5)}$:** Then $Z_1(t_N) = \varepsilon Z_5(1)$, so
   $$\varepsilon Z_3(1) = Z_3(t_N) = Z_1(t_N)^3 - 3Z_1(t_N) = \varepsilon\big(Z_5(1)^3 - 3Z_5(1)\big) = \varepsilon Z_{15}(1).$$
   Cancelling ε gives $Z_{15}(1) = Z_3(1)$, which factors identically in $\mathbf{Z}[z, z^{-1}]$ as
   $$0 = Z_{15}(1) - Z_3(1) = \big(z^{15} + z^{-15}\big) - \big(z^3 + z^{-3}\big) = z^{-15}\big(z^{12} - 1\big)\big(z^{18} - 1\big). \tag{5.10}$$
   Since $12 = 2^2 \cdot 3$ and $18 = 2 \cdot 3^2$, this forces $\gcd(N, 6) > 1$, again contradicting $\gcd(N, 30) = 1$ and $N > 1$.
3. **Step 3 (Conclusion $(t_3, t_5) \equiv (s, s)$, $\varepsilon = +1$, and $t_N \equiv \pm 1 \pmod N$):** By Step 2, we must have $(t_3, t_5) \equiv (s, s) \pmod{(3,5)}$, which gives $Z_1(t_N) = \varepsilon Z_1(1)$ and $Z_2(t_N) = \varepsilon Z_2(1)$. Using $Z_2(x) = Z_1(x)^2 - 2$:
   $$\varepsilon Z_2(1) = Z_2(t_N) = Z_1(t_N)^2 - 2 = \varepsilon^2 Z_1(1)^2 - 2 = Z_2(1).$$
   Since $N > 1$ is odd, $-1 \notin \mu_N$, so $z^4 \neq -1$ and $Z_2(1) = z^{-2}(z^4 + 1) \neq 0$. Dividing by $Z_2(1)$ gives $\varepsilon = +1$, and hence
   $$0 = Z_1(t_N) - Z_1(1) = z^{-t_N}\big(z^{t_N - 1} - 1\big)\big(z^{t_N + 1} - 1\big),$$
   which forces $t_N \equiv \pm 1 \pmod N$. Therefore $(t_3, t_5, t_N) \equiv (s, s, \pm 1) \pmod{(3, 5, N)}$, i.e., $t \in \pm H$. *(Remark: Alternatively, squaring (5.8) $Z_3(t_N) = \varepsilon Z_3(1)$ directly via $Z_6(x) = Z_3(x)^2 - 2$ gives $Z_6(t_N) - Z_6(1) = z^{-6t_N}(z^{6(t_N-1)} - 1)(z^{6(t_N+1)} - 1) = 0$, which since $\gcd(6, N) = 1$ immediately implies $t_N \equiv \pm 1 \pmod N$.)*

Case 2: Companion CRT Family (1.8) On (1.8), $t \in \pm H \iff t_3 \equiv s \pmod 3$, $t_5 \equiv \pm 1 \pmod 5$, and $t_N \equiv s \pmod N$ for some $s \in \{\pm 1\}$. Substituting $b_1 = \xi^{-t_5} z^{c_1 t_N}$, $b_2 = \xi^{-2t_5} z^{c_1 t_N}$, $b_4 = \xi^{2t_5} z^{c_1 t_N}$, $b_5 = \xi^{t_5} z^{c_1 t_N}$, $w = \omega^{2t_3}$, and $\delta(t) = \omega^{t_3} z^{2c_1 t_N}$ into (5.1)–(5.2) yields

$$\tau_t(W_1) = \big(2\omega^{2t_3} + \omega^{t_3}\phi_1(t_5)\big)z^{c_1 t_N} + \phi_2(t_5) + \big(\omega^{2t_3}\phi_2(t_5) + \omega^{t_3}\phi_1(t_5)\big)z^{-c_1 t_N}, \tag{5.11}$$

where $\phi_1(t_5) := \xi^{t_5} + \xi^{-t_5} = \frac{-1 + \delta_5(t_5)\sqrt{5}}{2}$ and $\phi_2(t_5) := \xi^{2t_5} + \xi^{-2t_5} = \frac{-1 - \delta_5(t_5)\sqrt{5}}{2}$ with $\delta_5(t_5) = +1$ if $t_5 \equiv \pm 1 \pmod 5$ and -1 if $t_5 \equiv \pm 2 \pmod 5$. Writing $\omega^{t_3} = \frac{-1 + \delta_3(t_3)\sqrt{-3}}{2}$ ($\delta_3(t_3) \in \{\pm 1\}$), $\tau_t(W_1)$ lies in the biquadratic subfield $\mathbf{Q}(\sqrt{5}, \sqrt{-3})(\zeta_N) \subset \mathbf{Q}(\zeta_{15N})$, which has the 4-element $\mathbf{Q}(\zeta_N)$-basis $\{1, \sqrt{5}, \sqrt{-3}, \sqrt{-15}\}$:

$$\begin{aligned}\tau_t(W_1) = {} & \left(-\frac{1}{2} - \frac{3}{4}z^{c_1 t_N} + \frac{1}{2}z^{-c_1 t_N}\right) \cdot 1 + \frac{\delta_5(t_5)}{2}\left(-1 - \frac{1}{2}z^{c_1 t_N}\right)\sqrt{5} \\ & + \frac{\delta_3(t_3)}{2}\left(-\frac{5}{2}z^{c_1 t_N}\right)\sqrt{-3} + \frac{\delta_3(t_3)\delta_5(t_5)}{4}\left(z^{c_1 t_N} + 2z^{-c_1 t_N}\right)\sqrt{-15}.\end{aligned} \tag{5.12}$$

Equating the coefficients of $\{1, \sqrt{5}, \sqrt{-3}, \sqrt{-15}\}$ over $\mathbf{Q}(\zeta_N)$ in $\tau_t(W_1) = \varepsilon\tau_s(W_1)$ ($s \in \{\pm 1\}$, where $\delta_5(s) = +1$ and $\delta_3(s) = s$): 1. The $\sqrt{5}$-coefficient gives

$$\delta_5(t_5)\Big(1 + \frac{1}{2}z^{c_1 t_N}\Big) = \varepsilon\Big(1 + \frac{1}{2}z^{c_1 s}\Big) \iff z^{c_1 t_N} - \big(\varepsilon\delta_5(t_5)\big)z^{c_1 s} = 2\big(\varepsilon\delta_5(t_5) - 1\big).$$

By the triangle inequality on the unit circle, $|z^{c_1 t_N} \mp z^{c_1 s}| \le 2 < 4$, so $\varepsilon\delta_5(t_5) = -1$ is impossible. Hence $\varepsilon\delta_5(t_5) = +1$ and $z^{c_1 t_N} = z^{c_1 s}$, which forces $t_N \equiv s \pmod N$ since $\gcd(c_1, N) = 1$. 2. Substituting $z^{c_1 t_N} = z^{c_1 s}$ into the $\sqrt{-3}$-coefficient $-\frac{5}{4}\delta_3(t_3)z^{c_1 t_N} = -\frac{5}{4}\varepsilon\delta_3(s)z^{c_1 s}$ gives $\delta_3(t_3) = \varepsilon\delta_3(s)$. 3. Substituting $z^{c_1 t_N} = z^{c_1 s}$ into the 1-coefficient gives $(1-\varepsilon)\big(-\frac{1}{2} - \frac{3}{4}z^{c_1 s} + \frac{1}{2}z^{-c_1 s}\big) = 0$. Since $3x^2 + 2x - 2 = 0$ has real roots $\frac{-1\pm\sqrt{7}}{3}$ of absolute value $\neq 1$, the second factor is non-zero; thus $\varepsilon = +1$, whence $\delta_5(t_5) = +1$ ($t_5 \equiv \pm 1 \pmod 5$), $\delta_3(t_3) = \delta_3(s)$ ($t_3 \equiv s \pmod 3$), and $t_N \equiv s \pmod N$, proving $t \in \pm H$. □

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6. Uniform Length-3 Separation of Local-Collision Multipliers for Arbitrary Odd $m = a \cdot m'$

6.1 Puncture-Spectrum Indistinguishability and Length-2 Trace Collisions

We now prove the second half of Theorem A(1) together with Theorem C, showing that for any odd composite conductor $m = a \cdot m'$ ($\gcd(a, m') = 1$), the symmetric local-collision multiplier $\upsilon \equiv (-1, +1) \pmod{(a, m')}$ is indistinguishable from ± 1 at all four punctures of S and on the length-2 braid words $g_1 g_0$ and $g_1 g_{-1}$, yet is separated uniformly in m by the length-3 word $W_1 = g_1^2 g_0$.

Theorem 6.1 (Uniform-in-m Separation of Symmetric Local-Collision Multipliers for Arbitrary Odd $m = a \cdot m'$). *Let $m = a \cdot m'$ be any odd integer factored into coprime odd integers $a \ge 3$ and $m' \ge 3$ with $\gcd(a, m') = 1$. Let $\mathbf{a} = (a_0, a_\infty, a_1, a_{-1}, a_\lambda, a_{-\lambda}) \in \{1, \dots, m-1\}^6$ be a $(2, 2, 1, 1)$-symmetric tuple under $H = \{1, h\}$ with $\sum a_i \equiv 0 \pmod m$, $\gcd(\mathbf{a}, m) = 1$, $a_\lambda \not\equiv a_1, a_{-1} \pmod m$, and*

$$a_1 + a_{-1} \equiv a_\lambda + a_{-\lambda} \not\equiv 0 \pmod m \qquad (\text{so } b_1 b_5 = b_2 b_4 =: \eta_+ \in \mu_{m'} \setminus \{1\} \iff \upsilon \notin \pm H). \tag{6.1}$$

Suppose $\upsilon \in (\mathbf{Z}/m\mathbf{Z})^\times \setminus (\pm H)$ is a local-monodromy collision multiplier with $\upsilon \equiv (-1, +1) \pmod{(a, m')}$ fixing $(a_1, a_{-1}, a_\lambda, a_{-\lambda}) \pmod m$ and inverting $w = \zeta_m^{e_0}$ ($w(\upsilon) = w^{-1} \neq w$, swapping $a_0 \leftrightarrow a_\infty \pmod m$, where $e_0 := a_0 + a_\lambda + a_{-\lambda}$ and $m' \mid e_0$). Then $a \mid a_1, a_{-1}, a_\lambda, a_{-\lambda}$ (so $b_1, b_2, b_4, b_5, \eta_+ \in \mu_{m'}$), $w = \omega \in \mu_a \setminus \{1\}$ has odd multiplicative order $d_a \mid a$ ($d_a \ge 3$), and: 1. **(Identical Puncture Spectra and Length-2 Trace Collisions):** *At all four punctures $p \in \{1, -1, 0, \infty\}$ of S, the operators $\rho_\upsilon(g_p)$ and $\rho_1(g_p)$ have identical projective spectra and Jordan canonical forms. Moreover, $\mathrm{tr}(\rho_t(g_1 g_{-1}))$ is independent of b_3 and equals*

$$\mathrm{tr}\big(\rho_t(g_1 g_{-1})\big) = b_1 b_2 b_4 + b_1 b_2 b_5 + b_1 b_4 b_5 + b_2 b_4 b_5 - (b_1 + b_5)(b_2 + b_4) + (b_1 + b_2 + b_4 + b_5), \tag{6.2}$$

and the length-2 braid words $g_1 g_0$ and $g_1 g_{-1}$ satisfy identically

$$I_\upsilon(g_1 g_0) = I_{-1}(g_1 g_0), \qquad I_\upsilon(g_1 g_{-1}) = I_1(g_1 g_{-1}) = I_{-1}(g_1 g_{-1}). \tag{6.3}$$

2. **(Uniform-in-m Length-3 Separation by $W_1 = g_1^2 g_0$):** *For every such odd $m = a \cdot m'$ and $\mathbf{a}$, the length-3 braid word $W_1 = g_1^2 g_0$ satisfies*

$$I_\upsilon(g_1^2 g_0) \notin \big\{I_1(g_1^2 g_0),\ I_{-1}(g_1^2 g_0)\big\}. \tag{6.4}$$

Proof. 1. **(Arithmetic Quantifiers and Part 1):** Since $\upsilon \equiv -1 \pmod a$ and a is odd, $\gcd(\upsilon - 1, a) = \gcd(-2, a) = 1$. Hence the congruences $(\upsilon - 1)x \equiv 0 \pmod m$ for $x \in \{a_1, a_{-1}, a_\lambda, a_{-\lambda}\}$ force $a \mid a_1, a_{-1}, a_\lambda, a_{-\lambda}$ and $b_1, b_2, b_4, b_5, \eta_+ \in \mu_{m'}$. Since $m' \mid e_0$ and $w(\upsilon) = w^{-1} \neq w$, the element $w = \zeta_m^{e_0} =: \omega \in \mu_a \setminus \{1\}$ has odd multiplicative order $d_a \ge 3$ dividing a. - At the punctures $p = 1$ and $p = -1$, the spectra $\mathrm{Spec}(g_1(t)) = \{b_1 b_2, b_4 b_5, 1, 1\}$ and $\mathrm{Spec}(g_{-1}(t)) = \{b_1 b_4, b_2 b_5, 1, 1\}$ depend only on $(b_1, b_2, b_4, b_5) \in \mu_{m'}^4$ and are therefore literally identical at $t = \upsilon$ and $t = 1$. - At $p = 0$, $\mathrm{Spec}(g_0(1)) = \{1, 1, \omega, \omega\}$ and $\mathrm{Spec}(g_0(\upsilon)) = \{1, 1, \omega^{-1}, \omega^{-1}\}$; scaling $g_0(1)$ by $\omega^{-1} \in \mathbf{C}^\times$ yields $\{\omega^{-1}, \omega^{-1}, 1, 1\} = \mathrm{Spec}(g_0(\upsilon))$. - At $p = \infty$, since $b_1 b_5 = b_2 b_4 = \eta_+$ by (6.1) and $w = b_2 b_3 b_4$, we have $b_\infty(t) = (b_1 b_5 w(t))^{-1} = \eta_+^{-1} w(t)^{-1}$. By Lemma 2.1(2), $g_\infty(t)$ is a semisimple rank-$(2, 2)$ bisection with $\mathrm{Spec}(g_\infty(1)) = \{\eta_+^{-1}, \eta_+^{-1}, \eta_+^{-1}\omega^{-1}, \eta_+^{-1}\omega^{-1}\}$ and $\mathrm{Spec}(g_\infty(\upsilon)) = \{\eta_+^{-1}, \eta_+^{-1}, \eta_+^{-1}\omega, \eta_+^{-1}\omega\}$; scaling $g_\infty(1)$ by $\omega \in \mathbf{C}^\times$ yields $\{\eta_+^{-1}\omega, \eta_+^{-1}\omega, \eta_+^{-1}, \eta_+^{-1}\} = \mathrm{Spec}(g_\infty(\upsilon))$. - For

the length-2 word $W_0 = g_1g_0$, Lemma 5.1(1) gives $\mathrm{tr}(g_1(t)g_0(t)) = (b_2+b_4)+(b_1+b_5)w(t)$. By (6.1), $\overline{b_2+b_4} = b_2^{-1}+b_4^{-1} = \eta_+^{-1}(b_2+b_4)$ and $\overline{b_1+b_5} = \eta_+^{-1}(b_1+b_5)$, so

$$\mathrm{tr}\big(g_1(\upsilon)g_0(\upsilon)\big) = (b_2+b_4)+(b_1+b_5)\overline{\omega} = \eta_+\big(\overline{b_2+b_4}+(\overline{b_1+b_5})\overline{\omega}\big) = \eta_+\,\mathrm{tr}\big(g_1(-1)g_0(-1)\big),$$

while $\det(g_1(\upsilon)g_0(\upsilon)) = \eta_+^2\overline{\omega}^2 = \eta_+^4(\eta_+^{-2}\overline{\omega}^2) = \eta_+^4\det(g_1(-1)g_0(-1))$, proving $I_\upsilon(g_1g_0) = I_{-1}(g_1g_0)$. - In (2.3), $D_1(t)$ has inverse $D_1(t)^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & -(b_3b_4)^{-1} & (b_3b_4)^{-1} \\ 1 & -b_4^{-1} & 0 & b_4^{-1} \\ 0 & 0 & 1 & \end{pmatrix}$, so the four diagonal entries of $g_1(t)g_{-1}(t) = g_1(t)D_1(t)^{-1}g_{1,\text{swapped}}(t)D_1(t)$ are $b_1b_2(b_1b_4+1-b_1)+(1-b_1)$, 1, $1-b_2(1-b_5)$, and $b_4(1-b_5)(b_5-1)+b_4b_5(b_2+b_5-b_2b_5)$, which are each independent of b_3 and sum directly to (6.2). Because (6.2) depends only on $(b_1,b_2,b_4,b_5) \in \mu_{m'}^4$ and $\upsilon \equiv +1 \pmod{m'}$ (with $\overline{\mathrm{tr}(g_1(1)g_{-1}(1))} = \eta_+^{-2}\,\mathrm{tr}(g_1(1)g_{-1}(1))$ under $b_1b_5 = b_2b_4 = \eta_+$), we have $\mathrm{tr}(g_1(\upsilon)g_{-1}(\upsilon)) = \mathrm{tr}(g_1(1)g_{-1}(1))$ and $\det(g_1(\upsilon)g_{-1}(\upsilon)) = \eta_+^4 = \det(g_1(1)g_{-1}(1))$, proving $I_\upsilon(g_1g_{-1}) = I_1(g_1g_{-1}) = I_{-1}(g_1g_{-1})$.

2. **($L(\omega)$-Basis Expansion and Binomial Factorizations for $W_1 = g_1^2g_0$):**
Set $L := \mathbf{Q}(\zeta_{m'})$, $D := b_1b_2b_4b_5 = \eta_+^2 \in \mu_{m'}$, and $r := \frac{a+1}{2} \in \mathbf{Z}$. Since a is odd and $m = a\cdot m'$ with m' odd, $2r = a+1 \equiv 1 \pmod a$ and $2\cdot\frac{m+1}{2} = am'+1 \equiv 1 \pmod a$, so $r \equiv \frac{m+1}{2} \pmod a$, $1-r \equiv r \pmod a$, and $\omega^{2r} = \omega$. Because $a \mid (a_1+a_{-1}+a_\lambda+a_{-\lambda})$ and $m' \mid e_0$, the normalization factor $\delta(t)$ in (5.2) evaluates at $t = 1, \upsilon, -1$ to

$$\delta(1) = \eta_+\omega^r, \qquad \delta(\upsilon) = \eta_+\omega^{-r}, \qquad \delta(-1) = \eta_+^{-1}\omega^{-r}.$$

Dividing $\mathrm{tr}(\rho_t(W_1)) = A_1(t)+B_1(t)w(t)$ by $\delta(t)$ and using $\omega\cdot\omega^{-r} = \omega^{1-r} = \omega^r$ yields in $L(\omega)$:

$$\tau_1(W_1) = \eta_+^{-1}\big(A_1\omega^{-r}+B_1\omega^r\big), \qquad \tau_\upsilon(W_1) = \eta_+^{-1}\big(B_1\omega^{-r}+A_1\omega^r\big), \qquad \tau_{-1}(W_1) = \eta_+\big(\overline{B_1}\omega^{-r}+\overline{A_1}\omega^r\big). \tag{6.5}$$

Since $\gcd(d_a,m') \mid \gcd(a,m') = 1$, the cyclotomic polynomial $\Phi_{d_a}(X)$ is irreducible over $L = \mathbf{Q}(\zeta_{m'})$, so $[L(\omega) : L] = \varphi(d_a) \geq \varphi(3) = 2$. Because $\omega^{2r} = \omega$ has degree ≥ 2 over L, the pair $\{\omega^{-r},\omega^r\} = \omega^{-r}\{1,\omega\}$ is **L-linearly independent** in $L(\omega)$.

Substituting $b_1 = \eta_+/b_5$ and $b_2 = \eta_+/b_4$ into (5.1) with $D = \eta_+^2$ gives four exact rational-function factorizations in L:

$$(A_1-B_1)+D(\overline{A_1}-\overline{B_1}) = \frac{(b_4-b_5)(\eta_++1)(b_4-b_1)(b_4b_5+\eta_+)^2}{b_4^2b_5\eta_+}, \tag{6.6}$$

$$(A_1+B_1)-D(\overline{A_1}+\overline{B_1}) = \frac{(\eta_+-1)(b_4+b_5)(b_4b_5+\eta_+)(b_4-b_1)^2}{b_4^2\eta_+}, \tag{6.7}$$

$$A_1-D\overline{A_1} = \frac{(\eta_+-1)(b_5-b_2)(b_4^2-b_1b_2)}{\eta_+}, \qquad B_1-D\overline{B_1} = \frac{(\eta_+-1)(b_4-b_1)(b_5^2-b_1b_2)}{\eta_+}. \tag{6.8}$$

(Identity (6.7) follows immediately by adding the two equations of (6.8) and substituting $b_5-b_2 = \frac{b_5}{b_4}(b_4-b_1)$ and $b_1b_2 = \frac{\eta_+^2}{b_4b_5}$.) We verify that every factor on the right-hand sides of (6.6) and (6.7) is non-zero in L:

- First, the equivalence $\eta_+ \neq 1 \iff \upsilon \notin \pm H$ in (6.1) holds unconditionally: if $\eta_+ = 1$, then $a_{-1} \equiv -a_1 \pmod m$ and $a_{-\lambda} \equiv -a_\lambda \pmod m$, which by $\sum a_i \equiv 0 \pmod m$ implies $a_\infty \equiv -a_0 \pmod m$, so $(-h)\mathbf{a} \equiv (a_\infty,a_0,a_1,a_{-1},a_\lambda,a_{-\lambda}) \equiv \upsilon\mathbf{a} \pmod m$ and $\gcd(\mathbf{a},m) = 1$ forces $\upsilon \equiv -h \in \pm H$; conversely, if $\eta_+ \neq 1$ ($a_1+a_{-1} \not\equiv 0 \pmod m$), then $\upsilon \equiv (-1,+1) \pmod{(a,m')}$ cannot be $\pm1 \pmod m$ (as $a,m' \geq 3$ are odd), cannot be $+h \pmod m$ (else $h \equiv -1 \pmod a$ in (1.2) forces $a \mid a_0, a_\infty$ and $a \mid \gcd(\mathbf{a},m) = 1$), and cannot be $-h \pmod m$ (else $h \equiv -1 \pmod{m'}$ gives $a_1+a_{-1} \equiv (1+h)a_1 \equiv 0 \pmod{m'}$ and hence $a_1+a_{-1} \equiv 0 \pmod m$, contradicting $\eta_+ \neq 1$).
- Second, since m' is odd, $-1 \notin \mu_{m'}$, so $\eta_+ + 1 \neq 0$, $b_4b_5+\eta_+ \neq 0$, and $b_4+b_5 \neq 0$.
- Third, the hypothesis $a_\lambda \not\equiv a_1, a_{-1} \pmod m$ implies $b_4 \neq b_5$, $b_4 \neq b_1$, and $b_5 \neq b_2$.

Consequently, the right-hand sides of both (6.6) and (6.7) are non-zero in L. Using the L-linear independence of $\{\omega^{-r},\omega^r\}$ in (6.5), we now rule out all four sign cases $\tau_\upsilon(W_1) = \pm\tau_{\pm1}(W_1)$:

- **If $\tau_\upsilon(W_1) = +\tau_1(W_1)$:** Equating coefficients of ω^{-r} gives $B_1 = A_1$ ($A_1-B_1 = 0$), which makes the left-hand side of (6.6) zero, a contradiction.

- **If $\tau_v(W_1) = -\tau_1(W_1)$:** Equating coefficients of ω^{-r} gives $B_1 = -A_1$ ($A_1+B_1 = 0$), which makes the left-hand side of (6.7) zero, a contradiction.
- **If $\tau_v(W_1) = -\tau_{-1}(W_1)$:** Equating coefficients of ω^{-r} and ω^r gives $B_1 = -D\overline{B_1}$ and $A_1 = -D\overline{A_1}$, so the left-hand side of (6.6) equals $(A_1 + D\overline{A_1}) - (B_1 + D\overline{B_1}) = 0$, a contradiction.
- **If $\tau_v(W_1) = +\tau_{-1}(W_1)$:** Equating coefficients of ω^{-r} and ω^r gives $B_1 - D\overline{B_1} = 0$ and $A_1 - D\overline{A_1} = 0$, which by (6.8) forces $b_4^2 = b_1 b_2 = b_5^2$, contradicting $b_4^2 - b_5^2 = (b_4 - b_5)(b_4 + b_5) \neq 0$.

By Lemma 5.1(3), $I_v(g_1^2 g_0) \notin \{I_1(g_1^2 g_0), I_{-1}(g_1^2 g_0)\}$. □

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7. Three-Factor Generalized Primary CRT Family $m = a \cdot b \cdot N$ and Method-Boundary Examples

We now prove Theorem D, extending both unconditional Zariski density and length-3 separation to the three-factor generalized primary CRT family (1.9) for all pairwise coprime odd integers $a \geq 3, b \geq 3, N \geq 7$ with $\gcd(N, 30) = 1$, and establishing the four-fold method-boundary examples (b.1)–(b.4) (where (b.1) at $(9, 5, 7)$ and (b.2) at $(3, 7, 11)$ evaluate companion-template branch weight tuples at admissible coprime parameter triples outside the generalized primary family (1.9) to show where complete-residue-block Kubert balance and companion (t_a, t_b)-independence break down, (b.3) at $(5, 3, 7)$ lies directly in the generalized primary family (1.9) of Theorem D(1) and shows where the global-average shortcut Mean > 16 fails, and (b.4) at $(5, 7, 3)$ violates $\gcd(N, 30) = 1$ on the primary template and shows that the 7-eigenvalue adjoint spectrum collapses).

7.1 Unconditional Zariski Density for $m = a \cdot b \cdot N$ via Rational Interval Bounds

Let $a \geq 3$, $b \geq 3$, and $N \geq 7$ be pairwise coprime odd integers with $\gcd(N, 30) = 1$, and let $m = a \cdot b \cdot N$ carry the generalized primary CRT tuple $\mathbf{a}_{\text{gen}}$ of (1.9). Following the CRT character coordinate convention (1.9a), for each unit $t \in (\mathbf{Z}/m\mathbf{Z})^\times$ with factored CRT coordinates $t_q \equiv t(m/q)^{-1} \pmod q$ ($q \in \{a, b, N\}$, so $\zeta_m^{t a_j} = \zeta_a^{t_a a_{j,a}} \zeta_b^{t_b a_{j,b}} z^{t_N a_{j,N}}$), write $\omega_a := \zeta_a^{t_a} \in \mu_a \setminus \{1\}$, $\eta_b := \zeta_b^{c_b t_b} \in \mu_b \setminus \{1\}$, and $u := z^{t_N} \in \mu_N$ (of exact primitive order N). Then (2.1) evaluates to

$$b_1 = \eta_b u^{-1}, \qquad b_2 = \eta_b u^{-2}, \qquad b_3 = \omega_a \eta_b^{-2}, \qquad b_4 = \eta_b u^2, \qquad b_5 = \eta_b u, \qquad w = b_2 b_3 b_4 = \omega_a. \tag{7.1}$$

Theorem 7.1 (Unconditional Zariski Density for the Generalized Primary CRT Family $m = a \cdot b \cdot N$). *Let $a \geq 3, b \geq 3, N \geq 7$ be pairwise coprime odd integers with $\gcd(N, 30) = 1$, and set $m = a \cdot b \cdot N$. Let $\alpha_a := \frac{c_a(1)}{\varphi(a)} = \frac{\mu(a)}{\varphi(a)}$ and $\beta_r := \frac{c_b(r)}{\varphi(b)}$ ($0 \leq r \leq 4$, $\beta_0 = 1$). Then:* 1. *Averaging $|\operatorname{tr}(W(t))|^2$ over $(t_a, t_b) \in (\mathbf{Z}/a\mathbf{Z})^\times \times (\mathbf{Z}/b\mathbf{Z})^\times$ gives the exact Chebyshev–Laurent polynomial in $X_k = u^k + u^{-k}$:*

$$M_{a,b}(u) := \frac{1}{\varphi(ab)} \sum_{t_a, t_b} \left|\operatorname{tr}(W(t))\right|^2 = (X_2^2 + X_4^2 + 2\alpha_a X_2 X_4) + 2\alpha_a (X_2 + X_4) \sum_{j=0}^{4} p_j \beta_{|j-2|} + \sum_{j,k=0}^{4} p_j p_k \beta_{|j-k|}, \tag{7.2}$$

where $(p_0, p_1, p_2, p_3, p_4) = 2(1, -(X_1 + X_2), 2 + X_1 X_2, -(X_1 + X_2), 1)$. 2. *For every triple (a, b, N) as above, there exists a unit $t^\star \in (\mathbf{Z}/m\mathbf{Z})^\times$ such that $|\operatorname{tr}(W(t^\star))|^2 > 16$. Consequently, for every $t \in (\mathbf{Z}/m\mathbf{Z})^\times$, $W(t)^m \in G_0(t)$ has infinite order and $\langle g_1(t), g_0(t) \rangle$ is Zariski-dense in $\mathrm{SU}(J(t))$ up to μ_m scalars.*

Proof. 1. Since a is odd, $\gcd(2, a) = 1$ and $c_a(1) = c_a(2) = \mu(a)$. Averaging $|\operatorname{tr}(W(t))|^2 = |C_0 + C_1 \omega_a + C_2 \omega_a^2|^2$ over $t_a \in (\mathbf{Z}/a\mathbf{Z})^\times$ gives

$$\frac{1}{\varphi(a)} \sum_{t_a \in (\mathbf{Z}/a\mathbf{Z})^\times} \left|\operatorname{tr}(W(t))\right|^2 = |C_0|^2 + |C_1|^2 + |C_2|^2 + 2\alpha_a \operatorname{Re}\big(C_0 \overline{C_1} + C_1 \overline{C_2} + C_2 \overline{C_0}\big), \tag{7.3}$$

and averaging (7.3) over $t_b \in (\mathbf{Z}/b\mathbf{Z})^\times$ via $\frac{1}{\varphi(b)} \sum_{t_b} \eta_b^{j-k} = \beta_{|j-k|}$ yields (7.2).

2. To exhibit $t^\star \in (\mathbf{Z}/m\mathbf{Z})^\times$ with $|\operatorname{tr}(W(t^\star))|^2 > 16$, we partition the admissible triples (a, b, N) into three complementary regimes:

- **Case 1 ($b \geq 5, N \geq 7$, and either $a = 3$ or $a \geq 5$):**
Specialize to factored coordinates $t_N^\star = 1$ ($u = e^{2\pi i/N}$) and $c_b t_b^\star \equiv \frac{b-1}{2} \pmod b$ ($\eta_b = -e^{-\pi i/b}$). Then $\eta_b + \eta_b^{-1} = -2\cos(\pi/b)$, so

$$\prod_{j\in\{1,2,4,5\}} |b_j(t^\star) - 1| = |\eta_b^2 - X_1\eta_b + 1|\,|\eta_b^2 - X_2\eta_b + 1| = \big(2\cos(\pi/b) + X_1\big)\big(2\cos(\pi/b) + X_2\big). \tag{7.4}$$

Since $\cos(\pi/b)$ is strictly increasing for $b \geq 5$ and $X_k = 2\cos(2\pi k/N)$ is strictly increasing for $N \geq 7$ ($k = 1, 2$), the product (7.4) achieves its minimum over all $b \geq 5, N \geq 7$ at $(b, N) = (5, 7)$, where $2\cos(\pi/5) = \frac{1+\sqrt{5}}{2}$. At $N = 7$, the minimal polynomial $f(X) := X^3 + X^2 - 2X - 1$ of $X_k = 2\cos(2\pi k/7)$ ($k = 1, 2, 3$) has derivative $f'(X) = 3X^2 + 2X - 2$, which satisfies $f'(X) \geq 3 > 0$ on $[1, 2]$ (f strictly increasing) and $f'(X) \leq -\frac{5}{4} < 0$ on $[-\frac{1}{2}, 0]$ (f strictly decreasing). Evaluating f at rational endpoints gives $f(\frac{623}{500}) = -\frac{631133}{125000000} < 0 < \frac{1}{64} = f(\frac{5}{4})$ on $[1, 2]$ and $f(-\frac{223}{500}) = \frac{274933}{125000000} > 0 > -\frac{37381}{15625000} = f(-\frac{111}{250})$ on $[-\frac{1}{2}, 0]$, isolating $X_1 = 2\cos(2\pi/7) \in (\frac{623}{500}, \frac{5}{4})$ (so $X_1 > \frac{623}{500}$) and $X_2 = 2\cos(4\pi/7) \in (-\frac{223}{500}, -\frac{111}{250})$ (so $X_2 > -\frac{223}{500}$ and $-X_2 = 2\cos(3\pi/7) > \frac{111}{250}$), while $\sqrt{5} > \frac{559}{250}$ (since $559^2 = 312481 < 312500 = 5 \cdot 250^2$). Hence for all $b \geq 5$ and $N \geq 7$:

$$\prod_{j\in\{1,2,4,5\}} |b_j(t^\star) - 1| \geq \left(\frac{1 + \frac{559}{250}}{2} + \frac{623}{500}\right)\left(\frac{1 + \frac{559}{250}}{2} - \frac{223}{500}\right) = \frac{1431}{500}\cdot\frac{587}{500} = \frac{839997}{250000} = \frac{3359988}{1000000} \approx 3.3600 > \frac{1679}{500} = 3.358.$$

  - If $a = 3$, subtracting $\operatorname{tr}(W(t))$ at $t_3 = 1$ and $t_3 = 2$ gives $\max_{t_3\in\{1,2\}} |\operatorname{tr}(W(t^\star))| \geq \sqrt{3}(\prod_{j\in\{1,2,4,5\}} |b_j(t^\star) - 1| - 1) > \frac{433}{250}(\frac{1679}{500} - 1) = \frac{1021014}{250000} = 4.084056 > 4$ (since $433^2 = 187489 < 187500 = 3 \cdot 250^2$).
  - If $a \geq 5$, then $\varphi(a) \geq 4$ and $|\alpha_a| = |\mu(a)|/\varphi(a) \leq \frac{1}{4}$. By Cauchy–Schwarz, $2|\operatorname{Re}(C_0\overline{C_1} + C_1\overline{C_2} + C_2\overline{C_0})| \leq 2(|C_0|^2 + |C_1|^2 + |C_2|^2)$, so (7.3) gives

$$\frac{1}{\varphi(a)} \sum_{t_a\in(\mathbf{Z}/a\mathbf{Z})^\times} \big|\operatorname{tr}(W(t^\star))\big|^2 \geq \big(1 - 2|\alpha_a|\big)\big(|C_0|^2 + |C_1|^2 + |C_2|^2\big) \geq \frac{1}{2}|C_1|^2 = 2\prod_{j\in\{1,2,4,5\}} \big|b_j(t^\star) - 1\big|^2 > 2(3.358)^2 = 22.552 > 16$$

- **Case 2 ($a \geq 5, b = 3, N \geq 11$):**
When $b = 3$, $\eta_3 \in \mu_3 \setminus \{1\}$ satisfies $\eta_3^2 + 1 = -\eta_3$, which collapses the degree-4 polynomial $C_1(\eta_3)$ in (4.5) to

$$C_1 = 2\eta_3^2(1 + X_1)(1 + X_2). \tag{7.5}$$

Specialize to factored coordinate $t_N^\star = 1$ ($u = e^{2\pi i/N}, N \geq 11$). Since $\pi < \frac{22}{7}$, we have $\pi^2 < \frac{484}{49} < 10$. By alternating Taylor series truncation, for $N \geq 11$,

$$X_1 = 2\cos(2\pi/N) \geq 2\cos(2\pi/11) > 2\left(1 - \frac{(2\pi)^2}{2 \cdot 11^2}\right) > 2\left(1 - \frac{20}{121}\right) = \frac{202}{121} > \frac{5}{3}, \qquad X_2 = X_1^2 - 2 > \frac{25}{9} - 2 = \frac{7}{9}.$$

Therefore $|C_1| = 2(1 + X_1)(1 + X_2) > 2\big(\frac{8}{3}\big)\big(\frac{16}{9}\big) = \frac{256}{27}$, and since $a \geq 5$ has $1 - 2|\alpha_a| \geq \frac{1}{2}$,

$$\frac{1}{\varphi(a)} \sum_{t_a\in(\mathbf{Z}/a\mathbf{Z})^\times} \big|\operatorname{tr}(W(t^\star))\big|^2 \geq \frac{1}{2}|C_1|^2 > \frac{32768}{729} > 44.9 > 16.$$

- **Case 3 ($a \geq 5, b = 3, N = 7$):**
Because $a, b = 3, N = 7$ are pairwise coprime odd integers, $a = 7$ is excluded by $\gcd(a, N) = 1$ and $a = 9$ is excluded by $\gcd(a, b) = 1$. Thus the only admissible odd integers $a \geq 5$ in Case 3 are $a = 5$ and $a \geq 11$ ($\gcd(a, 21) = 1$). At $b = 3, N = 7$, choosing factored coordinate $t_7^\star = 1$ ($u = e^{2\pi i/7}$) and using $\Phi_7(u) = 0 \iff 1 + X_1 + X_2 + X_3 = 0$ gives $(1 + X_1)(1 + X_2) = 1 + X_1 + X_2 + X_1X_2 = -X_3 + X_1 + X_3 = X_1$, so by (4.5) and (7.5):

$$\eta_3^{-2}C_0 = X_4 = X_3, \qquad \eta_3^{-2}C_1 = 2(1 + X_1)(1 + X_2) = 2X_1, \qquad \eta_3^{-2}C_2 = X_2,$$

$$\eta_3^{-2}\operatorname{tr}\big(W(t^\star)\big) = X_3 + 2X_1\omega_a + X_2\omega_a^2. \tag{7.6}$$

– **Subcase 3a ($a = 5$, so $(a, b, N) = (5, 3, 7), m = 105$):** Choose factored coordinate $t_5^\star = 2$ ($\omega_5 = e^{4\pi i/5}$, where $\omega_5 + \omega_5^{-1} = \frac{-1-\sqrt{5}}{2}$ and $\omega_5^2 + \omega_5^{-2} = \frac{-1+\sqrt{5}}{2}$) together with $t_3^\star = 1, t_7^\star = 1$, which by (1.9a) with $(e_5, e_3, e_7) = (21^{-1} \bmod 5, 35^{-1} \bmod 3, 15^{-1} \bmod 7) = (1, 2, 1)$ corresponds to $t^\star \equiv 92 \pmod{105}$ having ordinary CRT residues $(\bar{t}_5^\star, \bar{t}_3^\star, \bar{t}_7^\star) = (2, 2, 1) \pmod{(5, 3, 7)}$ and factored coordinates $(t_5^\star, t_3^\star, t_7^\star) = (2, 1, 1)$. Expanding $|X_3 + 2X_1\omega_5 + X_2\omega_5^2|^2$ in $\mathbf{Q}(\sqrt{5}, \cos(2\pi/7))$ using $X_1 + X_2 + X_3 = -1$ gives the exact algebraic identity

$$\left|\operatorname{tr}(W(92))\right|^2 = \frac{1}{2}\Big(25 + 6X_2 - X_3 + (1 - 3X_3)\sqrt{5}\Big) = \frac{7X_1^2 + X_1 + 12}{2} + \frac{3X_1^2 + 3X_1 - 2}{2}\sqrt{5}, \tag{7.7}$$

whose minimal polynomial over $\mathbf{Q}$ is the irreducible degree-6 polynomial

$$y^6 - 70y^5 + 1924y^4 - 26257y^3 + 185064y^2 - 630364y + 809201 = 0. \tag{7.8}$$

For $f(X) = X^3 + X^2 - 2X - 1$, $f'(X) = 3X^2 + 2X - 2 = 3(X+1)^2 - 4(X+1) - 1 \geq \frac{103}{25} > 0$ on $[-2, -\frac{9}{5}]$, so f is strictly increasing on $[-2, -\frac{9}{5}]$ with $f(-2) = -1 < 0 < \frac{1}{125} = f(-\frac{9}{5})$, isolating the smallest root $X_3 = 2\cos(6\pi/7) \in (-2, -\frac{9}{5})$. Together with $X_2 > -\frac{223}{500}$ and $\sqrt{5} > \frac{559}{250}$, (7.7) yields the exact rational lower bound

$$\left|\operatorname{tr}(W(92))\right|^2 > \frac{1}{2}\left(25 - \frac{1338}{500} + \frac{9}{5} + \left(1 + \frac{27}{5}\right)\frac{559}{250}\right) = \frac{48043}{2500} = 19.2172 > \frac{729}{38} > 16.$$

– **Subcase 3b ($a \geq 11$ with $\gcd(a, 21) = 1$):** Choose factored coordinate $t_a^\star \equiv \frac{a-1}{2} \pmod{a} \in (\mathbf{Z}/a\mathbf{Z})^\times$, so $\omega_a = -e^{-\pi i/a}$ and $\omega_a^2 = e^{-2\pi i/a}$. Using $\pi^2 < 10$ and alternating Taylor truncation for $a \geq 11$:

$$\cos(\pi/a) \geq \cos(\pi/11) > 1 - \frac{\pi^2}{2 \cdot 11^2} > 1 - \frac{10}{242} = \frac{116}{121}, \qquad \cos(2\pi/a) = 2\cos^2(\pi/a) - 1 > 2\left(\frac{116}{121}\right)^2 - 1 = \frac{12271}{14641}.$$

Since $-X_3 > \frac{9}{5}$, $X_1 > \frac{623}{500}$, and $-X_2 = 2\cos(3\pi/7) > \frac{111}{250}$ (from $X_2 \in (-\frac{223}{500}, -\frac{111}{250})$ in Case 1, where f is strictly decreasing on $[-\frac{1}{2}, 0]$ and $f(-\frac{111}{250}) = -\frac{37381}{15625000} < 0$), taking the negative real part of (7.6) gives

$$\begin{aligned}\left|\operatorname{tr}(W(t^\star))\right| &\geq -\operatorname{Re}\big(\eta_3^{-2}\operatorname{tr}(W(t^\star))\big) = (-X_3) + 2X_1\cos(\pi/a) + (-X_2)\cos(2\pi/a) \\ &> \frac{9}{5} + 2\left(\frac{623}{500}\right)\left(\frac{116}{121}\right) + \left(\frac{111}{250}\right)\left(\frac{12271}{14641}\right) = \frac{16694959}{3660250} \approx 4.5612 > \frac{9}{2} = 4.5 > 4,\end{aligned}$$

so $|\operatorname{tr}(W(t^\star))|^2 > \frac{81}{4} = 20.25 > 16$.

The Galois-conjugation argument of Theorem 4.2 (Step 5) in $\mathrm{Gal}(\mathbf{Q}(\zeta_m)/\mathbf{Q}) \simeq (\mathbf{Z}/m\mathbf{Z})^\times$ now implies that $W(t)^m \in G_0(t)$ has infinite order and $\langle g_1(t), g_0(t)\rangle$ is Zariski-dense in $\mathrm{SU}(J(t))$ up to μ_m scalars for all $t \in (\mathbf{Z}/m\mathbf{Z})^\times$. □

7.2 Length-3 Separation for $m = a \cdot b \cdot N$ via Four-Way Galois Involution Projections

When $a > 3$, the middle term $-(1 + \omega_a)Z_3(t_N)$ in $\operatorname{tr}(g_1^2 g_0) = A_1 + B_1\omega_a$ no longer simplifies via $-(1 + \omega_3) = \omega_3^2$ to a single root of unity, but remains a 2-term cyclotomic binomial $-(\omega_a^{-r_a} + \omega_a^{r_a})$ after normalization. We resolve this by combining Lemma 5.1's determinant normalization ($\varepsilon \in \{\pm 1\}$) with four commuting Galois involution projections on $\mathbf{Q}(\zeta_a) \otimes_{\mathbf{Q}} \mathbf{Q}(\zeta_b) \otimes_{\mathbf{Q}} \mathbf{Q}(\zeta_N)$.

Theorem 7.2 (Unconditional Length-3 Separation for the Generalized Primary CRT Family $m = a \cdot b \cdot N$). *Let $a \geq 3, b \geq 3, N \geq 7$ be pairwise coprime odd integers with $\gcd(N, 30) = 1$, and set $m = a \cdot b \cdot N$ with the generalized primary CRT branch weight tuple $\mathbf{a}_{\mathrm{gen}}$ of (1.9). Put $r_a := \frac{a+1}{2}$, $\delta(t) := \omega_a^{r_a}\eta_b^2 \in \mu_{ab}$, and $Z_k(t_N) := z^{kt_N} + z^{-kt_N} \in \mathbf{Q}(\zeta_N)$. Then:* 1. *For every unit $t \in (\mathbf{Z}/m\mathbf{Z})^\times$, $\det\rho_t(g_1^2 g_0) = \delta(t)^4$, and the SL_4-normalized trace $\tau_t(W_1) := \delta(t)^{-1}\operatorname{tr}\rho_t(g_1^2 g_0)$ is given by*

$$\tau_t(W_1) = \eta_b^{-1}\big(\omega_a^{-r_a}Z_2(t_N) + \omega_a^{r_a}Z_1(t_N)\big) - \big(\omega_a^{-r_a} + \omega_a^{r_a}\big)Z_3(t_N) + \eta_b\big(\omega_a^{-r_a}Z_5(t_N) + \omega_a^{r_a}Z_4(t_N)\big). \tag{7.9}$$

2. *For every unit* $t \in (\mathbf{Z}/m\mathbf{Z})^\times$,

$$I_t(g_1^2 g_0) \in \{I_1(g_1^2 g_0),\ I_{-1}(g_1^2 g_0)\} \iff \tau_t(g_1^2 g_0) = \pm\tau_{\pm 1}(g_1^2 g_0) \iff t \in \pm H. \tag{7.10}$$

Proof. 1. **Step 0 (Determinant Normalization $\varepsilon \in \{\pm 1\}$ and Derivation of (7.9)):**
Substituting (7.1) into Lemma 5.1(1)–(2) gives $D(t) = b_1 b_2 b_4 b_5 = \eta_b^4$ and $w(t) = \omega_a$, so $\det \rho_t(W_1) = D(t)^2 w(t)^2 = \eta_b^8 \omega_a^2$. Since $2r_a = a + 1 \equiv 1 \pmod a$, we have $\omega_a^{4r_a} = \omega_a^2$ and $\delta(t)^4 = (\omega_a^{r_a}\eta_b^2)^4 = \eta_b^8\omega_a^2 = \det\rho_t(W_1)$. Furthermore, (5.1) evaluates under (7.1) to

$$A_1 = \eta_b Z_2(t_N) - \eta_b^2 Z_3(t_N) + \eta_b^3 Z_5(t_N), \qquad B_1 = \eta_b Z_1(t_N) - \eta_b^2 Z_3(t_N) + \eta_b^3 Z_4(t_N).$$

Dividing $\operatorname{tr}\rho_t(W_1) = A_1 + B_1\omega_a$ by $\delta(t) = \omega_a^{r_a}\eta_b^2$ and using $\omega_a \cdot \omega_a^{-r_a} = \omega_a^{1-r_a} = \omega_a^{r_a}$ yields (7.9). By Lemma 5.1(3), since $m = abN$ is odd, any collision $I_t(W_1) = I_s(W_1)$ ($s \in \{\pm 1\}$) or projective equivalence $A\rho_t(W_1)A^{-1} = \chi(W_1)\rho_s(W_1)$ ($\chi(W_1) \in \mu_m$) satisfies $\delta(t)^4 = \chi(W_1)^4\delta(s)^4$, so the scalar $\varepsilon := \tau_t(W_1)/\tau_s(W_1) = \chi(W_1)\delta(s)\delta(t)^{-1}$ satisfies $\varepsilon^4 = 1$ and hence

$$\varepsilon \in \mu_4 \cap \mathbf{Q}(\zeta_m) = \{\pm 1\} \subset \mathbf{Q}^\times. \tag{7.11}$$

2. **Step 1 (Four-Way Galois Involution Projections and Universal Pure-Tensor Non-Degeneracy):**
Because a, b, N are pairwise coprime, the cyclotomic fields $\mathbf{Q}(\zeta_a), \mathbf{Q}(\zeta_b), \mathbf{Q}(\zeta_N)$ are mutually linearly disjoint over $\mathbf{Q}$, and multiplication gives a $\mathbf{Q}$-algebra isomorphism

$$\mathbf{Q}(\zeta_a) \otimes_{\mathbf{Q}} \mathbf{Q}(\zeta_b) \otimes_{\mathbf{Q}} \mathbf{Q}(\zeta_N) \xrightarrow{\sim} \mathbf{Q}(\zeta_m). \tag{7.12}$$

Let $\tau_{-1,a} \in \operatorname{Gal}(\mathbf{Q}(\zeta_m)/\mathbf{Q}(\zeta_{bN}))$ and $\tau_{-1,b} \in \operatorname{Gal}(\mathbf{Q}(\zeta_m)/\mathbf{Q}(\zeta_{aN}))$ be the partial complex conjugations acting via $\tau_{-1,a}(\zeta_a, \zeta_b, z) = (\zeta_a^{-1}, \zeta_b, z)$ and $\tau_{-1,b}(\zeta_a, \zeta_b, z) = (\zeta_a, \zeta_b^{-1}, z)$. Define the four commuting idempotent projectors

$$\Pi_{\epsilon_a,\epsilon_b} := \frac{1 + \epsilon_a\tau_{-1,a}}{2} \cdot \frac{1 + \epsilon_b\tau_{-1,b}}{2}, \qquad \Delta_{\epsilon_a\epsilon_b}(t) := \Pi_{\epsilon_a,\epsilon_b}\big(\tau_t(W_1)\big) \qquad (\epsilon_a, \epsilon_b \in \{+,-\}). \tag{7.13}$$

Applying $\frac{1\mp\tau_{-1,b}}{2}$ to (7.9) separates the $\tau_{-1,b}$-odd part

$$\frac{1 - \tau_{-1,b}}{2}\tau_t(W_1) = \frac{\eta_b - \eta_b^{-1}}{2}\Big[\omega_a^{r_a}\big(Z_4(t_N) - Z_1(t_N)\big) + \omega_a^{-r_a}\big(Z_5(t_N) - Z_2(t_N)\big)\Big] \tag{7.14}$$

from the $\tau_{-1,b}$-even part, and then applying $\frac{1\mp\tau_{-1,a}}{2}$ isolates three pure tensors $\Delta_{--}(t), \Delta_{+-}(t), \Delta_{-+}(t)$ in $\mathbf{Q}(\zeta_a) \otimes_{\mathbf{Q}} \mathbf{Q}(\zeta_b) \otimes_{\mathbf{Q}} \mathbf{Q}(\zeta_N)$:

$$\Delta_{--}(t) = \frac{1}{4}\big(\omega_a^{r_a} - \omega_a^{-r_a}\big) \otimes \big(\eta_b - \eta_b^{-1}\big) \otimes F_{--}(u), \tag{7.15}$$

$$\Delta_{+-}(t) = \frac{1}{4}\big(\omega_a^{r_a} + \omega_a^{-r_a}\big) \otimes \big(\eta_b - \eta_b^{-1}\big) \otimes F_{+-}(u), \tag{7.16}$$

$$\Delta_{-+}(t) = \frac{1}{4}\big(\omega_a^{r_a} - \omega_a^{-r_a}\big) \otimes \big(\eta_b + \eta_b^{-1}\big) \otimes F_{-+}(u), \tag{7.17}$$

where the $\mathbf{Q}(\zeta_N)$-factors $F_{--}(u), F_{+-}(u), F_{-+}(u) \in \mathbf{Z}[u, u^{-1}]$ ($u = z^{t_N}$) factor identically in $\mathbf{Z}[u, u^{-1}]$ as

$$\begin{aligned} F_{--}(u) &:= Z_4(t_N) - Z_1(t_N) - Z_5(t_N) + Z_2(t_N) = -u^{-5}(u-1)(u^3-1)(u^6+1), \\ F_{+-}(u) &:= Z_4(t_N) - Z_1(t_N) + Z_5(t_N) - Z_2(t_N) = u^{-5}(u+1)(u^3-1)(u^6-1), \\ F_{-+}(u) &:= Z_4(t_N) + Z_1(t_N) - Z_5(t_N) - Z_2(t_N) = -u^{-5}(u-1)(u^3+1)(u^6-1). \end{aligned} \tag{7.18}$$

We prove that **for every unit $t \in (\mathbf{Z}/m\mathbf{Z})^\times$, all three tensor factors of $\Delta_{--}(t), \Delta_{+-}(t), \Delta_{-+}(t)$ in (7.15)–(7.17) are strictly non-zero** in $\mathbf{Q}(\zeta_a)^\times, \mathbf{Q}(\zeta_b)^\times, \mathbf{Q}(\zeta_N)^\times$: - **First tensor factor in $\mathbf{Q}(\zeta_a)$:** Since $\gcd(t_a, a) = 1$ and $2r_a \equiv 1 \pmod a$, $\omega_a = \zeta_a^{t_a}$ has exact odd multiplicative order $a \geq 3$ and $\omega_a^{2r_a} = \omega_a$. Hence $\omega_a^{r_a} - \omega_a^{-r_a} = \omega_a^{-r_a}(\omega_a - 1) \neq 0$ (since $a \geq 3 > 1$) and $\omega_a^{r_a} + \omega_a^{-r_a} = \omega_a^{-r_a}(\omega_a + 1) \neq 0$ (since a is odd, so $-1 \notin \langle\omega_a\rangle$). - **Second tensor factor in $\mathbf{Q}(\zeta_b)$:** Since $\gcd(c_b t_b, b) = 1$, $\eta_b = \zeta_b^{c_b t_b}$ has exact odd multiplicative order $b \geq 3$. Hence $\eta_b - \eta_b^{-1} = \eta_b^{-1}(\eta_b^2 - 1) \neq 0$ (as $b \nmid 2$) and $\eta_b + \eta_b^{-1} = \eta_b^{-1}(\eta_b^2 + 1) \neq 0$ (as $b \nmid 4$). - **Third tensor factor in $\mathbf{Q}(\zeta_N)$:** Since $\gcd(t_N, N) = 1$, $u = z^{t_N}$ has exact multiplicative order $N \geq 7$. By (7.18), $F_{--}(u) = 0 \implies u^{12} = 1 \implies N \mid 12$, whereas $F_{+-}(u) = 0$ or

$F_{-+}(u) = 0 \implies u^6 = 1 \implies N \mid 6$. Since $\gcd(N, 30) = 1$ and $N \geq 7 > 1$, we have $\gcd(N, 12) = \gcd(N, 6) = 1$, so $N \nmid 12$ and $N \nmid 6$. Therefore $F_{--}(u) \neq 0$, $F_{+-}(u) \neq 0$, and $F_{-+}(u) \neq 0$ in $\mathbf{Q}(\zeta_N)^\times$.

(Remark on $N = 7$ and the Role of Step 0: At $N = 7$, $u^7 = 1$ implies $Z_5(t_7) - Z_2(t_7) = u^{-5}(u^7 - 1)(u^3 - 1) = 0$, which makes $F_{--}(u)$ and $F_{+-}(u)$ equal to $Z_4(t_7) - Z_1(t_7) = u^{-4}(u^5 - 1)(u^3 - 1) \neq 0$, so neither $\Delta_{--}(t)$ nor $\Delta_{+-}(t)$ vanishes. Because $Z_5 - Z_2 = 0$ collapses the bracket in (7.14) to the single monomial $\omega_a^{r_a}(Z_4 - Z_1)$, Step 0's determinant reduction $\varepsilon \in \{\pm 1\} \subset \mathbf{Q}^\times$ ensures that ε commutes with $\Pi_{\varepsilon_a, \varepsilon_b}$ and prevents any non-trivial root-of-unity shift before projecting.)

3. **Step 2 (Pure-Tensor Rational-Scalar Uniqueness and Field-Norm Proof that $t_a \equiv \pm 1 \pmod{a}$ and $t_b \equiv \pm 1 \pmod{b}$):**
   By the CRT coordinate convention (1.9a), applying the factorwise Galois automorphism $\sigma_{e_a}^{-1} \otimes \sigma_{e_b}^{-1} \otimes \sigma_{e_N}^{-1} \in \mathrm{Gal}(\mathbf{Q}(\zeta_m)/\mathbf{Q})$ (which sends the CRT-compatible primitive roots $\zeta_a^{e_a} \mapsto \zeta_a$, $\zeta_b^{e_b} \mapsto \zeta_b$, $z^{e_N} \mapsto z$, fixes $\varepsilon \in \{\pm 1\} \subset \mathbf{Q}^\times$, and commutes with $\tau_{-1,a}, \tau_{-1,b}$) replaces the factored coordinates $(t_a, t_b, t_N) \equiv (e_a t, e_b t, e_N t)$ by the ordinary CRT residues $(\bar{t}_a, \bar{t}_b, \bar{t}_N) = (t \bmod a, t \bmod b, t \bmod N)$, under which $s \in \{\pm 1\}$ has coordinates (s, s, s) $(\mathrm{mod}\ (a, b, N))$ and $z^{\bar{s}_N} = z^{\pm 1}$; for notational simplicity we write (t_a, t_b, t_N) for $(\bar{t}_a, \bar{t}_b, \bar{t}_N)$ in Steps 2–3. Since $\varepsilon \in \{\pm 1\} \subset \mathbf{Q}^\times$ by (7.11), applying $\Pi_{\varepsilon_a, \varepsilon_b}$ to $\tau_t(W_1) = \varepsilon \tau_s(W_1)$ ($s \in \{\pm 1\}$) yields the non-zero pure-tensor equalities in $\mathbf{Q}(\zeta_a) \otimes_{\mathbf{Q}} \mathbf{Q}(\zeta_b) \otimes_{\mathbf{Q}} \mathbf{Q}(\zeta_N)$:

$$\Delta_{--}(t) = \varepsilon \Delta_{--}(s) \neq 0, \qquad \Delta_{+-}(t) = \varepsilon \Delta_{+-}(s) \neq 0, \qquad \Delta_{-+}(t) = \varepsilon \Delta_{-+}(s) \neq 0. \tag{7.19}$$

   In any tensor product $V_1 \otimes_{\mathbf{Q}} V_2 \otimes_{\mathbf{Q}} V_3$ of $\mathbf{Q}$-vector spaces, if $x_1 \otimes x_2 \otimes x_3 = y_1 \otimes y_2 \otimes y_3 \neq 0$, then each factor x_i is a unique non-zero rational scalar multiple of y_i. Comparing the first tensor factor of $\Delta_{--}(t) = \varepsilon \Delta_{--}(s)$ and $\Delta_{+-}(t) = \varepsilon \Delta_{+-}(s)$ in $\mathbf{Q}(\zeta_a)^\times$ gives $\kappa_-, \kappa_+ \in \mathbf{Q}^\times$ such that

$$\omega_a(t_a)^{r_a} - \omega_a(t_a)^{-r_a} = \kappa_-\big(\omega_a(s)^{r_a} - \omega_a(s)^{-r_a}\big), \qquad \omega_a(t_a)^{r_a} + \omega_a(t_a)^{-r_a} = \kappa_+\big(\omega_a(s)^{r_a} + \omega_a(s)^{-r_a}\big). \tag{7.20}$$

   Because $\gcd(t_a, a) = \gcd(s, a) = 1$, the Galois automorphism $\sigma_{t_a s^{-1}} \in \mathrm{Gal}(\mathbf{Q}(\zeta_a)/\mathbf{Q})$ sends $\omega_a(s) \mapsto \omega_a(t_a)$. Thus the left-hand side of each equation in (7.20) is a $\mathrm{Gal}(\mathbf{Q}(\zeta_a)/\mathbf{Q})$-conjugate of the parenthesized factor on the right-hand side. Taking the field norm $N_{\mathbf{Q}(\zeta_a)/\mathbf{Q}} : \mathbf{Q}(\zeta_a)^\times \to \mathbf{Q}^\times$ of both sides yields

$$\kappa_\pm^{\varphi(a)} = 1 \qquad \text{and} \qquad \kappa_+^{\varphi(a)} = 1.$$

   Since $\kappa_-, \kappa_+ \in \mathbf{Q}^\times$, this forces $\kappa_- \in \{\pm 1\}$ **and** $\kappa_+ \in \{\pm 1\}$. Adding the two equations of (7.20) gives

$$2\omega_a(t_a)^{r_a} = (\kappa_+ + \kappa_-)\omega_a(s)^{r_a} + (\kappa_+ - \kappa_-)\omega_a(s)^{-r_a} \in \big\{\pm 2\omega_a(s)^{r_a},\ \pm 2\omega_a(s)^{-r_a}\big\}.$$

   Squaring both sides and using $2r_a \equiv 1 \pmod{a}$ gives $\omega_a(t_a) = \omega_a(s)^{\pm 1}$, i.e., $t_a \equiv \pm s \equiv \pm 1 \pmod{a}$.

   By the identical argument on the second tensor factor in $\mathbf{Q}(\zeta_b)^\times$ using $\Delta_{--}(t) = \varepsilon \Delta_{--}(s) \neq 0$ and $\Delta_{-+}(t) = \varepsilon \Delta_{-+}(s) \neq 0$, there exist $\lambda_-, \lambda_+ \in \mathbf{Q}^\times$ such that

$$\eta_b(t_b) - \eta_b(t_b)^{-1} = \lambda_-\big(\eta_b(s) - \eta_b(s)^{-1}\big), \qquad \eta_b(t_b) + \eta_b(t_b)^{-1} = \lambda_+\big(\eta_b(s) + \eta_b(s)^{-1}\big),$$

   and taking $N_{\mathbf{Q}(\zeta_b)/\mathbf{Q}}$ forces $\lambda_-, \lambda_+ \in \{\pm 1\}$, hence $\eta_b(t_b) = \pm \eta_b(s)^{\pm 1}$. Since η_b has odd order $b \geq 3$, $-1 \notin \langle \eta_b \rangle$, so $\eta_b(t_b) = \eta_b(s)^{\pm 1}$ and $t_b \equiv \pm s \equiv \pm 1 \pmod{b}$, with $\kappa_- = \epsilon_1 \in \{\pm 1\}$ ($t_a \equiv \epsilon_1 s \pmod{a}$), $\kappa_+ = +1$, $\lambda_- = \epsilon_2 \in \{\pm 1\}$ ($t_b \equiv \epsilon_2 s \pmod{b}$), and $\lambda_+ = +1$ (note that at $b = 3$, $\eta_3(t_3) + \eta_3(t_3)^{-1} = -1 \in \mathbf{Q}$ for both $t_3 \in \{1, 2\}$, which gives $\lambda_+ = +1$ and $\lambda_- = \epsilon_2 \in \{\pm 1\}$ immediately without needing any linear independence of $\{\eta_3^{-1}, 1, \eta_3\}$).

4. **Step 3 (Pure-Tensor Ratio Reduction in $\mathbf{Q}(\zeta_N)^+$ Valid for All Odd $b \geq 3$ Including $b = 3$):**
   Substituting $(\kappa_-, \kappa_+, \lambda_-, \lambda_+) = (\epsilon_1, +1, \epsilon_2, +1)$ into the pure-tensor equalities (7.19) and cancelling the non-zero $\mathbf{Q}(\zeta_a)^\times$ and $\mathbf{Q}(\zeta_b)^\times$ factors yields three exact equations in $\mathbf{Q}(\zeta_N)^\times$ (where $u = z^{t_N}$ and $z^s = z^{\pm 1}$, so $F_{\bullet\bullet}(z^s) = F_{\bullet\bullet}(z)$):

$$F_{--}(u) = \varepsilon \epsilon_1 \epsilon_2 F_{--}(z), \qquad F_{+-}(u) = \varepsilon \epsilon_2 F_{+-}(z), \qquad F_{-+}(u) = \varepsilon \epsilon_1 F_{-+}(z). \tag{7.20a}$$

   Put $x := u + u^{-1} = Z_1(t_N) \in (-2, 2)$ and $x_0 := z + z^{-1} = 2\cos(2\pi/N) \in (1, 2)$ (since $N \geq 7 > 6$). From the factored forms (7.18), the ratios $F_{+-}(u)/F_{-+}(u)$ and $F_{--}(u)/F_{+-}(u)$ simplify in $\mathbf{Q}(x)$ to

$$\frac{F_{+-}(u)}{F_{-+}(u)} = -\frac{u^2 + u + 1}{u^2 - u + 1} = -\frac{x + 1}{x - 1}, \qquad \frac{F_{--}(u)}{F_{+-}(u)} = -\frac{(u - 1)(u^6 + 1)}{(u + 1)(u^6 - 1)} = -\frac{x(x^2 - 3)}{(x + 2)(x^2 - 1)}. \tag{7.20b}$$

Dividing the equations in (7.20a) gives

$$\frac{x+1}{x-1} = \epsilon_1\epsilon_2\frac{x_0+1}{x_0-1} \qquad \text{and} \qquad \frac{x(x^2-3)}{(x+2)(x^2-1)} = \epsilon_1\frac{x_0(x_0^2-3)}{(x_0+2)(x_0^2-1)}. \tag{7.20c}$$

- **If $\epsilon_1\epsilon_2 = -1$:** The first equation of (7.20c) gives $\frac{x+1}{x-1} = -\frac{x_0+1}{x_0-1} \iff x = \frac{1}{x_0}$. Substituting $x = 1/x_0$ into the second equation of (7.20c) and clearing the positive denominator $(2x_0+1)(x_0+2)(x_0^2-1) > 0$ on $x_0 \in (1,2)$ yields $-2(x_0^4 - x_0^3 - 6x_0^2 - x_0 + 1) = 0$ for $\epsilon_1 = +1$ (impossible since $x_0^4 - x_0^3 - 6x_0^2 - x_0 + 1 = x_0^2(x_0^2-6) - (x_0^3 + x_0 - 1) < 1(4-6) - (1+1-1) = -3 < 0$ on $x_0 \in (1,2)$) and $2(x_0-1)(x_0+1)^3 = 0$ for $\epsilon_1 = -1$ (impossible since $x_0 > 1$).
- **Hence $\epsilon_1\epsilon_2 = +1$:** The Möbius map $x \mapsto \frac{x+1}{x-1}$ is injective, so the first equation of (7.20c) forces $x = x_0$, i.e., $z^{t_N} + z^{-t_N} = z + z^{-1} \implies t_N \equiv \pm 1 \pmod N$. Since $N \geq 7$ and $\gcd(N, 30) = 1$, $x_0(x_0^2 - 3) \neq 0$, so substituting $x = x_0$ into the second equation of (7.20c) forces $\epsilon_1 = +1$, whence $\epsilon_2 = +1$, and substituting $(\epsilon_1, \epsilon_2) = (+1, +1)$ and $u = z^{\pm 1}$ into $F_{--}(u) = \varepsilon\epsilon_1\epsilon_2 F_{--}(z) \neq 0$ forces $\varepsilon = +1$. Thus $(t_a, t_b, t_N) \equiv (s, s, \pm 1) \pmod{(a, b, N)}$ and $t \in \pm H$. □

7.3 Proof of Theorem D(2): Four-Fold Method-Boundary Examples

We now prove Part 2 of Theorem D by establishing the four explicit method-boundary examples (b.1)–(b.4). Here (b.1) at $(a, b, N) = (9, 5, 7)$ and (b.2) at $(a, b, N) = (3, 7, 11)$ have admissible coprime parameter triples (a, b, N) but evaluate companion-type branch weight tuples outside the generalized primary family (1.9) to show where complete-residue-block Kubert balance and companion (t_a, t_b)-independence break down; (b.3) at $(a, b, N) = (5, 3, 7)$ lies directly on the generalized primary family (1.9) of Theorems 7.1–7.2 and shows where the global-average shortcut Mean > 16 fails; and (b.4) at $(a, b, N) = (5, 7, 3)$ violates $\gcd(N, 30) = 1$ on the primary template and shows that the 7-eigenvalue adjoint spectrum of Theorem 3.2 collapses:

1. **Method-Boundary (b.1) — Arithmetic Rigidity of $(a, b) = (3, 5)$ for Kubert $(2, 2)$-Balance:**
   In any $(2, 2, 1, 1)$-symmetric tuple $\mathbf{a} = (a_0, a_\infty, a_1, a_{-1}, a_\lambda, a_{-\lambda})$, the involution h fixes the 2 weights $\{a_0, a_\infty\}$ and pairs the 4 weights $\{a_1, a_{-1}, a_\lambda, a_{-\lambda}\}$. For characterwise $(2, 2)$-balance ($d_t = \sum_{i=1}^{6}\{ta_i/m\} = 3$ across all $t \in (\mathbf{Z}/m\mathbf{Z})^\times$) to arise from Kubert's distribution identity

   $$\sum_{k=1}^{d-1}\left\{x + \frac{k}{d}\right\} = \frac{d-1}{2} + \{dx\} - \{x\}, \tag{7.21}$$

   the 2 fixed weights $\{a_0, a_\infty\}$ (which are $0 \pmod b$ and non-zero $\pmod a$) must form a complete set of non-zero residues $\{1/a, \ldots, (a-1)/a\}$ modulo a, which forces

   $$a - 1 = 2 \iff a = 3 \qquad \left(\text{contributing } \frac{a-1}{2} = 1 \text{ to } d_t\right),$$

   and the 4 paired weights $\{a_1, a_{-1}, a_\lambda, a_{-\lambda}\}$ (which are $0 \pmod a$ and non-zero $\pmod b$) must form a complete set of non-zero residues $\{1/b, \ldots, (b-1)/b\}$ modulo b, which forces

   $$b - 1 = 4 \iff b = 5 \qquad \left(\text{contributing } \frac{b-1}{2} = 2 \text{ to } d_t\right),$$

   while universal cancellation of $\{3x\} - \{x\} + \{5y\} - \{y\}$ across all t requires $N \mid (3 \cdot 5 - 1) = 14 \implies N = 7$ $(m = 105)$.
   Explicit Counterexample at $(a, b, N) = (9, 5, 7)$ ($m = 315$): Taking $a = 3^2 = 9, b = 5, N = 7$ with $\mathbf{a} \equiv (2, 7, 0, 0, 0, 0) \pmod 9$, $(0, 0, 1, 4, 2, 3) \pmod 5$, $(1, 1, 3, 3, 3, 3) \pmod 7$, the two fixed weights have mod-9 residues $\{2, 7\} \subsetneq (\mathbf{Z}/9\mathbf{Z})^\times$. Across the $\varphi(315) = 144$ units $t \in (\mathbf{Z}/315\mathbf{Z})^\times$, direct evaluation gives $d_t = 3$ on 80 units, $d_t = 2$ (signature $(1, 3)$) on 32 units, and $d_t = 4$ (signature $(3, 1)$) on 32 units.

2. **Method-Boundary (b.2) — Breakdown of Companion (t_a, t_b)-Independence for $b \neq 5$:**
   In (4.8)–(4.9) on the companion family (1.8), the independence of $|\operatorname{tr}(W(t))|^2$ from (t_3, t_5) relies on the fact

that when $b = 5$, the four roots $\xi^{\pm t_5}, \xi^{\pm 2t_5}$ exhaust $\mu_5 \setminus \{1\}$ for every $t_5 \in (\mathbf{Z}/5\mathbf{Z})^\times$, so $\prod_{j\in\{1,2,4,5\}}(b_j - 1) = \prod_{r=1}^{4}(u - \xi^r) = \Phi_5(u)$ and $a_0 + a_2 = \sum_{r=1}^{4} \xi^r = -1$ are independent of t_5. For any odd $b > 5$, the four weights $\{\pm t_b, \pm 2t_b\} \pmod b$ omit $b - 5 > 0$ non-trivial b-th roots of unity.
Explicit Counterexample at $(a, b, N) = (3, 7, 11)$ **(**$m = 231$**):** With $b = 7$ and $\mathbf{a} \equiv (0, 0, 1, 6, 2, 5) \pmod 7$, the four paired weights omit $\{3t_7, 4t_7\} \pmod 7$, so $a_0 + a_2 = -1 - (\zeta_7^{3t_7} + \zeta_7^{-3t_7}) \neq -1$ and $|\operatorname{tr}(W(t))|^2$ takes 3 distinct values as t_7 varies over $\{1, 2, 3\} \pmod 7$ for fixed t_{11}.

3. **Method-Boundary (b.3) — Breakdown of the Global Galois-Mean Inequality Mean** > 16 **at** $(a, b, N) = (5, 3, 7)$**:**
When $b = 3$ on the generalized primary family (1.9), $\beta_1 = \beta_2 = \beta_4 = -\frac{1}{2}$ and $\beta_3 = +1$ in (7.2), corresponding to the cyclotomic collapse $\eta_3^2 + 1 = -\eta_3$ in (7.5). At $(a, b, N) = (5, 3, 7)$ ($m = 105$, with the roles of 3 and 5 swapped relative to (1.7)), evaluating (7.2) and averaging over $t_7 \in (\mathbf{Z}/7\mathbf{Z})^\times$ gives the exact global Galois mean
$$\frac{1}{48} \sum_{t\in(\mathbf{Z}/105\mathbf{Z})^\times} \left|\operatorname{tr}(W(t))\right|^2 = \frac{35}{3} = 11.\overline{6} < 16,$$
demonstrating that the global Galois-mean inequality fails at $(b, N) = (3, 7)$ and that the specialized point evaluation at $t^\star \equiv 92 \pmod{105}$ in Theorem 7.1 (Case 3a, where $|\operatorname{tr}(W(92))|^2 > \frac{48043}{2500} > 16$) is indispensable.

4. **Method-Boundary (b.4) — Collapse of the** 7**-Eigenvalue Adjoint Spectrum at** $N = 3$**:**
In Theorem 3.2, the distinctness of the 7 eigenvalues of $\mathrm{Ad}(g_1(t))$ in (3.10) on the primary template requires $u^3 \neq u^{-3}$ in μ_N, where $u = z^{t_N}$ and $\lambda_1 = \eta_b^2 u^{-3}, \lambda_2 = \eta_b^2 u^{+3}$.
Explicit Counterexample at $(a, b, N) = (5, 7, 3)$ **(**$m = 105$**, with** $N = 3$ **in the third slot violating** $\gcd(N, 30) = 1$**):** Since $z \in \mu_3$, we have $z^{\pm 3t_3} = 1$ for all $t \in (\mathbf{Z}/105\mathbf{Z})^\times$, which forces
$$\lambda_1 = \lambda_2 = \eta_7^2 \implies \lambda_1\lambda_2^{-1} = \lambda_2\lambda_1^{-1} = 1.$$
Consequently, since $g_1(t)$ has spectrum $(\eta_7^2, \eta_7^2, 1, 1)$ with multiplicities $(2, 2)$, the spectrum of $\mathrm{Ad}(g_1(t))$ on $\mathfrak{sl}_4(\mathbf{C})$ collapses from 7 distinct eigenvalues to only 3 distinct eigenvalues $\{1 \text{ (mult. } 7), \eta_7^2 \text{ (mult. } 4), \eta_7^{-2} \text{ (mult. } 4)\}$ (as the ratio 1 has multiplicity $2^2 + 2^2 - 1 = 7$ on $\mathfrak{sl}_4(\mathbf{C})$ and $\eta_7^{\pm 2}$ each have multiplicity $2 \times 2 = 4$), merging the root spaces $\mathbf{C}E_{1,2}$ and $\mathbf{C}E_{2,1}$ into the trivial eigenspace $\mathfrak{g}_1$ and invalidating the Lagrange-interpolation root isolation of Theorem 3.2. (By contrast, for higher multiples of 3 such as $N = 9$ at $(a, b, N) = (5, 7, 9), m = 315$, $z^{\pm 3t_9}$ has order 3 rather than 1, so the collapse to 3 eigenvalues is specific to $N = 3$.) □

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8. Consequences for Generic Mumford–Tate Groups and All-Powers Hodge Classes on One-Dimensional Slices

8.1 Full Pairwise Projective Non-Equivalence via Galois Transitivity

We first deduce from Theorems 5.2 and 7.2 that the single length-3 braid word $W_1 = g_1^2 g_0$ separates not only $[1]$ from every $[t] \neq [\pm 1]$, but **all** $\binom{|\mathcal{T}_{\mathrm{prim}}/\{\pm 1\}|}{2}$ pairs of primitive character orbits on $\pi_1(S)$.

Proposition 8.1 (Pairwise Projective Non-Equivalence Across All Primitive Character Orbits on S**).**
Let $m = 15N$ *(*$\gcd(N, 30) = 1, N > 1$*) on the primary or companion CRT family (1.7)–(1.8), or let* $m = a \cdot b \cdot N$ *(pairwise coprime odd* $a \geq 3, b \geq 3, N \geq 7, \gcd(N, 30) = 1$*) on the generalized primary CRT family (1.9). For any two units* $t, s \in (\mathbf{Z}/m\mathbf{Z})^\times$*, the following are equivalent:* 1. *The projective representations* $\mathbf{P}\rho_t$ *and* $\mathbf{P}\rho_s$ *(or* $\mathbf{P}\rho_t$ *and* $\mathbf{P}\rho_s^\vee \simeq \mathbf{P}\rho_{-s}$*) are isomorphic on* $\pi_1(S)$*.* 2. *The length-*3 *braid word* $W_1 = g_1^2 g_0$ *satisfies* $I_t(W_1) \in \{I_s(W_1), I_{-s}(W_1)\}$*.* 3. $\tau_t(W_1) = \pm\tau_{\pm s}(W_1)$ *in* $\mathbf{Z}[\zeta_m]$*.* 4. $t \in \pm sH$*, i.e.,* $[t] = [\pm s]$ *in* $\mathcal{T}_{\mathrm{prim}} = (\mathbf{Z}/m\mathbf{Z})^\times/H$*.*

Proof. If $t \in \pm sH$, the Prym involution τ_h and complex conjugation intertwine ρ_t with ρ_s or ρ_{-s}, so (4) $\implies$ (1), and (1) $\implies$ (2) because $I_t(W_1)$ is a $\mathrm{PGL}_4(\mathbf{C})$-invariant; moreover, (2) $\iff$ (3) holds by Lemma 5.1(3). It remains to prove (3) $\implies$ (4). Suppose $\tau_t(W_1) = \varepsilon\tau_{\pm s}(W_1)$ with $\varepsilon \in \{\pm 1\} \subset \mathbf{Q}^\times$. Since $s \in (\mathbf{Z}/m\mathbf{Z})^\times$, there is a unique Galois automorphism $\sigma_{s^{-1}} \in \mathrm{Gal}(\mathbf{Q}(\zeta_m)/\mathbf{Q})$ sending $\zeta_m \mapsto \zeta_m^{s^{-1}}$. Because the entries of $g_1(u), g_0(u)$ and the normalization factor $\delta(u)$ in (5.2) are polynomial expressions in ζ_m^u, applying $\sigma_{s^{-1}}$ gives
$$\sigma_{s^{-1}}\big(\tau_u(W_1)\big) = \tau_{us^{-1}}(W_1) \qquad \text{for all } u \in (\mathbf{Z}/m\mathbf{Z})^\times.$$

Since $\varepsilon \in \{\pm 1\} \subset \mathbf{Q}$ is fixed by $\sigma_{s^{-1}}$, applying $\sigma_{s^{-1}}$ to $\tau_t(W_1) = \varepsilon\tau_{\pm s}(W_1)$ yields

$$\tau_{ts^{-1}}(W_1) = \varepsilon\tau_{\pm 1}(W_1) \qquad (\varepsilon \in \{\pm 1\}).$$

By Theorem 5.2 (for $m = 15N$) and Theorem 7.2 (for $m = a \cdot b \cdot N$), this forces $ts^{-1} \in \pm H$, and hence $t \in \pm sH$. □

8.2 Goursat Reduction and Proof of Corollary E

Proof of Corollary E.
Let $\pi : B_S \to S$ be the smooth proper Prym abelian scheme over $S = \mathbf{P}^1_\lambda \setminus \{0, \infty, 1, -1\}$ with generic fiber B_η, and let $B_{\eta,\mathrm{prim}}$ be the rational primitive isotypic summand of B_η (defined over $\mathbf{Q}$) corresponding to the faithful characters of μ_m. The H-fixed cyclotomic field $L := \mathbf{Q}(\zeta_m)^H$ of degree $[L : \mathbf{Q}] = \varphi(m)/2$ acts on $B_{\eta,\mathrm{prim}}$ by $\mathbf{Q}$-endomorphisms, so $H_1(B_{\eta,\mathrm{prim}}, \mathbf{Q})$ is a single L-module of rank 4 over L (and dimension $2\varphi(m)$ over $\mathbf{Q}$). After extending scalars from $\mathbf{Q}$ to $\mathbf{C}$, the complex local system $R^1\pi_*\mathbf{C}_{\mathrm{prim}}$ decomposes into the 4-dimensional embedding eigenspaces $E_{[t]} := H^1(B_{\eta,\mathrm{prim}}, \mathbf{C})_{[t]}$ indexed by $[t] \in \mathcal{T}_{\mathrm{prim}} = (\mathbf{Z}/m\mathbf{Z})^\times/H$, while over $\mathbf{R}$ conjugate embeddings pair into the 8-dimensional real local systems $(\mathbf{V}_{[t]} \oplus \mathbf{V}_{[-t]})_{\mathbf{R}}$ indexed by $[t] \in \mathcal{T}_{\mathrm{prim}}/\{\pm 1\} = (\mathbf{Z}/m\mathbf{Z})^\times/(\pm H)$. Once the slice-specific inputs of Step 1 below are established on S, the group-theoretic Goursat decoupling, derived Mumford–Tate comparison, and balanced invariant-theoretic upgrade in Steps 2–5 follow the same abstract mechanism as on the full configuration space in [3, Lemma 6.1, Proposition 6.3, Theorem 6.5, and Theorem 7.1]; we record the self-contained deduction over S and the distinction between unbalanced and characterwise balanced families below.

1. By Theorem 4.2 (for $m = 15N$) and Theorem 7.1 (for $m = a \cdot b \cdot N$), for each $[t] \in \mathcal{T}_{\mathrm{prim}}/\{\pm 1\}$ the connected algebraic monodromy group $G_0(t)^\circ$ of $\mathbf{V}_{[t]}$ over $\mathbf{C}$ is $\mathrm{SL}_4(\mathbf{C})$, and the connected real algebraic monodromy group on $(\mathbf{V}_{[t]} \oplus \mathbf{V}_{[-t]})_{\mathbf{R}}$ is the special unitary group $\mathrm{SU}(J(t)) \simeq \mathrm{SU}(d_t - 1, 5 - d_t)$, whose complexified Lie algebra is the simple Lie algebra $\mathfrak{sl}_4(\mathbf{C})$.
2. The only outer automorphism of $\mathfrak{sl}_4(\mathbf{C})$ is the contragredient involution $X \mapsto -X^T$, which sends the standard representation $\mathbf{V}_{[t]}$ to its dual $\mathbf{V}_{[t]}^\vee \simeq \mathbf{V}_{[-t]}$ (up to determinant twist). By Proposition 8.1, for any two distinct classes $[t] \neq [\pm s]$ in $\mathcal{T}_{\mathrm{prim}}/\{\pm 1\}$, the projective representations $\mathbf{P}\rho_t$ and $\mathbf{P}\rho_{\pm s}$ on $\pi_1(S)$ are non-isomorphic, so the Lie-algebra representations of $\mathrm{Lie}(G^\circ_{\mathrm{mon}})$ on $\mathbf{V}_{[t]}$ and on $\mathbf{V}_{[s]}, \mathbf{V}_{[s]}^\vee$ are non-isomorphic.
3. By Goursat's lemma for semisimple Lie algebras (see [3, Lemma 6.1 and Proposition 6.3]), a subalgebra of a direct sum of simple Lie algebras $\bigoplus_{i=1}^r \mathfrak{sl}_4(\mathbf{C})$ that projects surjectively onto each factor and has no isomorphism or contragredient isomorphism between distinct projections must be the full direct sum. Therefore, the connected algebraic monodromy group G°_{mon} of $R^1\pi_*\mathbf{Q}_{\mathrm{prim}}$ over S has no diagonal identifications between distinct $\pm H$ orbits and equals the full semisimple product over $\mathbf{R}$:
$$G^\circ_{\mathrm{mon}}(\mathbf{R}) = \prod_{[t] \in (\mathbf{Z}/m\mathbf{Z})^\times/(\pm H)} \mathrm{SU}\big(J(t)\big). \tag{8.1}$$
4. By the theorem of Deligne and André on the monodromy of polarized variations of Hodge structure ([4, Proposition 7.5], [5, Theorem 1]; cf. [6, Corollary 2.16 and Proposition 3.4], [2, Theorem 3.1.3]), G°_{mon} is a normal subgroup of (in particular, contained in) the generic **derived** Mumford–Tate (Hodge) group $\mathrm{Hg}(B_{\eta,\mathrm{prim}})_{\mathrm{der}}$. On the other hand, $\mathrm{Hg}(B_{\eta,\mathrm{prim}})_{\mathrm{der}}$ commutes with the endomorphism field $L = \mathbf{Q}(\zeta_m)^H$ and preserves the Hermitian polarization form $J(t)$ on each complex eigenspace $E_{[t]}$, so $\mathrm{Hg}(B_{\eta,\mathrm{prim}})_{\mathrm{der}}(\mathbf{R}) \subseteq \prod_{[t] \in (\mathbf{Z}/m\mathbf{Z})^\times/(\pm H)} \mathrm{SU}(J(t))$ (cf. [3, Theorem 6.5]). Combining both inclusions gives equality for the derived Hodge group:
$$\mathrm{Hg}(B_{\eta,\mathrm{prim}})_{\mathrm{der}}(\mathbf{R}) = G^\circ_{\mathrm{mon}}(\mathbf{R}) = \prod_{[t] \in (\mathbf{Z}/m\mathbf{Z})^\times/(\pm H)} \mathrm{SU}\big(J(t)\big). \tag{8.2}$$
   By Schur's lemma applied to the pairwise non-isomorphic irreducible $\mathrm{Hg}(B_{\eta,\mathrm{prim}})_{\mathrm{der}}(\mathbf{C})$-modules $E_{[t]}$, the endomorphism algebra of $B_{\eta,\mathrm{prim}}$ over $\overline{\mathbf{Q}(S)}$ is exhausted by L:
$$\mathrm{End}_{\overline{\mathbf{Q}(S)}}(B_{\eta,\mathrm{prim}}) \otimes_{\mathbf{Z}} \mathbf{Q} \simeq \mathbf{Q}(\zeta_m)^H,$$
   ruling out any cross-factor endomorphisms between distinct $[t] \neq [s]$ in $\mathcal{T}_{\mathrm{prim}}$.

5. **Distinction Between Unbalanced Families (Cross-Orbit Determinant Classes) and Characterwise $(2,2)$-Balanced Families:**
Because each factor $\mathrm{SL}_4(\mathbf{C})$ of $\mathrm{Hg}(B_{\eta,\mathrm{prim}})_{\mathrm{der}}(\mathbf{C})$ acts by the identity on its 1-dimensional determinant line $D_{[t]} := \bigwedge^4 E_{[t]} \subset H^4(B_{\eta,\mathrm{prim}}, \mathbf{C})$ (which has Hodge type $(d_t - 1, 5 - d_t)$, up to exchanging the two entries according to homology/cohomology convention), the equality (8.2) for the **derived** Hodge group does not by itself determine the central torus of $\mathrm{Hg}(B_{\eta,\mathrm{prim}})$ or exclude Hodge classes built from products of determinant lines across distinct orbits on unbalanced families:
   - **Explicit Cross-Orbit Determinant Hodge Classes on the Primary Family at $m = 105, N = 7$:** Consider the primary family (1.7) at $N = 7, m = 105$ with concrete branch weight tuple $\mathbf{a} = (91, 56, 57, 27, 72, 12)$ and $h = 76 \pmod{105}$. Here $d_t = \frac{1}{105}\sum_{j=1}^{6}(ta_j \bmod 105)$ satisfies the exact finite identity
$$d_{8g} = d_{13g} \in \{2, 3, 4\}, \qquad d_{8g} + d_{-13g} = 6 \qquad \text{for all } g \in (\mathbf{Z}/105\mathbf{Z})^\times$$
(on the 12 representatives $g = (1, 2, 4, 8, 13, 16, 17, 19, 23, 31, 34, 38)$ of $(\mathbf{Z}/105\mathbf{Z})^\times/(\pm H)$, $d_{8g} = d_{13g} = (3, 4, 3, 3, 3, 4, 4, 2, 2, 3, 3, 2)$, and the remaining units follow from $d_{ht} = d_t$ and $d_{-t} = 6 - d_t$). Since $\pm 8H = \{8, 22, 83, 97\} \pmod{105}$ does not contain 13, the orbits $[8]$ and $[\pm 13]$ are distinct, and the individual determinant system $\bigoplus_{[t]} D_{[t]}$ is not of pure Hodge type $(2,2)$ (since d_{8g} takes values 2 and 4). Nevertheless, because $(d_{8g} - 1, 5 - d_{8g}) + (d_{-13g} - 1, 5 - d_{-13g}) = (d_{8g} + d_{-13g} - 2, 10 - (d_{8g} + d_{-13g})) = (4, 4)$ for every $g \in (\mathbf{Z}/105\mathbf{Z})^\times$, every Galois conjugate of $D_{[8]} \otimes D_{[-13]} \subset H^8(B_{\eta,\mathrm{prim}}, \mathbf{C})$ has Hodge type $(4,4)$ and is fixed by $\mathrm{Hg}(B_{\eta,\mathrm{prim}})_{\mathrm{der}}$. Consequently, the direct sum of the Galois conjugates of $D_{[8]} \otimes D_{[-13]}$ descends to a non-zero rational Hodge substructure of type $(4,4)$ in $H^8(B_{\eta,\mathrm{prim}}, \mathbf{Q})$ coupling the distinct orbits $[8]$ and $[-13]$ that is not generated by divisor classes (which pair $E_{[t]}$ with $E_{[-t]}$).
   - **Full Mumford–Tate Group and All-Powers Reduction on Characterwise $(2,2)$-Balanced Subfamilies:** On any characterwise $(2,2)$-balanced subfamily—where $d_t = 3$ and $\mathrm{SU}(J(t)) \simeq \mathrm{SU}(2,2)$ for **every** $t \in (\mathbf{Z}/m\mathbf{Z})^\times$, such as the companion family (1.8) at $m = 105, N = 7$ in Proposition 2.2(3)—every individual determinant line $D_{[t]} = \bigwedge^4 E_{[t]}$ already has Hodge type $(d_t - 1, 5 - d_t) = (2,2)$ across the entire Galois orbit. Consequently, the Hodge cocharacter acts by the identity on every $D_{[t]}$, the central torus of $\mathrm{Hg}(B_{\eta,\mathrm{prim}})$ is trivial, and the full generic Hodge group equals its derived group:
$$\mathrm{Hg}(B_{\eta,\mathrm{prim}})(\mathbf{R}) = \mathrm{Hg}(B_{\eta,\mathrm{prim}})_{\mathrm{der}}(\mathbf{R}) = \prod_{[t] \in (\mathbf{Z}/m\mathbf{Z})^\times/(\pm H)} \mathrm{SU}(2,2).$$
By the first fundamental theorem for SL_4 as in [3, Theorem 7.1 and Lemma 7.2] (with $F = L = \mathbf{Q}(\zeta_m)^H$ and $n = 4, q = 2$), the space of generic Hodge classes on every self-product $B^k_{\eta,\mathrm{prim}}$ $(k \geq 1)$ over S is generated as an algebra by divisor classes in $M_k(L)^{\dagger=1} \subseteq \mathrm{NS}(B^k_{\eta,\mathrm{prim}})_{\mathbf{Q}}$ together with the individual L-determinant volume subspace $W_L(B_{\eta,\mathrm{prim}}) := \bigwedge^4_L H^1(B_{\eta,\mathrm{prim}}, \mathbf{Q})$ (and its mixed-copy polarizations); thus all-powers rational Hodge algebraicity on $B^k_{\eta,\mathrm{prim}}$ reduces to the separate cycle-construction input $W_L(B_{\eta,\mathrm{prim}}) \subset \mathrm{cl}\ CH^2(B_{\eta,\mathrm{prim}})_{\mathbf{Q}}$ (Hypothesis (S) and equation (7.2) in [3, Theorem 1.1 and Theorem 7.1]), which is not claimed from monodromy alone. □

8.3 Methodological Comparison with Prior Work

We situate Theorems A–D and Corollary E relative to existing methods in the literature: 1. **Full Configuration Spaces and Rank-1 Reflection Generation ([1], [2], [3], [7], [8], [9], [10], [11], [12]):**
Works on the full configuration space $\mathcal{M}_{0,n}$ ([1, Proposition 9.2], [2, Proposition 3.3.5 and Theorem 5.1.1], [8, Theorems 4.1 and 4.4], [9, Theorem 1], [11, Theorems A and B], [3, Lemma 6.2]) or on 1-dimensional Jordan–Pochhammer slices where a single branch point moves against $n-1$ fixed branch points ([7, Proposition 2.10], [10, Theorem 2.1 and Propositions 3.4–3.5], [12, Section 2, Lemma 2.10]) rely on local rank-1 pseudoreflections around simple collisions and on induction along positive-dimensional boundary strata. On the symmetric 1-dimensional slice $S \subset \mathcal{M}_{0,6}$, Lemma 2.1 proves that all local puncture generators $g_p(t)$ $(p \in \{1, -1, 0, \infty\})$ and their non-scalar powers $g_p(t)^k \notin \mathbf{C}^\times I_4$ have $\mathrm{rank}(g_p(t)^k - cI_4) \geq 2$ under the non-resonance conditions of Lemma 2.1, and all boundary strata are 0-dimensional simultaneous collisions. Theorem 3.2 replaces reflection generation by an explicit 7-eigenvalue adjoint root-space dichotomy in the $g_1(t)$-eigenbasis combined with the linear disjointness $\mathbf{Q}(\zeta_a) \cap \mathbf{Q}(\zeta_{bN}) = \mathbf{Q}$. 2. **Signature-Injectivity**

vs. Equal-Signature Cyclotomic Separation ([13]):
Criteria for maximal Mumford–Tate groups based on distinct Hodge signatures ([13, Definition 4.3 and Theorem 4.4]) require the unordered signature pairs $\{f_t, f_{-t}\}$ to be pairwise distinct across CM orbits, which for 6 branch points restricts the conductor to $\varphi(m) \le 6$. On our families $m = 15N$ and $m = a \cdot b \cdot N$, $\varphi(m) \ge 48 \to \infty$ while every 4-dimensional block V_t falls into one of only two unordered signature buckets $\{\{(1,3),(3,1)\},\{(2,2)\}\}$ on the $m = 15N$ families (1.7)–(1.8) (and at most three buckets including $\{(0,4),(4,0)\}$ on the generalized family (1.9)). Moreover, Theorem 6.1 shows that even local puncture spectra and length-2 braid traces collide at $v \equiv (-1,+1) \pmod{(a,m')}$. Theorems 5.2, 6.1, and 7.2 achieve complete separation of all equal-signature and locally isospectral orbits using a single length-3 braid word $W_1 = g_1^2 g_0$ via the dual $\mathcal{B}_{15}$-basis over $\mathbf{Q}(\zeta_N)$ and four-way Galois involution projections. 3. **Unramified Curve Covers and Static Fermat Varieties ([14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25]):**
Classical works on Fermat varieties and cyclic covers ([14], [15, Theorem I], [16, Theorem 5.2], [17, Theorem 4.9], [18, Theorem A], [19, Theorem 2-1 and Corollary 2-3], [20, Theorem 1.2], [21, Theorems 2.0 and 3.2], [22, Theorem 3.4], [23, Proposition 2.2], [24, Theorem 3.1 and Corollary 3.2]) treat static hypersurfaces, rigid CM abelian factors associated to 3-point covers of $\mathbf{P}^1$, or rigid Weil-type Hodge substructures, while Patel–Zhang [25, Theorems 1.1 and 1.2] construct Weil-type cycles on static unramified abelian covers of curves of genus $g(C') \ge 2$. By contrast, Theorems A–D establish full special unitary monodromy $\prod \mathrm{SU}(J(t))$ (spanning all three indefinite real forms $\mathrm{SU}(1,3), \mathrm{SU}(2,2), \mathrm{SU}(3,1)$ on $m = 15N$, together with compact SU(4) when $d_t \in \{1,5\}$ on $m = a \cdot b \cdot N$) for ramified cyclic covers over the genus-zero base $\mathbf{P}^1$ moving along the 1-dimensional symmetric slice S. 4. **Characteristic-Zero Braid Invariants vs. Finite-Field Specialization and Vanishing-Sum Methods ([26], [27], [28], [29], [30], [31], [32]):**
Whereas arithmetic specialization methods certify endomorphism rigidity for individual conductors m by computing Frobenius characteristic polynomials at good-reduction primes (invoking Tate's theorem [26, Main Theorem], Shimura's reduction theory [27], and Néron–Ogg–Shafarevich specialization in SGA 7 I [28, Exposé IX]), and sheaf-theoretic or cyclotomic relation methods rely on middle convolution ([29]) or the classification of vanishing sums of roots of unity (Mann [30], Conway–Jones [31, Theorems 1 and 2], Lam–Leung [32]), the proofs in Sections 2–8 work directly with characteristic-zero braid matrices, Ramanujan-sum trace averages, and tensor-product Galois projections across the infinite families $m = 15N$ ($\gcd(N,30) = 1$) and $m = a \cdot b \cdot N$.

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9. Representative Conductors and Explicit Cyclotomic Evaluations

We record in Tables 9.1 and 9.2 the explicit cyclotomic invariants, Jacobi principal-minor and adjoint-spectrum counts, Ramanujan-sum Galois means $\mathcal{M}_{\mathrm{prim}}(N)$ and $\mathcal{M}_{\mathrm{comp}}(N)$, length-3 separation counts in $\mathbf{Z}[\zeta_m]$, and boundary counterexample distributions across representative prime, prime-power, and composite conductors for Theorems A–D.

9.1 Representative Conductors for Theorem B ($m = 15N, \gcd(N,30) = 1$)

N	Factorization Type	Conductor $m = 15N$	$\varphi(m)$	Non-Trivial Cosets in $\mathcal{T}_{\mathrm{prim}}/\pm 1$	Units with $P_{2..5} \neq 1$ & $\lvert\mathrm{Spec}(\mathrm{Ad}\, g_1)\rvert = 7$	Primary Galois Mean $\mathcal{M}_{\mathrm{prim}}(N)$	Companion Galois Mean $\mathcal{M}_{\mathrm{comp}}(N)$	Non-Trivial Cosets Separated by $W_1 = g_1^2 g_0$
7	Prime	105	48	11	48/48	$21 > 16$	34/3 ($\max\lvert\mathrm{tr}\rvert^2 = 20.59 > 16$)	11/11
11	Prime	165	80	19	80/80	$41 > 16$	$86/5 = 17.20 > 16$	19/19
13	Prime	195	96	23	96/96	$269/6 \approx 44.83 > 16$	$56/3 \approx 18.67 > 16$	23/23

N	Factorization Type	Conductor $m = 15N$	$\varphi(m)$	Non-Trivial Cosets in $\mathcal{T}_{\text{prim}}/\pm 1$	Units with $P_{2..5} \neq 1$ & $\lvert\operatorname{Spec}(\operatorname{Ad} g_1)\rvert = 7$	Primary Galois Mean $\mathcal{M}_{\text{prim}}(N)$	Companion Galois Mean $\mathcal{M}_{\text{comp}}(N)$	Non-Trivial Cosets Separated by $W_1 = g_1^2 g_0$
17	Prime	255	128	31	128/128	$397/8 = 49.625 > 16$	$41/2 = 20.50 > 16$	31/31
19	Prime	285	144	35	144/144	$461/9 \approx 51.22 > 16$	$190/9 \approx 21.11 > 16$	35/35
23 ... 73	13 Primes	345 ... 1095	176 ... 576	43 ... 143	100%	$\frac{64p-294}{p-1} \in [53.5, 60.8]$	$\frac{26p-114}{p-1} \in [22.0, 24.8]$	100%
49	**Prime Square** 7^2	735	336	83	336/336	$194/3 \approx 64.67 > 16$	$26 > 16$	83/83
121	**Prime Square** 11^2	1815	880	219	880/880	$64 > 16$	$26 > 16$	219/219
169	**Prime Square** 13^2	2535	1248	311	1248/1248	$64 > 16$	$26 > 16$	311/311
343	**Prime Cube** 7^3	5145	2352	587	2352/2352	$64 > 16$	$26 > 16$	587/587
77	**Composite** $7 \cdot 11$	1155	480	119	480/480	$683/10 = 68.30 > 16$	$412/15 \approx 27.47 > 16$	119/119

9.2 Representative Conductors for Theorems C and D ($m = a \cdot m'$ and $m = a \cdot b \cdot N$)

Theorem	Parameters (a, m') or (a, b, N)	Conductor m	$\varphi(m)$	Cyclotomic Invariants in $\mathbf{Z}[\zeta_m]$
Theorem C	$(3, 35), (5, 21), (7, 55), (9, 35), (11, 105), (11, 57), (25, 21), (27, 35)$	105 ... 1155	48 ... 480	$I_v(g_1 g_0) = I_{-1}(g_1 g_0)$, $I_v(g_1 g_{-1}) = I_{\pm 1}(g_1 g_{-1})$, and $I_v(g_1^2 g_0) \notin \{I_1(g_1^2 g_0), I_{-1}(g_1^2 g_0)\}$
Theorem D(1)	$(3, 7, 11)$	231	120	$P_{2..5} \neq 1$, $\lvert\operatorname{Spec}(\operatorname{Ad} g_1)\rvert = 7$, $\Delta_{--}, \Delta_{+-} \neq 0$, Mean $= 142/3 > 16$, 29/29 cosets separated

Theorem	Parameters (a, m') or (a, b, N)	Conductor m	$\varphi(m)$	Cyclotomic Invariants in $\mathbf{Z}[\zeta_m]$
Theorem D(1)	$(5, 3, 7)$	105	48	$P_{2..5} \neq 1$, $\lvert \operatorname{Spec}(\operatorname{Ad} g_1)\rvert = 7$, $\Delta_{--}, \Delta_{+-} \neq 0$ (48/48), $\lvert \operatorname{tr}(W(92))\rvert^2 > \frac{48043}{2500} > 16$, 11/11 cosets separated
Theorem D(1)	$(5, 7, 11)$	385	240	$P_{2..5} \neq 1$, $\lvert \operatorname{Spec}(\operatorname{Ad} g_1)\rvert = 7$, $\Delta_{--}, \Delta_{+-} \neq 0$, Mean $= 676/15 > 16$, 59/59 cosets separated
Theorem D(1)	$(7, 3, 11)$	231	120	$P_{2..5} \neq 1$, $\lvert \operatorname{Spec}(\operatorname{Ad} g_1)\rvert = 7$, $\Delta_{--}, \Delta_{+-} \neq 0$, Mean $= 442/15 > 16$, 29/29 cosets separated
Theorem D(1)	$(7, 5, 11)$	385	240	$P_{2..5} \neq 1$, $\lvert \operatorname{Spec}(\operatorname{Ad} g_1)\rvert = 7$, $\Delta_{--}, \Delta_{+-} \neq 0$, Mean $= 191/5 > 16$, 59/59 cosets separated
Theorem D(1)	$(9, 5, 7)$ & $(5, 9, 7)$	315	144	$P_{2..5} \neq 1$, $\lvert \operatorname{Spec}(\operatorname{Ad} g_1)\rvert = 7$, $\Delta_{--}, \Delta_{+-} \neq 0$ (144/144), Mean $= 52/3 > 16$ at $(9, 5, 7)$ and $112/3 > 16$ at $(5, 9, 7)$, 35/35 cosets separated
Theorem D(1)	$(5, 3, 13)$ & $(11, 3, 7)$	195, 231	96, 120	$P_{2..5} \neq 1$, $\lvert \operatorname{Spec}(\operatorname{Ad} g_1)\rvert = 7$, $\Delta_{--}, \Delta_{+-} \neq 0$ (100%), all non-trivial cosets separated

Theorem	Parameters (a, m') or (a, b, N)	Conductor m	$\varphi(m)$	Cyclotomic Invariants in $\mathbf{Z}[\zeta_m]$
Theorem D(2)	$(9,5,7), (3,7,11), (5,3,7), (5,3,15)$	$315, 231, 105, 105$	$144, 120, 48, 48$	Companion-template examples (b.1) $d_t \in \{2:32, 3:80, 4:32\}$ at $(9,5,7)$ and (b.2) 3 distinct t_7-values at $(3,7,11)$; primary-family example (b.3) Mean $= 35/3 < 16$ at $(5,3,7)$; and (b.4) $\lvert\operatorname{Spec}(\operatorname{Ad} g_1)\rvert = 3$ at $(5,7,3)$ outside $\gcd(N,30) = 1$

Data availability

All proofs of Theorems A, B, C, D, and Corollary E are self-contained algebraic and topological arguments in Sections 2–8. The finite rational and cyclotomic polynomial ring evaluations in $\mathbf{Z}[X]/(\Phi_m(X))$ summarized in Tables 9.1 and 9.2 can be reproduced directly from the explicit matrix formulas (2.2)–(2.3), the trace identities (4.1), (4.7), (4.9), (5.1), (7.2), and (7.9), and the factorizations (6.6)–(6.8) and (7.18). The accompanying supplement provides the standalone verification script `verify_slice_monodromy.py` (`python3 verify_slice_monodromy.py`, using Python 3 and SymPy), its output log `verify_slice_monodromy.out`, and checksum file `CONTENTS.sha256` to reproduce: (i) the symbolic matrix identities over $\mathbf{Z}[b_1^{\pm1}, \dots, b_5^{\pm1}]$ (Check 1), (ii) the exact rational `fractions.Fraction` Ramanujan-sum means, Kubert signature distributions, and Corollary E(3) determinant-character identities (Checks 2, 3, and 6), (iii) the 100% floating-point-free exact cyclotomic polynomial ring evaluations verifying complete coordinate separation $\tau_t(g_1^2 g_0) \neq \pm\tau_{\pm1}(g_1^2 g_0)$ in $\mathbf{Z}[z]/(\Phi_N(z))$ across all 23 conductors N in Table 9.1 (both Primary and Companion families) and in the tensor-product cyclotomic ring $\mathbf{Z}[x_a, x_b, z]/(\Phi_a(x_a), \Phi_b(x_b), \Phi_N(z))$ across all 12 three-factor triples (a, b, N) in Table 9.2 (Check 7), and (iv) independent numerical `cmath` cross-checks (Checks 4, 5, and 6). These finite checks serve as supporting symbolic, exact-ring, and numerical corroboration rather than a substitute for the general proofs in Sections 2–8.

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